

The Escauriaza–Seregin–Šverák theorem: version for the Lean formalization

Abstract

We state a suitable form of Leray’s global existence theorem for the three-dimensional Navier–Stokes equations on $\mathbb{R}^3$, which is proved in the companion Caffarelli–Kohn–Nirenberg paper, and we prove the endpoint regularity theorem of Escauriaza, Seregin, and Šverák. Every divergence-free initial datum in $L^2(\mathbb{R}^3)$ has a global Leray–Hopf solution that is suitable and has a canonical pressure in $L^{5/3}(\mathbb{R}^3 \times (0, T))$ on each finite time interval. The formalized Caffarelli–Kohn–Nirenberg theorem then gives zero one-dimensional parabolic Hausdorff measure for its singular set. Every Leray–Hopf solution with finite $L^\infty(0, T; L^3(\mathbb{R}^3))$ norm is regular at every point of $\mathbb{R}^3 \times (0, T)$, belongs to $L^5(\mathbb{R}^3 \times (0, T))$, and is unique among Leray–Hopf solutions with the same datum. The endpoint argument combines an open-top smallness criterion, vorticity regularity, and explicit Carleman estimates for unique continuation and backward uniqueness. We also prove the Ladyzhenskaya–Prodi–Serrin theorem for the full range $3 < s \leq \infty$, smoothness under the critical $L_t^\infty L_x^3$ bound, and the resulting regularity criterion for $3 \leq s \leq \infty$. These results, together with the unique continuation, backward uniqueness and Carleman theorems, are formally verified in Lean 4.

Contents

1	Introduction	3
2	Notation and conventions	7
3	Solution classes and main results	7
4	Carleman inequalities	11
5	Unique continuation across spatial boundaries	19
6	Backward uniqueness on a half-space	23
7	Leray existence and the associated pressure	29
8	Local regularity near small-energy points	30
9	Stability of suitable weak solutions and the blow-up limit	44
10	Proof of the local theorem	53
11	Proof of the global theorem	59
12	The trace and heat flow	62
13	The forced Stokes response	63

14	A solution near the initial time	65
15	Weak–strong comparison	66
16	Positive times and the global conclusion	68
17	Statements	70
18	Energy equality and comparison	71
19	Strong solutions and continuation	75
20	Smooth representatives	79
A	Corrections to the sources	82

1 Introduction

The three-dimensional Navier–Stokes equations pose two basic questions for finite-energy data: whether a global weak solution exists, and whether such a solution can develop singularities. Leray’s construction gives global weak solutions on $\mathbb{R}^3$ for divergence-free initial velocities in $L^2(\mathbb{R}^3)$ [7]. The Caffarelli–Kohn–Nirenberg theorem gives a geometric answer to the second question for *suitable* weak solutions: their singular set has zero one-dimensional parabolic Hausdorff measure [3]. The corresponding theorem is available in the formalized CKN library [2]. Thus suitability is the condition that lets Leray’s existence result carry the CKN partial-regularity conclusion.

Escauriaza, Seregin, and Šverák proved a sharper endpoint statement: a Leray–Hopf solution whose velocity belongs to $L^\infty(0, T; L^3(\mathbb{R}^3))$ has no singular points in the open time interval [4, Theorems 1.3–1.4]. They also proved that this bound forces the velocity into space-time L^5 and gives uniqueness among Leray–Hopf solutions with the same datum. This norm is invariant under Navier–Stokes scaling, so the result settles regularity at a critical integrability threshold. The proof here gives the endpoint L^5 and uniqueness conclusions by constructing a small-time L^5 solution and proving a weak–strong comparison; it also fixes the associated pressure and makes the endpoint regularity steps explicit.

Main results. Section 3 fixes the solution notions and contains the precise definitions used below. Statements (A)–(E) and (G) repeat results stated in Section 3; (A)–(C) and (G) are proved in the Caffarelli–Kohn–Nirenberg paper [2] (Section 7); statement (F) is Theorem 16.3, stated and proved in Part V. Statements (H)–(J) are Theorem 17.1 and Corollaries 17.2 and 17.3, stated and proved in Part VI.

- (A) **Leray existence.** For every $a \in J$ there are u , Du and p such that (u, Du) is a global Leray–Hopf solution with datum a , and $(u, Du, p, 0)$ is a suitable weak solution on $\mathbb{R}^3 \times (0, \infty)$ for every $q > 5/2$ (Theorem 3.5).
- (B) **The CKN headline.** For every $a \in J$ there is a global Leray–Hopf solution (u, Du) with datum a whose singular set in $\mathbb{R}^3 \times (0, \infty)$ has zero one-dimensional parabolic Hausdorff measure (Corollary 3.6).
- (C) **Associated pressure.** Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$. There is $p \in L^{5/3}(\mathbb{R}^3 \times (0, T))$ such that (S3) of Definition 3.1 holds on $\mathbb{R}^3 \times (0, T)$ with $f = 0$. If in addition $\text{ess sup}_{t \in (0, T)} \|u(\cdot, t)\|_{L^3(\mathbb{R}^3)} < \infty$, then this p also belongs to $L^\infty((0, T); L^{3/2}(\mathbb{R}^3))$ (Theorem 3.7).
- (D) **Local endpoint regularity.** Let u , Du , p be measurable on $Q = B_1 \times (-1, 0)$ with

$$\begin{aligned} \text{ess sup}_{t \in (-1, 0)} \int_{B_1} |u(x, t)|^2 dx < \infty, \quad \iint_Q |u|^2 + |Du|^2 < \infty, \quad p \in L^{3/2}(Q), \\ \text{ess sup}_{t \in (-1, 0)} \int_{B_1} |u(x, t)|^3 dx < \infty, \end{aligned}$$

such that $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ on B_1 for a.e. $t \in (-1, 0)$, and (S2) and (S3) of Definition 3.1 hold on Q with $f = 0$. Then there is $\gamma \in (0, 1]$ and a function $\tilde{u}$ equal to u almost everywhere on $B_{1/2} \times (-1/4, 0)$ that is parabolically Hölder continuous with exponent γ on the closure of $B_{1/2} \times (-1/4, 0)$ (Theorem 3.12).

- (E) **Global endpoint regularity.** Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$, and suppose $\text{ess sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty$. Then every point of $\mathbb{R}^3 \times (0, T)$ is a regular point of u (Theorem 3.13).
- (F) **L^5 and uniqueness.** Under the same hypotheses, $u \in L^5(\mathbb{R}^3 \times (0, T))$, and every other Leray–Hopf solution (v, Dv) on $[0, T]$ with datum a satisfies $u = v$ almost everywhere on $\mathbb{R}^3 \times (0, T)$ (Theorem 16.3).
- (G) **Forced Leray existence.** For data and forces in the class of Theorem 3.9, there is one global forced Leray–Hopf solution that is suitable for every $q > 5/2$ for which f is locally in L^q ; its singular set has zero one-dimensional parabolic Hausdorff measure, and its associated pressure has the local split stated in Theorems 3.10 and 3.11.
- (H) **Ladyzhenskaya–Prodi–Serrin.** Every Leray–Hopf solution in $L^\ell(0, T; L^s(\mathbb{R}^3))$ with $3/s + 2/\ell = 1$ and $3 < s \leq \infty$ is unique among Leray–Hopf solutions with its datum and has a representative smooth on $\mathbb{R}^3 \times (0, T]$, one-sided at T (Theorem 17.1).
- (I) **Smooth critical endpoint.** Every Leray–Hopf solution in $L^\infty(0, T; L^3(\mathbb{R}^3))$ has the same smoothness and uniqueness, as well as a finite space-time L^5 norm (Corollary 17.2).
- (J) **Regularity criterion.** Combining (H) and (I): every Leray–Hopf solution in $L^\infty(0, T; L^3(\mathbb{R}^3))$, in $L^\ell(0, T; L^s(\mathbb{R}^3))$ with $3 < s < \infty$ and $\ell = 2s/(s - 3)$, or in $L^2(0, T; L^\infty(\mathbb{R}^3))$ is unique among Leray–Hopf solutions with its datum and has a representative smooth on $\mathbb{R}^3 \times (0, T]$, one-sided at T (Corollary 17.3).

Smoothness. Part VI proves the Ladyzhenskaya–Prodi–Serrin theorem in the full range $3 < s \leq \infty$, including smoothness through the top time. At $(s, \ell) = (5, 5)$ it turns the L^5 conclusion of Theorem 16.3 into Corollary 17.2. Theorem 16.3 itself states only integrability and uniqueness; smoothness is stated separately, in Theorem 17.1 and Corollaries 17.2 and 17.3.

Part I: linear continuation. The ESS argument ultimately applies backward uniqueness to vorticity, but its linear inputs must match the regularity actually obtained from a suitable weak solution. We therefore formulate the linear fields in the space-time weak-derivative class $W_2^{2,1}$ and require continuity only where a trace is evaluated. Propositions 4.1 and 4.2 first establish the Gaussian and anisotropic Carleman estimates for smooth compactly supported fields, then Lemma 4.4 extends them by density to the fields used in the continuation arguments. The first estimate propagates infinite-order vanishing across a spatial boundary through an L^2 Gaussian bound on parabolic boxes; this replaces the pointwise local regularity estimate used in ESS Lemma 4.3. The half-space estimate gives decay away from the boundary plane. Its initial-time cutoff is controlled by a trace inequality, and its noncompact spatial cutoff is justified by collar estimates, so no zero extension across the initial slice is needed. The final large-growth case of backward uniqueness is handled on finitely many rescaled time slabs. The constants are explicit: the Gaussian estimate uses $c_0 = e^{4/3}(9 + 2\sqrt{6})$, and the half-space estimate uses $a_0(\alpha) = 2$ and $c_*(\alpha) = 5 + 7/(2(2\alpha - 1))$. These details make the linear results applicable in the weak vorticity class without adding smoothness assumptions.

Parts II and III: existence and the associated pressure. The Leray–Hopf weak equation is initially tested only against divergence-free vector fields, whereas suitability requires the momentum identity for arbitrary compactly supported vector tests. The proofs that every $a \in J$ has a global

Leray–Hopf solution that is suitable, that every Leray–Hopf solution has an associated pressure defined by the whole-space double Riesz transform of $u \otimes u$, and their versions with a force, do not use Parts I and IV–VI. They were moved to the last part of the Caffarelli–Kohn–Nirenberg paper [2], where the statements of Section 3 keep their labels; Section 7 records this and the whole-space pressure used in Parts IV–VI. The construction there uses spatially regularized solutions given by an L^2 Fourier mild formulation, in place of the Oseen-kernel construction of Ożański–Pooley [8, Theorem 4.2], and passes to the limit with global tail control by a compactness argument that does not rely on a Helly selection theorem, Rellich compactness, or Aubin–Lions. Since the regularized pairs do not solve the unregularized equations, the limit is taken directly in their momentum and local-energy identities. The Caffarelli–Kohn–Nirenberg theorem then gives the stated measure bound for the singular set.

Part IV: the $L^\infty L^3$ endpoint. The local endpoint theorem assumes the weak momentum equation and a uniform L^3 bound but does not assume the local energy inequality. A local L^4 estimate and space-time mollification recover the local energy equality; a momentum-pairing modulus supplies a weakly continuous L^3 representative up to the top time, and a fixed whole-space pressure split controls the harmonic remainder under rescaling. We then measure smallness on open past cylinders by

$$r^{-2} \int_{t_0-r^2}^{t_0} \int_{B_r(x_0)} (|u|^3 + |p|^{3/2}) \, dx \, dt.$$

If this scale-invariant quantity is below the CKN threshold, Theorem A gives interior Hölder regularity. To reach the terminal face, shifted centers and overlapping representatives give a one-sided Hölder extension; the top point is not treated as an interior CKN regular point. The good-point lemma makes this criterion open and yields a lower bound at every bad point. Thus failure of Hölder regularity on the target compact cylinder produces the quantitative lower bound needed for a blow-up contradiction.

After rescaling a bad point, compactness gives an ancient suitable solution with zero terminal velocity trace and a pressure equal to the whole-space Riesz pressure of its velocity tensor. Far from the origin, the L^3 and pressure tails are small; CKN epsilon regularity makes the velocity bounded there. We then bootstrap the weak vorticity equation only as far as needed: a localized vorticity energy estimate, local heat gains, product bounds, and div–curl estimates give $W_2^{2,1}$ control and continuity of vorticity. This finite weak-to-regular argument replaces the all-orders Stokes/maximal-regularity bounds and local strong-solvability input used in the exterior part of ESS’s proof [4, Theorem 1.4, §§3–5]. Half-space backward uniqueness forces the vorticity to vanish outside a ball. At almost every time the CKN singular-set theorem supplies a time interval of regular points; unique continuation spreads the vanishing across the ball, and an L^3 Liouville argument makes the entire blow-up velocity zero. That contradicts the bad-point lower bound. For the global theorem, parabolic rescaling applies the local result at a later center, placing each interior point strictly inside the resulting Hölder cylinder.

Part V: L^5 and uniqueness. The critical bound gives only an almost-every-time L^3 estimate, so the first step is to identify the initial datum in L^3 using good times and the strong L^2 trace. A heat-flow approximation then makes the linear solution small in $L^5 \cap L^4$ on a short interval. The forced Stokes estimates control the quadratic term in those two norms, producing a local Leray–Hopf solution; the $(5, 5)$ energy comparison identifies it with the given weak solution. Away from zero, the pressure tail and the local regularity estimate give a uniform velocity bound over

the whole space, and the multiplicative inequality converts that bound and the finite dissipation into L^5 . The final comparison applies to every Leray–Hopf solution with the same datum.

Lean formalization. The main results of this manuscript, together with the linear continuation results used in their proofs, are formally verified in the proof assistant Lean 4 with its mathematical library Mathlib, building on the Lean formalization of the Caffarelli–Kohn–Nirenberg theorem [2]. The formalization accompanies this manuscript [1]. Ten of the results in the table below are stated there as Lean theorems, each in its own file of the directory `ESS/Statements`; the theorem `ESS.essGlobal`, for example, is stated in `ESS/Statements/EssGlobal.lean`, and the other files are named in the same way. The six results in the first part of the table, on Leray existence and the associated pressure, are proved in the Caffarelli–Kohn–Nirenberg paper [2] (Section 7) and are formalized in the CKN library, each as a Lean theorem in its own file of the directory `CKN/Statements`; the ESS library imports them.

Result	Content	Lean theorem
Theorem 3.5	Leray existence	<code>CKN.leray_existence</code>
Corollary 3.6	Leray existence, singular set	<code>CKN.leray_existence_singularSet</code>
Theorem 3.7	associated pressure	<code>CKN.associatedPressure</code>
Theorem 3.9	forced Leray existence	<code>CKN.lerayExistenceForced</code>
Theorem 3.10	forced existence, singular set	<code>CKN.lerayExistenceForcedSingularSet</code>
Theorem 3.11	forced associated pressure	<code>CKN.associatedPressureForced</code>
Theorem 3.12	local endpoint regularity	<code>ESS.essLocal</code>
Theorem 3.13	global endpoint regularity	<code>ESS.essGlobal</code>
Proposition 4.1	Gaussian Carleman inequality	<code>ESS.carlemanGaussian</code>
Proposition 4.2	half-space Carleman inequality	<code>ESS.carlemanHalfSpace</code>
Theorem 5.2	unique continuation	<code>ESS.uniqueContinuation</code>
Theorem 6.7	backward uniqueness	<code>ESS.backwardUniqueness</code>
Theorem 16.3	L^5 and uniqueness	<code>ESS.essL5Unique</code>
Theorem 17.1	mixed-norm smoothness and uniqueness	<code>ESS.ladyzhenskayaProdiSerrin</code>
Corollary 17.2	smooth critical endpoint	<code>ESS.essSmooth</code>
Corollary 17.3	criterion for $3 \leq s \leq \infty$	<code>ESS.serrinCriterion</code>

The definitions of J , finite-time and global Leray–Hopf solutions and their forced versions, the force classes, the forced quadratic tensor, and the space-time weak-derivative class are stated in `CKN/Statements`. The ten ESS theorems are stated in `ESS/Statements`; they use these CKN definitions, and suitable weak solutions and regular points are also those of [2]. The six CKN theorems and the ten ESS theorems are proved in full, without `sorry`, and their proofs use only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The formal proofs follow the arguments of this manuscript and of the CKN paper, although they are not a line-by-line transcription of them. As an independent check of the statements, the directory `comparators` of the ESS library contains two pairs of files, `Linear` and `Regularity`. In each pair, the file `Challenge.lean` imports only Mathlib and restates its theorems with every definition written directly in terms of Mathlib, and the file `Solution.lean` derives each restatement from the corresponding theorem of the formalization. Together they restate all ten ESS theorems. The comparator pair for the Leray results is part of the CKN library; it restates five of the six theorems above, and Theorem 3.11 is not restated, because its pressure constructions are too long for a self-contained restatement.

Outline. Section 2 sets the notation, and Section 3 gives the solution classes and main statements. Part I (Sections 4–6) proves the linear continuation results; Parts II and III (Section 7) record

the existence and associated-pressure results, which are proved in the CKN paper [2]; Part IV (Sections 8–11) proves the regularity endpoint; Part V (Sections 12–16) proves the L^5 and uniqueness conclusions; and Part VI (Sections 17–20) proves the mixed-norm theorem, the smooth endpoint corollary and the regularity criterion for $3 \leq s \leq \infty$. The appendix records corrections and proof details in the comparison sources.

2 Notation and conventions

Space, time and norms. Space is $\mathbb{R}^3$ and time is $\mathbb{R}$; a space-time point is $z = (x, t)$. On vectors and matrices, $|\cdot|$ is the Euclidean (Frobenius) norm. For a field $w: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$ with components w_i we write $\partial_j w_i$ for spatial and $\partial_t w_i$ for time derivatives, $|\nabla w|^2 = \sum_{i,j} (\partial_j w_i)^2$ and $|\nabla^2 w|^2 = \sum_{i,j,k} (\partial_j \partial_k w_i)^2$. The ball $B_r(x)$ is open; $B_r = B_r(0)$.

Open past cylinders. For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$\begin{aligned} Q_r^-(z_0) &= B_r(x_0) \times (t_0 - r^2, t_0), \\ \overline{Q_r^-(z_0)} &= \overline{B_r(x_0)} \times [t_0 - r^2, t_0]. \end{aligned}$$

Space-time weak derivatives. Let $U \subset \mathbb{R}^3 \times \mathbb{R}$ be open and $w \in L^1_{\text{loc}}(U; \mathbb{R}^3)$. Fields $Dw = (Dw_{ij})$, $D^2w = (D^2w_{ijk})$ and $D_t w = (D_t w_i)$ in $L^1_{\text{loc}}(U)$ are the *space-time weak derivatives* $\partial_j w_i$, $\partial_j \partial_k w_i$ and $\partial_t w_i$ of w on U if, for every $\varphi \in C_c^\infty(U)$ and all indices,

$$\int_U w_i \partial_j \varphi = - \int_U Dw_{ij} \varphi, \quad \int_U Dw_{ij} \partial_k \varphi = - \int_U D^2w_{ijk} \varphi, \quad \int_U w_i \partial_t \varphi = - \int_U D_t w_i \varphi.$$

Such data are unique almost everywhere. We then write ∇w , $\nabla^2 w$, $\partial_t w$ for them and $Lw = \partial_t w + \Delta w$ for the field $D_t w_i + \sum_j D^2w_{ijj}$; inequalities involving these are understood almost everywhere. The class $W_2^{2,1}(U)$ consists of those w with $|w| + |\nabla w| + |\nabla^2 w| + |\partial_t w| \in L^2(U)$.

3 Solution classes and main results

This section fixes the solution notions and states the main results. The suitable weak solutions are exactly those of the formalized Caffarelli–Kohn–Nirenberg theorem [2]; the Leray–Hopf solutions transcribe (1.3)–(1.7) of [4].

Test functions and suitable weak solutions. For open $\Omega \subset \mathbb{R}^3$ and an open interval $I \subset \mathbb{R}$ let $C_c^\infty(\Omega \times I)$ denote the smooth functions with compact support in $\Omega \times I$. A velocity is a map $u: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$, and its weak spatial gradient is carried as separate data $Du: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^{3 \times 3}$ with $Du_{ij} = \partial_j u_i$.

Definition 3.1 (Suitable weak solution; [2, Definition 1.5]). Let $q > 5/2$, $\Omega \subset \mathbb{R}^3$ open, $I \subset \mathbb{R}$ an open interval, and $f \in L^q_{\text{loc}}(\Omega \times I; \mathbb{R}^3)$. The quadruple (u, Du, p, f) is a *suitable weak solution* on $\Omega \times I$ if the following hold.

- (S1) For every open Ω' with compact closure $\overline{\Omega'} \subset \Omega$ and every interval J with compact closure $\overline{J} \subset I$: u, Du, p, f are measurable on $\Omega' \times J$; $\text{ess sup}_{t \in J} \int_{\Omega'} |u(x, t)|^2 dx < \infty$; $\iint_{\Omega' \times J} |u|^2 + |Du|^2 < \infty$; $p \in L^{3/2}(\Omega' \times J)$; $f \in L^q(\Omega' \times J)$; and for a.e. $t \in J$, $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ on Ω' .

(S2) $\iint u \cdot \nabla \psi = 0$ for every $\psi \in C_c^\infty(\Omega \times I)$.

(S3) For every $\varphi \in C_c^\infty(\Omega \times I; \mathbb{R}^3)$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i - p \operatorname{div} \varphi - f \cdot \varphi \right] = 0.$$

(S4) For every $\psi \in C_c^\infty(\Omega \times I)$ with $\psi \geq 0$,

$$2 \iint |Du|^2 \psi \leq \iint \left[|u|^2 (\partial_t \psi + \Delta \psi) + (|u|^2 + 2p) u \cdot \nabla \psi + 2(f \cdot u) \psi \right].$$

A point $z_0 \in \Omega \times I$ is *regular* if u agrees almost everywhere on some open neighborhood $N \subset \Omega \times I$ of z_0 with a parabolically Hölder continuous function [2, Definition 2.2]; the *singular set* is the set of points of $\Omega \times I$ that are not regular.

Divergence-free data. Following [4], J is the closure in $L^2(\mathbb{R}^3; \mathbb{R}^3)$ of the smooth compactly supported divergence-free fields. We use the equivalent description by the weak divergence.

Lemma 3.2. *A field $a \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ belongs to J if and only if $\int_{\mathbb{R}^3} a \cdot \nabla \psi = 0$ for every $\psi \in C_c^\infty(\mathbb{R}^3)$.*

Proof of Lemma 3.2. If $a \in J$, choose $a_k \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ with $\operatorname{div} a_k = 0$ and $a_k \rightarrow a$ in L^2 . For $\psi \in C_c^\infty(\mathbb{R}^3)$, integration by parts gives $\int a_k \cdot \nabla \psi = 0$, and $|\int (a_k - a) \cdot \nabla \psi| \leq \|a_k - a\|_2 \|\nabla \psi\|_2 \rightarrow 0$. This proves the weak-divergence identity.

Conversely, suppose that $a \in L^2$ satisfies that identity. With the Fourier convention $\widehat{f}(\xi) = \int e^{-2\pi i x \cdot \xi} f(x) dx$, for $\delta > 0$ define a vector potential A_δ and its curl b_δ by

$$\widehat{A}_\delta(\xi) = \frac{i \xi \times \widehat{a}(\xi)}{2\pi(|\xi|^2 + \delta^2)}, \quad \widehat{b}_\delta(\xi) = \frac{|\xi|^2}{|\xi|^2 + \delta^2} \widehat{a}(\xi). \quad (3.1)$$

The weak divergence identity gives $\xi \cdot \widehat{a}(\xi) = 0$ almost everywhere, so $b_\delta = \operatorname{curl} A_\delta$. The multipliers show that A_δ and its first weak derivatives belong to L^2 . Spatially mollifying this H^1 potential gives smooth L^2 potentials whose curls converge to b_δ in L^2 ; cutting off each smooth potential on expanding balls shows that its curl belongs to J . Thus $b_\delta \in J$.

For $\delta_n = 1/(n+1)$ the curl multiplier is bounded by one and converges to one almost everywhere. Plancherel and dominated convergence therefore give

$$\|b_{\delta_n} - a\|_2 \rightarrow 0. \quad (3.2)$$

The closedness of J in L^2 now gives $a \in J$. This route regularizes the Fourier potential and passes to its curl in L^2 ; it does not require an L^6 potential estimate or a prescribed cutoff-curl sequence for a . $\square$

Leray–Hopf solutions. The velocity of a Leray–Hopf solution is a function defined at every time, and it is this function, not an equivalence class, that the definition constrains at every time. The same function is used as the velocity of a suitable weak solution.

Definition 3.3 (Leray–Hopf solution). Let $0 < T < \infty$ and $a \in J$. The pair (u, Du) is a *Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with initial datum a* if the following hold.

- (LH1) *Energy class*, [4, (1.3)]. u and Du are measurable on $\mathbb{R}^3 \times (0, T)$; $\operatorname{ess\,sup}_{t \in (0, T)} \int |u(x, t)|^2 dx < \infty$; $\iint_{\mathbb{R}^3 \times (0, T)} |u|^2 + |Du|^2 < \infty$; and for a.e. $t \in (0, T)$, $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ on $\mathbb{R}^3$ and $\int u(x, t) \cdot \nabla \psi(x) dx = 0$ for every $\psi \in C_c^\infty(\mathbb{R}^3)$.
- (LH2) *Weak continuity*, [4, (1.4)]. For every $w \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ the map $t \mapsto \int u(x, t) \cdot w(x) dx$ is continuous on $[0, T]$.
- (LH3) *Weak equation*, [4, (1.5)]. For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$ with $\operatorname{div} \varphi = 0$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i \right] = 0.$$

- (LH4) *Energy inequality*, [4, (1.6)]. For every $t_0 \in [0, T]$,

$$\frac{1}{2} \int |u(x, t_0)|^2 dx + \int_0^{t_0} \int |Du|^2 dx dt \leq \frac{1}{2} \int |a|^2 dx.$$

- (LH5) *Initial datum*, [4, (1.7)]. $\int |u(x, t) - a(x)|^2 dx \rightarrow 0$ as $t \downarrow 0$.

A *global Leray–Hopf solution* with datum a is a pair (u, Du) that is a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ for every $T > 0$.

Remark 3.4 (The global notion). [4] define the global notion by replacing $[0, T]$ with $[0, \infty)$ in (1.3)–(1.7). Read literally, (1.3) would then require $u \in L^2(0, \infty; W^{1,2})$, that is, $\int_0^\infty \int |u|^2 < \infty$, which Leray’s solutions need not satisfy; the energy inequality controls only $\sup_t \int |u|^2$ and $\int_0^\infty \int |\nabla u|^2$. We therefore define global solutions through every finite interval, which is the reading under which [4, Theorem 1.1] holds.

Main results. The first theorem is Leray’s existence theorem, with the constructed solution suitable in the sense of Definition 3.1.

Theorem 3.5 (Existence). *For every $a \in J$ there are u , Du and p such that (u, Du) is a global Leray–Hopf solution with datum a , and $(u, Du, p, 0)$ is a suitable weak solution on $\mathbb{R}^3 \times (0, \infty)$ for every $q > 5/2$.*

Combined with the Caffarelli–Kohn–Nirenberg theorem [2, Theorem 2.10], this gives the headline of [3].

Corollary 3.6. *For every $a \in J$ there is a global Leray–Hopf solution (u, Du) with datum a whose singular set in $\mathbb{R}^3 \times (0, \infty)$ has zero one-dimensional parabolic Hausdorff measure.*

Every Leray–Hopf solution has a pressure.

Theorem 3.7 (Associated pressure). *Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$. There is $p \in L^{5/3}(\mathbb{R}^3 \times (0, T))$ such that (S3) of Definition 3.1 holds on $\mathbb{R}^3 \times (0, T)$ with $f = 0$. If in addition $\operatorname{ess\,sup}_{t \in (0, T)} \|u(\cdot, t)\|_{L^3(\mathbb{R}^3)} < \infty$, then this p also belongs to $L^\infty((0, T); L^{3/2}(\mathbb{R}^3))$.*

Forced Leray–Hopf solutions. The same all-time velocity representative is used in the forced equation, the work term, and the initial trace.

Definition 3.8 (Forced Leray–Hopf solution). Let $0 < T < \infty$, $a \in J$, and $f \in L^2(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. The pair (u, Du) is a *forced Leray–Hopf solution* on $\mathbb{R}^3 \times [0, T]$ with datum a and force f if it satisfies (LH1), (LH2), and (LH5) of Definition 3.3, and also the following two clauses:

(FLH1) For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$ with $\operatorname{div} \varphi = 0$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i - f \cdot \varphi \right] dx dt = 0.$$

(FLH2) For every $t \in [0, T]$,

$$\frac{1}{2} \int_{\mathbb{R}^3} |u(x, t)|^2 dx + \int_0^t \int_{\mathbb{R}^3} |Du|^2 dx ds \leq \frac{1}{2} \int_{\mathbb{R}^3} |a|^2 dx + \int_0^t \int_{\mathbb{R}^3} f \cdot u dx ds.$$

A *global forced Leray–Hopf solution* is one pair (u, Du) that is a forced Leray–Hopf solution on every finite interval, with the same representative and force.

The force work is an ordinary Lebesgue integral: on every finite slab, $u \in L_t^\infty L_x^2$ and $f \in L_{t,x}^2$, so $f \cdot u \in L_{t,x}^1$.

Theorem 3.9 (Forced suitable existence). *Let $a \in J$ and let $f: \mathbb{R}^3 \times (0, \infty) \rightarrow \mathbb{R}^3$ satisfy $f \in L^2(\mathbb{R}^3 \times (0, T))$ for every finite T and $f \in L_{\text{loc}}^{q_0}(\mathbb{R}^3 \times (0, \infty))$ for some $q_0 > 5/2$. There are one velocity u , weak gradient Du , and pressure p such that (u, Du) is a global forced Leray–Hopf solution with datum a . For every $q > 5/2$ for which $f \in L_{\text{loc}}^q(\mathbb{R}^3 \times (0, \infty))$, the same quadruple (u, Du, p, f) is a suitable weak solution on $\mathbb{R}^3 \times (0, \infty)$ in the sense of Definition 3.1. In particular, its local energy inequality contains the term $2(f \cdot u)\psi$.*

Theorem 3.10 (Forced singular-set conclusion). *Under the hypotheses of Theorem 3.9, for every $q > 5/2$ for which $f \in L_{\text{loc}}^q$, the witness there satisfies*

$$\mathcal{H}_{\text{par}}^1(\operatorname{Sing}(u)) = 0.$$

Theorem 3.11 (Forced associated pressure). *Let (u, Du) be a forced Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$ and $f \in L^2(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. Let $p_N = P[u \otimes u]$ and let p_f be the force pressure of [2, Lemma 13.1]. Then*

$$p_N \in L^{5/3}(\mathbb{R}^3 \times (0, T)), \quad p_f \in L^2((0, T); L^6(\mathbb{R}^3)), \quad p = p_N + p_f \in L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T)).$$

For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i - p \operatorname{div} \varphi - f \cdot \varphi \right] dx dt = 0.$$

When $f = 0$, this pressure is the pressure of Theorem 3.7 and has all of its stated global and conditional integrability properties.

The next theorem is the local result [4, Theorem 1.4]. Its hypotheses are those of a suitable weak solution without the local energy inequality, imposed on the whole cylinder $B_1 \times (-1, 0)$ rather than on compactly contained subcylinders, together with the $L^\infty L^3$ bound.

Theorem 3.12 (Local regularity). *Let u, Du, p be measurable on $Q = B_1 \times (-1, 0)$ with*

$$\begin{aligned} \operatorname{ess\,sup}_{t \in (-1, 0)} \int_{B_1} |u(x, t)|^2 dx &< \infty, & \iint_Q |u|^2 + |Du|^2 &< \infty, & p &\in L^{3/2}(Q), \\ \operatorname{ess\,sup}_{t \in (-1, 0)} \int_{B_1} |u(x, t)|^3 dx &< \infty, \end{aligned}$$

such that $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ on B_1 for a.e. $t \in (-1, 0)$, and (S2) and (S3) of Definition 3.1 hold on Q with $f = 0$. Then there is $\gamma \in (0, 1]$ and a function $\tilde{u}$ equal to u almost everywhere on $B_{1/2} \times (-1/4, 0)$ that is parabolically Hölder continuous with exponent γ on the closure of $B_{1/2} \times (-1/4, 0)$.

The global result is [4, Theorem 1.3] in the form of its step (3.5): the solution has no singular points.

Theorem 3.13 (Regularity of $L^\infty L^3$ Leray–Hopf solutions). *Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$, and suppose $\text{ess sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty$. Then every point of $\mathbb{R}^3 \times (0, T)$ is a regular point of u .*

Remark 3.14 (Smoothness on positive-time intervals). Theorem 16.3 records the L^5 and uniqueness conclusions of [4, Theorem 1.3]. The Prodi–Serrin theorem [10, Theorem 8.17, printed p. 173] further implies that its L^5 solution is smooth on every closed interval $[\delta, T]$ with $0 < \delta < T$; smoothness is a consequence, not a separate clause of the theorem.

Part I. Carleman inequalities, unique continuation and backward uniqueness

4 Carleman inequalities

The weighted estimates below convert control of the backward heat operator into control of a field and its gradient. The first weight amplifies small times while remaining Gaussian in space; the second also grows rapidly in the normal variable x_3 on the translated half-space.

Throughout this section, $L = \partial_t + \Delta$. All vector and matrix norms are Euclidean, and derivatives and integrations are componentwise. The calculations are the compact-support versions of (6.2)–(6.11) and (6.13)–(6.26) in [4].

Proposition 4.1. *There is an absolute constant*

$$c_0 = e^{4/3}(9 + 2\sqrt{6}).$$

For every $a > 0$ and every $w \in C_c^\infty(\mathbb{R}^3 \times (0, 2); \mathbb{R}^3)$, with $h(t) = te^{(1-t)/3}$, one has

$$\begin{aligned} & \int_{\mathbb{R}^3 \times (0, 2)} h^{-2a}(t) e^{-|x|^2/(4t)} \left(\frac{a}{t} |w|^2 + |\nabla w|^2 \right) dx dt \\ & \leq c_0 \int_{\mathbb{R}^3 \times (0, 2)} h^{-2a}(t) e^{-|x|^2/(4t)} |Lw|^2 dx dt. \end{aligned} \tag{4.1}$$

Proof. Set $q = a + 1$, $\phi(x, t) = -|x|^2/(8t) - q \log h(t)$, and $v = e^\phi w$. The field v is smooth and compactly supported in $\mathbb{R}^3 \times (0, 2)$. Define the conjugated operator

$$\mathcal{L}_\phi v := e^\phi Lw = \partial_t v - \text{div}(v \otimes \nabla \phi) - \nabla v \nabla \phi + \Delta v + (|\nabla \phi|^2 - \partial_t \phi) v.$$

Here $\text{div}(v \otimes \nabla \phi)_i = \sum_j \partial_j (v_i \partial_j \phi)$ and $(\nabla v \nabla \phi)_i = \sum_j (\partial_j v_i) (\partial_j \phi)$. In particular,

$$\mathcal{L}_\phi v = (\partial_t + \Delta)v - 2\nabla \phi \cdot \nabla v + (|\nabla \phi|^2 - \partial_t \phi - \Delta \phi)v.$$

For this compactly supported field, write $t\mathcal{L}_\phi = S + A$, where

$$Sv = t\Delta v + t(|\nabla\phi|^2 - \partial_t\phi)v - \frac{1}{2}v, \quad (4.2)$$

$$Av = \frac{1}{2}(\partial_t(tv) + t\partial_tv) - t(\operatorname{div}(v \otimes \nabla\phi) + \nabla v \nabla\phi). \quad (4.3)$$

For compactly supported v , the product rule and the flux identity below compute the integrated cross term directly. The time integrations have no endpoint terms because v is supported in a compact subset of $(0, 2)$, and the spatial integrations have no boundary terms because v has compact spatial support.

All integrals in this proof are over $\mathbb{R}^3 \times (0, 2)$ unless specified otherwise. Write $\mathcal{D}_\phi(v)$ for the density in the following direct product-rule computation, with $i, j \in \{1, 2, 3\}$ summed:

$$\begin{aligned} I := \int \mathcal{D}_\phi(v) dx dt &= 4 \int t^2 (\phi_{,ij} \partial_i v \cdot \partial_j v + \phi_{,ij} \phi_{,i} \phi_{,j} |v|^2) \\ &\quad + \int t^2 |v|^2 (\partial_t^2 \phi - 2\partial_t |\nabla\phi|^2 - \Delta^2 \phi) + \int t |\nabla v|^2 - \int t |v|^2 (|\nabla\phi|^2 - \partial_t \phi). \end{aligned} \quad (4.4)$$

Consequently, expanding the square and using the integrated flux identity below gives

$$P := \int t^2 |\mathcal{L}_\phi v|^2 dx dt = \int |Sv|^2 dx dt + \int |Av|^2 dx dt + I. \quad (4.5)$$

For vector fields, sum the scalar fluxes over components:

$$\begin{aligned} F_0 &= t^2 (|\nabla\phi|^2 - \partial_t \phi) |v|^2 - t^2 |\nabla v|^2 - \frac{1}{2} t |v|^2, \\ F_i &= 2t^2 \partial_i v \cdot \partial_t v + t v \cdot \partial_i v - 4t^2 \partial_i v \cdot \left(\sum_{j=1}^3 \phi_{,j} \partial_j v \right) + 2t^2 \phi_{,i} |\nabla v|^2 \\ &\quad - 2t^2 (\Delta\phi) v \cdot \partial_i v + t^2 \partial_i (\Delta\phi) |v|^2 - 2t^2 (|\nabla\phi|^2 - \partial_t \phi) \phi_{,i} |v|^2 + t \phi_{,i} |v|^2, \quad i = 1, 2, 3. \end{aligned}$$

Expanding $2(Sv) \cdot (Av)$ and differentiating the displayed fluxes by the product rule, with equality of mixed space-time derivatives, gives the pointwise identity

$$2(Sv) \cdot (Av) = \mathcal{D}_\phi(v) + \partial_t F_0 + \sum_{i=1}^3 \partial_i F_i.$$

The fluxes and their derivatives are supported in $\operatorname{supp} v$, a compact subset of the open cylinder, so their time and spatial integrals vanish. Thus $I = 2 \int Sv \cdot Av$. Expanding $|t\mathcal{L}_\phi v|^2 = |Sv + Av|^2$ pointwise gives the displayed identity for P ; nonnegativity of the two square integrals yields $I \leq P$. The explicit value below gives $I \geq 0$. This integrated estimate requires no separate space-time pairing identities for S and A .

For the present ϕ , let $r = h'/h = 1/t - 1/3$. Then

$$\nabla\phi = -\frac{x}{4t}, \quad \phi_{,ij} = -\frac{\delta_{ij}}{4t}, \quad |\nabla\phi|^2 = \frac{|x|^2}{16t^2}, \quad |\nabla\phi|^2 - \partial_t \phi = -|\nabla\phi|^2 + qr,$$

while $\Delta^2 \phi = 0$ and

$$\partial_t^2 \phi - 2\partial_t |\nabla\phi|^2 = \frac{q}{t^2}.$$

Substituting these identities into (4.4), the ∇v terms cancel, as do the two terms containing $t|\nabla\phi|^2|v|^2$. The remainder is

$$I = q \int (1 - tr)|v|^2 dx dt = \frac{q}{3} \int t|v|^2 dx dt. \quad (4.6)$$

In particular, (4.5) gives $0 \leq I \leq P$.

The pointwise identity

$$|\nabla v|^2 = \frac{1}{2}(\partial_t + \Delta)|v|^2 - v \cdot (\partial_t v + \Delta v) \quad (4.7)$$

holds componentwise. To integrate it, use the expression for $\mathcal{L}_\phi v$ above. Spatial integration by parts yields $\int v \cdot (2\nabla\phi \cdot \nabla v) = -\int (\Delta\phi)|v|^2$; this has no boundary term because v is compactly supported. The $\Delta\phi$ terms then cancel, giving

$$\int t^2 |\nabla v|^2 = -\int t|v|^2 - \int t^2 v \cdot \mathcal{L}_\phi v + \int t^2 |v|^2 (|\nabla\phi|^2 - \partial_t \phi). \quad (4.8)$$

Here the time integration by parts has no endpoint terms because the support is compact in $(0, 2)$, and the spatial Laplacian integration by parts has no boundary term because the support is compact in $\mathbb{R}^3$.

Let

$$E = \int t^2 (|\nabla v|^2 + |v|^2 |\nabla\phi|^2).$$

Combining (4.8) with $|\nabla\phi|^2 - \partial_t \phi = -|\nabla\phi|^2 + qh'/h$ gives

$$E = -\int t|v|^2 - \int t^2 v \cdot \mathcal{L}_\phi v + q \int t^2 \frac{h'}{h} |v|^2 \leq 3I + \left| \int t^2 v \cdot \mathcal{L}_\phi v \right|.$$

Indeed, the coefficient of $|v|^2$ before the last cross term is $-t + qt^2 h'/h = at - qt^2/3 \leq qt$, and $q \int t|v|^2 = 3I$ by (4.6). Since $0 < t < 2$, (4.6) and Cauchy–Schwarz imply

$$\left| \int t^2 v \cdot \mathcal{L}_\phi v \right| \leq \left(\int t^2 |v|^2 \right)^{1/2} P^{1/2} \leq (6I)^{1/2} P^{1/2} \leq \sqrt{6} P.$$

It follows that

$$E \leq (3 + \sqrt{6})P. \quad (4.9)$$

Finally, $e^\phi |\nabla w| \leq |\nabla v| + |v| |\nabla\phi|$, so $e^{2\phi} |\nabla w|^2 \leq 2(|\nabla v|^2 + |v|^2 |\nabla\phi|^2)$. Since $e^{2\phi} = h^{-2(a+1)} e^{-|x|^2/(4t)}$ and $(th^{-1})^2 = e^{-2(1-t)/3}$ satisfies $e^{-2/3} \leq (th^{-1})^2 \leq e^{2/3}$ on $(0, 2)$, (4.9) yields

$$\begin{aligned} \int h^{-2a} e^{-|x|^2/(4t)} \left((a+1) \frac{|w|^2}{t} + |\nabla w|^2 \right) &\leq e^{2/3} (9 + 2\sqrt{6}) P \\ &\leq e^{4/3} (9 + 2\sqrt{6}) \int h^{-2a} e^{-|x|^2/(4t)} |Lw|^2. \end{aligned}$$

The first inequality uses $(a+1) \geq a$ and the lower bound on $(th^{-1})^2$; the second uses its upper bound. This proves (4.1) with the stated c_0 . $\square$

Proposition 4.2. *For every $\alpha \in (1/2, 1)$ there are constants $a_0(\alpha) > 0$ and $c_*(\alpha) > 0$ such that, for every $a > a_0(\alpha)$ and every $w \in C_c^\infty(\{x_3 > 1\} \times (0, 1); \mathbb{R}^3)$, the following estimate holds with $x' = (x_1, x_2)$ and*

$$\begin{aligned} \phi(x, t) &= -\frac{|x'|^2}{8t} + a(1-t)x_3^{2\alpha}t^{-\alpha} : \\ &\int_{\{x_3 > 1\} \times (0, 1)} t^2 e^{2\phi} \left(a \frac{|w|^2}{t^2} + \frac{|\nabla w|^2}{t} \right) dx dt \\ &\leq c_*(\alpha) \int_{\{x_3 > 1\} \times (0, 1)} t^2 e^{2\phi} |Lw|^2 dx dt. \end{aligned} \quad (4.10)$$

One may take $a_0(\alpha) = 2$ and

$$c_*(\alpha) = 5 + \frac{7}{2(2\alpha - 1)}.$$

Proof. Fix $\alpha \in (1/2, 1)$, $a \geq 2$, and a test field as in the statement. Set $Q = \{x_3 > 1\} \times (0, 1)$, $v = e^\phi w$, and $\mathcal{L}_\phi v = e^\phi Lw$. The function $x_3^{2\alpha}$ is smooth on $x_3 > 1$, and v has compact support in Q . Apply the conjugation and commutator identities above on Q , with all their integrations by parts justified by this compact support: there are no contributions from spatial infinity, the plane $x_3 = 1$, or the time endpoints $t = 0, 1$.

The gradients and Hessians of the two parts of ϕ are

$$\nabla \phi^{(1)} = -\frac{x'}{4t}, \quad \nabla \phi^{(2)} = 2\alpha a(1-t)x_3^{2\alpha-1}t^{-\alpha}e_3, \quad \phi^{(1)} = -\frac{|x'|^2}{8t}, \quad \phi^{(2)} = a(1-t)x_3^{2\alpha}t^{-\alpha}. \quad (4.11)$$

Their gradients are orthogonal, so

$$\nabla \phi^{(1)} \cdot \nabla \phi^{(2)} = 0, \quad |\nabla \phi|^2 = |\nabla \phi^{(1)}|^2 + |\nabla \phi^{(2)}|^2. \quad (4.12)$$

The only nonzero entries of the Hessians are

$$\phi_{,ij}^{(1)} = -\frac{\delta_{ij}}{4t} \quad (1 \leq i, j \leq 2), \quad \phi_{,33}^{(2)} = 2\alpha(2\alpha - 1)a(1-t)x_3^{2\alpha-2}t^{-\alpha}. \quad (4.13)$$

It follows that

$$\phi_{,ij}\phi_{,i}\phi_{,j} = -\frac{1}{4t}|\nabla \phi^{(1)}|^2 + \frac{2\alpha(2\alpha - 1)a(1-t)}{t^\alpha}x_3^{2\alpha-2}|\nabla \phi^{(2)}|^2 \geq -\frac{|x'|^2}{64t^3}. \quad (4.14)$$

To split the commutator (4.4), define, for $s = 1, 2$,

$$\begin{aligned} I_s &= 4 \int_Q t^2 \left(\phi_{,ij}^{(s)} \partial_i v \cdot \partial_j v + \phi_{,ij}^{(s)} \phi_{,i}^{(s)} \phi_{,j}^{(s)} |v|^2 \right) dx dt \\ &\quad + \int_Q t^2 |v|^2 \left(\partial_t^2 \phi^{(s)} - 2\partial_t |\nabla \phi^{(s)}|^2 - \Delta^2 \phi^{(s)} - \frac{1}{t} |\nabla \phi^{(s)}|^2 + \frac{1}{t} \partial_t \phi^{(s)} \right) dx dt. \end{aligned}$$

The Hessians and gradients in (4.11)–(4.13) have disjoint tangential and normal coordinates. Thus the mixed terms vanish, and (4.4) becomes

$$I = I_1 + I_2 + \int_Q t |\nabla v|^2 dx dt. \quad (4.15)$$

For $\phi^{(1)}$, direct differentiation gives

$$\begin{aligned} 4t^2 \phi_{,ij}^{(1)} \partial_i v \cdot \partial_j v &= -t |\nabla' v|^2, \\ 4t^2 \phi_{,ij}^{(1)} \phi_{,i}^{(1)} \phi_{,j}^{(1)} &= -t |\nabla \phi^{(1)}|^2, \\ t^2 \left(\partial_t^2 \phi^{(1)} - 2\partial_t |\nabla \phi^{(1)}|^2 - \Delta^2 \phi^{(1)} - t^{-1} |\nabla \phi^{(1)}|^2 + t^{-1} \partial_t \phi^{(1)} \right) &= t |\nabla \phi^{(1)}|^2. \end{aligned}$$

Therefore $I_1 = -\int_Q t |\nabla' v|^2$ and

$$I = \int_Q t |\partial_3 v|^2 dx dt + I_2. \quad (4.16)$$

For the second part of the weight, $\phi_{,33}^{(2)} > 0$ on Q , and $\phi_{,33}^{(2)} (\partial_3 \phi^{(2)})^2 \geq 0$. Dropping these nonnegative terms from I_2 gives

$$I_2 \geq \int_Q t^2 |v|^2 (A_1 + A_2 + A_3) dx dt, \quad (4.17)$$

where

$$A_1 = -\partial_t |\nabla \phi^{(2)}|^2, \quad A_2 = A_1 - \Delta^2 \phi^{(2)} - \frac{1}{t} |\nabla \phi^{(2)}|^2, \quad A_3 = \partial_t^2 \phi^{(2)} + \frac{1}{t} \partial_t \phi^{(2)}.$$

Differentiating $x_3^{2\alpha}$ four times yields

$$\Delta^2 \phi^{(2)} = 2\alpha(2\alpha-1)(2\alpha-2)(2\alpha-3)a(1-t)x_3^{2\alpha-4}t^{-\alpha}.$$

Using the exact derivative of $|\nabla \phi^{(2)}|^2$, and $2t + (2\alpha-1)(1-t) \geq 2\alpha-1$, we obtain

$$A_2 \geq \frac{1-t}{t^\alpha} x_3^{2\alpha-4} a(2\alpha-1) \left[\frac{4\alpha^2 a x_3^{2\alpha+2}}{t^{\alpha+1}} - 2\alpha(2\alpha-2)(2\alpha-3) \right]. \quad (4.18)$$

On Q , $x_3^{2\alpha+2} t^{-(\alpha+1)} \geq 1$. The bracket in (4.18) is therefore at least

$$8\alpha^2 - 2\alpha(2\alpha-2)(2\alpha-3) = -4\alpha(2\alpha-1)(\alpha-3) > 0,$$

where the last inequality uses $1/2 < \alpha < 1$. Hence $A_2 > 0$ on Q for every $a \geq 2$.

A direct differentiation also gives

$$A_3 = a x_3^{2\alpha} t^{-\alpha-2} (\alpha^2 - (1-\alpha)^2 t) \geq a(2\alpha-1) \frac{x_3^{2\alpha}}{t^{\alpha+2}}. \quad (4.19)$$

For A_1 , direct calculation gives

$$\begin{aligned} A_1 - \frac{1}{t} |\nabla \phi^{(2)}|^2 &= 4\alpha^2 a^2 x_3^{4\alpha-2} (1-t) t^{-2\alpha-1} (2t + (2\alpha-1)(1-t)) \\ &\geq (2\alpha-1) 4\alpha^2 a^2 (1-t) x_3^{4\alpha-2} t^{-2\alpha-1} \geq 0. \end{aligned}$$

Thus $A_1 \geq t^{-1} |\nabla \phi^{(2)}|^2$. From (4.16)–(4.19), the commutator energy identity, and $x_3^{2\alpha} t^{-\alpha} \geq 1$, we obtain

$$\begin{aligned} P &:= \int_Q t^2 |\mathcal{L}_\phi v|^2 dx dt \geq I \geq a(2\alpha-1)U + G, \\ U &:= \int_Q \frac{x_3^{2\alpha}}{t^\alpha} |v|^2 dx dt, \quad G := \int_Q t |v|^2 |\nabla \phi^{(2)}|^2 dx dt. \end{aligned} \quad (4.20)$$

In particular, for $\theta = 2\alpha - 1 > 0$ and $V = \int_Q |v|^2 dx dt$, we have $V \leq U$ and $P \geq a\theta U + G$.

Integrating (4.7) with the factor t gives

$$\int_Q t |\nabla v|^2 dx dt = -\frac{1}{2} \int_Q |v|^2 dx dt - \int_Q tv \cdot \mathcal{L}_\phi v dx dt + \int_Q t |v|^2 (|\nabla \phi|^2 - \partial_t \phi) dx dt. \quad (4.21)$$

The time integration by parts has no endpoint terms because v is compactly supported in $(0, 1)$; the spatial integrations by parts have no boundary terms because v is compactly supported in $\{x_3 > 1\}$. Since $|\nabla \phi^{(1)}|^2 - \partial_t \phi^{(1)} = -|\nabla \phi^{(1)}|^2$, rearranging (4.21) yields

$$\int_Q t (|\nabla v|^2 + |v|^2 |\nabla \phi|^2) dx dt = -\frac{1}{2} \int_Q |v|^2 dx dt - \int_Q tv \cdot \mathcal{L}_\phi v dx dt \quad (4.22)$$

$$+ 2 \int_Q t |v|^2 |\nabla \phi^{(2)}|^2 dx dt - \int_Q t |v|^2 \partial_t \phi^{(2)} dx dt. \quad (4.23)$$

Moreover,

$$-t \partial_t \phi^{(2)} = a \frac{x_3^{2\alpha}}{t^\alpha} (\alpha + (1 - \alpha)t) \leq a \frac{x_3^{2\alpha}}{t^\alpha},$$

since $0 < t < 1$ and $0 < \alpha < 1$. The identity $e^\phi \nabla w = \nabla v - v \nabla \phi$ implies $\frac{1}{2} e^{2\phi} |\nabla w|^2 \leq |\nabla v|^2 + |v|^2 |\nabla \phi|^2$. Therefore (4.22) gives

$$D := \frac{1}{2} \int_Q t e^{2\phi} |\nabla w|^2 dx dt \leq \left| \int_Q tv \cdot \mathcal{L}_\phi v dx dt \right| + 2G + aU. \quad (4.24)$$

It remains to bound the cross term with constants uniform in a . Cauchy–Schwarz, followed by Young’s inequality with equal weights, gives the explicit absorption

$$\begin{aligned} \left| \int_Q tv \cdot \mathcal{L}_\phi v \right| &\leq (VP)^{1/2} \leq (UP)^{1/2} = \left(\frac{a\theta U P}{a\theta} \right)^{1/2} \\ &\leq \frac{1}{2} a\theta U + \frac{1}{2a\theta} P \leq \left(\frac{1}{2} + \frac{1}{2a\theta} \right) P \leq \left(\frac{1}{2} + \frac{1}{4\theta} \right) P, \end{aligned}$$

where the penultimate inequality uses $a\theta U \leq P$ from (4.20), and the last uses $a \geq 2$. The same bound gives $aV \leq P/\theta$, while (4.20) gives $G \leq P$ and $aU \leq P/\theta$. Consequently,

$$a \int_Q e^{2\phi} |w|^2 dx dt + \int_Q t e^{2\phi} |\nabla w|^2 dx dt = aV + 2D \leq \left(5 + \frac{7}{2\theta} \right) P.$$

Since $P = \int_Q t^2 e^{2\phi} |Lw|^2 dx dt$ and $\theta = 2\alpha - 1$, this is (4.10) with the stated constants. $\square$

Remark 4.3 (Departure from [4]). The printed proof of Proposition 6.2 [4, pp. 240–243] ends by saying that Cauchy–Schwarz and (6.23), (6.26) imply the claim, without displaying the uniform constant or the absorption. The estimates above supply that omitted step: $a \geq 2$ makes A_2 positive, and Young’s inequality bounds the cross term uniformly, giving $a_0(\alpha) = 2$ and $c_*(\alpha) = 5 + 7/(2(2\alpha - 1))$. The proposition’s statement is unchanged.

Extension to Sobolev fields and a Caccioppoli inequality. The uniqueness theorems apply the two inequalities to cut-off fields of class $W_2^{2,1}$, and they need two elementary estimates for the differential inequality $|Lw| \leq c_1(|w| + |\nabla w|)$.

Lemma 4.4. *Set $U_G = \mathbb{R}^3 \times (0, 2)$ and $U_H = \{x_3 > 1\} \times (0, 1)$. The following extensions retain the constants of the stated propositions:*

1. *For every $a > 0$, the inequality in Proposition 4.1 holds for all $w \in W_2^{2,1}(U_G; \mathbb{R}^3)$ compactly supported in U_G , with the same absolute constant c_0 .*
2. *For every $\alpha \in (1/2, 1)$ and every $a > a_0(\alpha)$, the inequality in Proposition 4.2 holds for all $w \in W_2^{2,1}(U_H; \mathbb{R}^3)$ compactly supported in U_H , with the same constant $c_*(\alpha)$.*

Proof. Fix one of the two open sets U in the statement and let $K = \text{supp } w \Subset U$. Extend w by zero to $\mathbb{R}^3 \times \mathbb{R}$. Because w vanishes in a neighborhood of ∂U , this extension has as its weak derivatives the zero extensions of Dw , D^2w , and $D_t w$: testing against a smooth function on $\mathbb{R}^3 \times \mathbb{R}$ reduces, after inserting a cutoff equal to one near K , to the weak-derivative identities on U . Thus, for a standard space-time mollifier ρ_δ ,

$$\partial_j(w * \rho_\delta) = (Dw)_{\cdot j} * \rho_\delta, \quad \partial_k \partial_j(w * \rho_\delta) = (D^2w)_{\cdot jk} * \rho_\delta, \quad \partial_t(w * \rho_\delta) = (D_t w) * \rho_\delta.$$

These identities use only the stated spatial weak derivatives through order two and the time weak derivative; no mixed derivative is needed. If δ is smaller than the distance from K to ∂U , then $w * \rho_\delta \in C_c^\infty(U; \mathbb{R}^3)$. All these mollifications are supported in a fixed compact set $K' \Subset U$, on which every scalar weight and coefficient in the two inequalities is bounded. The approximate-identity theorem gives convergence of $w * \rho_\delta$, its first and second spatial derivatives, and its time derivative to the corresponding fields in $L^2(K')$. Hence the weighted L^2 norms of the fields and of $L(w * \rho_\delta)$ converge to those of w and Lw . Apply the appropriate proposition to $w * \rho_\delta$ and pass to the limit. The constants do not depend on δ . $\square$

Remark 4.5 (Departure from [4]). Propositions 6.1 and 6.2 of [4], printed pp. 238–243, are stated for smooth compactly supported fields. The argument above supplies the density passage needed to use those estimates for compactly supported $W_2^{2,1}$ fields.

Lemma 4.6. *Let $0 < r \leq 1$, $c_1 \geq 0$, $x \in \mathbb{R}^3$, and $t \in \mathbb{R}$. Set $Q = B_{2r}(x) \times (t, t + 4r^2)$. Suppose that $w \in W_2^{2,1}(Q; \mathbb{R}^3)$ and*

$$|Lw| \leq c_1(|w| + |\nabla w|) \quad \text{a.e. on } Q.$$

Then, for the absolute constant $K = 256$,

$$\iint_{B_r(x) \times (t, t+r^2)} |\nabla w|^2 \leq K(1 + c_1^2 + r^{-2}) \iint_{B_{2r}(x) \times (t, t+4r^2)} |w|^2.$$

The same estimate holds when t is the initial time of the domain, provided w has a continuous extension to $\overline{B_{2r}(x)} \times \{t\}$; no derivative at that slice is assumed.

Proof. Choose $0 < 4\varepsilon < r^2$. Take the smooth spatial weight η supported in $B_{2r}(x)$, equal to one on $B_r(x)$, with $|\nabla \eta| \leq 400/(99r)$. Let θ be the product of a lower time transition from zero at $t + \varepsilon$ to one at $t + 2\varepsilon$ and an upper transition from one through $t + (101/100)r^2$ to zero at $t + 3r^2$. It is supported in $[t + \varepsilon, t + 3r^2]$ and equals one on $[t + 2\varepsilon, t + r^2]$. Its lower transition is

nondecreasing and finishes before $t + r^2$; after $t + r^2$, θ is nonincreasing and $|\theta'| \leq 800/(199r^2)$. Set $\psi(y, s) = \eta(y)^2\theta(s)^2$. Its product derivatives satisfy

$$\sum_j |\partial_j \psi|^2 \leq \frac{4(400/99)^2}{r^2} \psi, \quad \partial_s \psi \geq -\frac{1600}{199r^2}.$$

The weak derivatives give the integrated identity

$$\iint_Q Lw \cdot w \psi = -\frac{1}{2} \iint_Q |w|^2 \partial_s \psi - \iint_Q |\nabla w|^2 \psi - \iint_Q \sum_{i,j} w_i \partial_j w_i \partial_j \psi,$$

where $Q = B_{2r}(x) \times (t, t + 4r^2)$. The time cutoff is supported inside the open interval, so this identity uses no trace or derivative at t . Applying the differential inequality and Young's inequality gives

$$\iint_Q |\nabla w|^2 \psi \leq 2 \left(c_1 + c_1^2 + \frac{4(400/99)^2 + 800/199}{r^2} \right) \iint_Q |w|^2.$$

The coefficient is at most $256(1 + c_1^2 + r^{-2})$. Since $\psi = 1$ on $B_r(x) \times (t + 2\varepsilon, t + r^2)$, let $\varepsilon \downarrow 0$ and use monotone convergence. This proves the estimate, including at an initial time, without using a derivative there. $\square$

Lemma 4.7. *Let $B \subset \mathbb{R}^3$ be open and $\tau > 0$. Suppose that $w : B \times (0, \tau) \rightarrow \mathbb{R}^3$ has space-time weak time derivative $D_t w \in L^2(B \times (0, \tau))$, extends continuously to $B \times [0, \tau)$, and satisfies $w(x, 0) = 0$ for every $x \in B$. Then, for every $0 < t < \tau$,*

$$\int_B |w(x, t)|^2 dx \leq t \int_0^t \int_B |D_t w(x, s)|^2 dx ds.$$

Proof. Fix $\Omega \Subset B$ and $T < \tau$. Continuity makes w square-integrable on $\Omega \times (0, T)$, and $D_t w$ is square-integrable there by hypothesis. Testing the weak time-derivative identity with products $\varphi(x)\zeta(s)$ from countable dense families and applying Fubini shows that, for almost every $x \in \Omega$, the section $s \mapsto w(x, s)$ has weak derivative $D_t w(x, s)$ on $(0, T)$. Both sections are in $L^2((0, T))$ for almost every such x , so this section has an absolutely continuous representative. That representative agrees almost everywhere in s with the given continuous section; hence they agree for every $s \in (0, T)$, and continuity at zero identifies its value there as zero. Therefore, for almost every $x \in \Omega$ and every $t < T$,

$$w(x, t) = \int_0^t D_t w(x, s) ds, \quad |w(x, t)|^2 \leq t \int_0^t |D_t w(x, s)|^2 ds$$

by Cauchy–Schwarz. Integrate over Ω , then exhaust B by relatively compact open subsets and use monotone convergence. Taking T with $t < T < \tau$ proves the claim. $\square$

Remark 4.8 (Departure from [4]). In §5, Lemma 5.2 of [4], printed pp. 234–236, the zero extension across the initial slice is used without specifying a trace topology; Equation (5.7) in that proof and (5.19) in the proof of Lemma 5.3, printed pp. 234 and 236, are pointwise gradient bounds invoked from parabolic regularity. The estimates above instead provide a spacetime L^2 gradient bound in the weak-derivative class and the small-time trace inequality under the stated continuous representative convention.

5 Unique continuation across spatial boundaries

The integral form of Lemma 4.3 replaces the maximum in (4.4) by the space-time L^2 norms already present in (4.1). Its proof keeps the Gaussian weight in Proposition 6.1 and estimates the cutoff errors directly in L^2 ; in particular, it does not use the local L^2 -to-pointwise estimate (4.18) of [4].

Lemma 5.1 (Gaussian estimate from integral vanishing). *Let $R, T, c_1 > 0$ with $T \leq 1$, let $x_0 \in \mathbb{R}^3$, and let $w: B_R(x_0) \times [0, T] \rightarrow \mathbb{R}^3$ be continuous. Suppose that w has space-time weak derivatives ∇w , $\nabla^2 w$, and $\partial_t w$ on $B_R(x_0) \times (0, T)$ and*

$$|w| + |\nabla w| + |\nabla^2 w| + |\partial_t w| \in L^2(B_R(x_0) \times (0, T)), \quad |Lw| \leq c_1(|w| + |\nabla w|) \quad \text{a.e.}$$

Assume also that for every $m \in \mathbb{N}$ there are $C_m, r_m > 0$ such that

$$\int_0^{r^2} \int_{B_r(x_0)} |w(y, s)|^2 dy ds \leq C_m r^m \quad (0 < r < \min\{r_m, R, \sqrt{T}\}). \quad (5.1)$$

Put

$$\mathcal{N}_{x_0}(R, T; w) = \int_0^{3T/4} \int_{B_{3R/4}(x_0)} (|w|^2 + T|\nabla w|^2) dy ds.$$

There are absolute constants

$$\beta = \frac{1}{100}, \quad \beta_1 = \frac{3}{8}\sqrt{2\beta}, \quad \beta_2 = 16\beta,$$

and a number $\gamma = \gamma(c_1) \in (0, 3/16)$ such that, whenever $0 < t \leq \gamma T$, $|x - x_0| \leq \beta_1 R$, and $\beta_2 t \leq |x - x_0|^2$, one has

$$\int_t^{2t} \int_{B_{\sqrt{2t}}(x)} |w(y, s)|^2 dy ds \leq C(c_1) \mathcal{N}_{x_0}(R, T; w) \exp\left(-\frac{|x - x_0|^2}{2t}\right), \quad (5.2)$$

where $C(c_1)$ depends only on c_1 and the absolute constants in Proposition 4.1. Equivalently, the average of $|w|^2$ over this box is bounded by the right-hand side of (5.2) divided by $2^{3/2}|B_1|t^{5/2}$.

Proof. Translation allows us to take $x_0 = 0$. Fix an output point x satisfying the conclusion's spatial conditions. The spatial condition and the definitions of β_1, β_2 imply

$$|x| + \sqrt{2t} \leq \left(\beta_1 + \sqrt{\frac{2}{\beta_2}}\beta_1\right)R = \left(\beta_1 + \frac{3}{16}\right)R < R, \quad 2t \leq 2\gamma T < \frac{3T}{8} < T.$$

Thus the box in (5.2) lies inside the domain of w . Set $\mu = \sqrt{2\beta}$ and $X = x/\mu$, and put

$$\lambda = \sqrt{2t}, \quad \rho = \frac{2|X|}{\lambda}, \quad v(y, s) = w(\lambda y, \lambda^2 s).$$

The affine change of variables in the definition of weak derivatives gives $D_y v = \lambda D w$, $D_y^2 v = \lambda^2 D^2 w$, and $D_s v = \lambda^2 D_t w$, evaluated at $(\lambda y, \lambda^2 s)$. Consequently, on $Q(\rho, 2) = B_\rho \times (0, 2)$,

$$|Lv| \leq c_1 \lambda (|v| + |\nabla v|) \quad \text{a.e.}$$

Here we used $\lambda < 1$, which follows from $t \leq \gamma T$, $T \leq 1$, and $\gamma < 3/16$. The output point conditions give

$$|X| \leq \frac{\beta_1 R}{\mu} = \frac{3R}{8}, \quad |X|^2 \geq \frac{\beta_2 t}{\mu^2} = 8t.$$

Consequently,

$$\rho \geq 4, \quad \lambda\rho = 2|X| \leq \frac{3R}{4}, \quad 2\lambda^2 = 4t \leq \frac{3T}{4}.$$

Thus $Q(\rho, 2)$ is mapped into $B_{3R/4}(0) \times (0, 3T/4)$. The integral vanishing also transfers to v : for every m and all sufficiently small r ,

$$\int_0^{r^2} \int_{B_r} |v|^2 \leq \lambda^{-5} C_m (\lambda r)^m.$$

Let $h(s) = se^{(1-s)/3}$ and

$$a = \frac{\beta\rho^2}{2\log h(3/2)}, \quad \omega_a(y, s) = h(s)^{-2a} e^{-|y|^2/(4s)}.$$

Since $h(3/2) > 1$, $a > 0$. Choose smooth cutoffs $\theta_\rho(y)$ and $\eta(s)$ with $\theta_\rho = 1$ on $B_{\rho-1}$ and supported in B_ρ , and $\eta = 1$ on $(0, 3/2)$ and $\eta = 0$ for $s \geq 7/4$. Their derivatives through order two are bounded independently of $\rho \geq 4$. Choose $\chi_\varepsilon \in C^\infty(\mathbb{R})$ equal to zero for $s \leq \varepsilon$, equal to one for $s \geq 2\varepsilon$, and satisfying $|\chi'_\varepsilon| \leq C\varepsilon^{-1}$. For sufficiently small $\varepsilon > 0$, set $z_\varepsilon = \chi_\varepsilon \theta_\rho \eta v$, extending it by zero outside $Q(\rho, 2)$. The weak product rule, obtained by applying the weak derivative identities to the product of a test function and a smooth cutoff, shows that z_ε and its first spatial, second spatial, and time weak derivatives are in L^2 and that it has compact support in $\mathbb{R}^3 \times (0, 2)$. The density extension Lemma 4.4 therefore permits applying Proposition 4.1 to z_ε .

Let

$$K = (B_\rho \times (0, 2)) \setminus (B_{\rho-1} \times (0, 3/2)).$$

Using the weak product rule twice and the inequality for Lv , one obtains

$$|Lz_\varepsilon|^2 \leq Cc_1^2\lambda^2(|z_\varepsilon|^2 + |\nabla z_\varepsilon|^2) + C(c_1)\mathbf{1}_K(|v|^2 + |\nabla v|^2) + C\varepsilon^{-2}\mathbf{1}_{(0, 2\varepsilon)}|v|^2.$$

Indeed, the derivatives of $\theta_\rho \eta$ are supported in K ; in the leading term, replace $\chi_\varepsilon \theta_\rho \eta \nabla v$ by ∇z_ε minus the cutoff-gradient term, which is again supported in K . The last term contains the derivative of χ_ε .

For $\rho \geq 4$, $a \geq a_* := 8\beta/\log h(3/2) > 0$. The left side of the Carleman inequality controls

$$\kappa \int \omega_a(|z_\varepsilon|^2 + |\nabla z_\varepsilon|^2), \quad \kappa = \min\{1, a_*/2\},$$

because $s < 2$ on the support. Let C be the absolute coefficient of $c_1^2\lambda^2$ in the preceding bound and let c_0 be the constant in Proposition 4.1. Since $\lambda^2 = 2t \leq 2\gamma$, choose

$$0 < \gamma(c_1) < \min\left\{\frac{3}{16}, \frac{\kappa}{4Cc_0c_1^2}\right\}.$$

Then the first term on the right is absorbed by the Carleman left side. The result is

$$\begin{aligned} \int \omega_a\left(\frac{a}{s}|z_\varepsilon|^2 + |\nabla z_\varepsilon|^2\right) &\leq C(c_1) \int_K \omega_a(|v|^2 + |\nabla v|^2) \\ &\quad + C\varepsilon^{-2} \int_0^{2\varepsilon} \int_{B_\rho} \omega_a |v|^2. \end{aligned} \tag{5.3}$$

We verify that the last term tends to zero. Fix an integer $m > 4a + 4$ and a radius $r_* > 0$ small enough that the scaled form of (5.1) with this exponent m holds whenever $0 < r < r_*$. For

$0 < s < 1$, $h(s) \geq cs$ with an absolute $c > 0$. Put $\tau_j = 2^{1-j}\varepsilon$ and $I_j = (\tau_j/2, \tau_j)$, for $j = 0, 1, \dots$; these intervals cover $(0, 2\varepsilon)$ up to endpoints. Partition space into $S_{j,0} = B_{\sqrt{\tau_j}}$ and, for $k \geq 1$,

$$S_{j,k} = \{y : 2^{k-1}\sqrt{\tau_j} \leq |y| < 2^k\sqrt{\tau_j}\}.$$

If $2^k\sqrt{\tau_j} < r_*$, the set $S_{j,k} \times I_j$ is contained in $B_{2^k\sqrt{\tau_j}} \times]0, (2^k\sqrt{\tau_j})^2[$. On this shell the Gaussian is at most $e^{-4^k/16}$ for $k \geq 1$ (and at most one for $k = 0$), so its weighted integral is bounded by

$$C_m \tau_j^{m/2-2a} 2^{km} e^{-4^k/16},$$

with the exponential factor omitted at $k = 0$. Summing these shells gives at most $C_{m,a} \tau_j^{m/2-2a}$. On the remaining shells, $|y| \geq r_*/2$; their total contribution is at most

$$C \|v\|_{L^2(Q(\rho,2))}^2 \tau_j^{-2a} e^{-r_*^2/(16\tau_j)}.$$

Consequently the final term of (5.3) is bounded by

$$C_{m,a} \varepsilon^{-2} \sum_{j \geq 0} \tau_j^{m/2-2a} + C \varepsilon^{-2} \|v\|_{L^2(Q(\rho,2))}^2 \sum_{j \geq 0} \tau_j^{-2a} e^{-r_*^2/(16\tau_j)}.$$

The first sum is $O(\varepsilon^{m/2-2a-2}) = o(1)$. For every $q > 0$ the exponential dominates τ^{-q} as $\tau \downarrow 0$; choose $q > 2a + 2$. The second sum after multiplication by ε^{-2} is then $O(\varepsilon^{q-2a-2}) = o(1)$. This proves the required limit of the initial-time error.

For the remaining cutoff term, we claim

$$\sup_K \omega_a \leq e^{-\beta\rho^2}. \quad (5.4)$$

On the part where $s \geq 3/2$, monotonicity of h on $[3/2, 2]$ gives $h(s)^{-2a} \leq h(3/2)^{-2a} = e^{-\beta\rho^2}$. On the spatial collar $|y| \geq \rho - 1$, $s < 3/2$, and $(\rho - 1)^2 \geq 9\rho^2/16$. The function

$$g(s) = h(s)^{-2a} e^{-\rho^2/(16s)}$$

is increasing on $(0, 2)$: differentiating its logarithm and substituting $a = \beta\rho^2/(2\log h(3/2))$ reduces the sign to $1/16 + (\beta/\log h(3/2))(-s + s^2/3) \geq 0$. Since $s - s^2/3 \leq 3/4$ for all s , with equality at $s = 3/2$, this follows from $\beta \leq \log h(3/2)/12$; here $\log h(3/2) = \log(3/2) - 1/6 > 0.23$. Moreover, $g(2) \leq h(2)^{-2a} \leq h(3/2)^{-2a} = e^{-\beta\rho^2}$, since $h(2) > h(3/2)$. On the spatial collar, $|y|^2 \geq 9\rho^2/16$ gives $e^{-|y|^2/(4s)} \leq e^{-9\rho^2/(64s)} \leq e^{-\rho^2/(16s)}$, so $\omega_a(y, s) \leq g(s) \leq g(2) \leq e^{-\beta\rho^2}$. This proves (5.4). Thus the weighted collar integral is at most

$$e^{-\beta\rho^2} \int_{Q(\rho,2)} (|v|^2 + |\nabla v|^2).$$

Fatou's lemma applied to the nonnegative Carleman integrand in (5.3), together with the vanishing initial-time term, now yields

$$D := \int_0^{3/2} \int_{B_{\rho-1}} \omega_a(|v|^2 + |\nabla v|^2) \leq C(c_1) e^{-\beta\rho^2} \int_{Q(\rho,2)} (|v|^2 + |\nabla v|^2). \quad (5.5)$$

The change of variables gives

$$\int_{Q(\rho,2)} (|v|^2 + |\nabla v|^2) = \lambda^{-5} \int_{\lambda Q(\rho,2)} |w|^2 + \lambda^{-3} \int_{\lambda Q(\rho,2)} |\nabla w|^2 \leq 2\lambda^{-5} \mathcal{N}_0(R, T; w),$$

where $\lambda Q(\rho, 2)$ denotes $B_{\lambda\rho} \times (0, 2\lambda^2)$ and we used $\lambda^2 \leq 2T$. Put $\mu = \sqrt{2\beta}$ and $\Omega = B_{\mu X/\lambda} \times (1/2, 1)$. The condition $\rho \geq 4$ implies $B_{\mu X/\lambda}(1) \subset B_{\rho-1}$, since $\mu \leq 1$ and $|X|/\lambda = \rho/2$. On Ω , $h(s) \leq 1$ and $|y|^2 \leq 2\mu^2 |X|^2 / \lambda^2 + 2$, so

$$\int_{\Omega} |v|^2 \leq e^{\mu^2 |X|^2 / \lambda^2 + 1} D.$$

Rescaling this box back to the original variables, using $\rho = 2|X|/\lambda$ and $\lambda^2 = 2t$, gives

$$\int_t^{2t} \int_{B_{\lambda}(\mu X)} |w(y, s)|^2 dy ds \leq C(c_1) \mathcal{N}_0(R, T; w) e^{-\beta |X|^2 / t}.$$

By construction $x = \mu X$, so the target ball is $B_{\lambda}(x)$ and $\beta |X|^2 / t = |x|^2 / (2t)$. This is (5.2). The average statement follows because the box has measure $2^{3/2} |B_1| t^{5/2}$. $\square$

Theorem 5.2 (Unique continuation across a spatial boundary). *Let $R, T, c_1 > 0$ and let $w: B_R \times [0, T) \rightarrow \mathbb{R}^3$ be continuous. Suppose that w has space-time weak derivatives ∇w , $\nabla^2 w$, and $\partial_t w$ on $Q(R, T) = B_R \times (0, T)$ and*

$$|w| + |\nabla w| + |\nabla^2 w| + |\partial_t w| \in L^2(Q(R, T)), \quad |Lw| \leq c_1(|w| + |\nabla w|) \quad \text{a.e.} \quad (5.6)$$

Assume that for every $k \in \mathbb{N}$ there is $C_k > 0$ such that

$$|w(x, t)| \leq C_k(|x| + \sqrt{t})^k \quad ((x, t) \in Q(R, T)). \quad (5.7)$$

Then $w(x, 0) = 0$ for every $x \in B_R$.

Proof. Set $T_0 = \min\{T, 1\}$ and restrict w to $B_R \times [0, T_0)$. For $0 < r < \min\{1, R, \sqrt{T_0}\}$, (5.7) gives $|w(y, s)|^2 \leq C_k^2 (2r)^{2k}$ on $B_r \times (0, r^2)$. This cylinder has measure $|B_1| r^5$, so

$$\int_0^{r^2} \int_{B_r} |w|^2 \leq |B_1| C_k^2 2^{2k} r^{2k+5}.$$

Given any integer m , choose k with $2k + 5 \geq m$; since $r < 1$, this proves the integral vanishing hypothesis of Lemma 5.1 at the origin.

Apply that lemma on $B_R \times (0, T_0)$. If $x \neq 0$ and $|x| < \beta_1 R$, then for all sufficiently small $t > 0$ its conditions hold. The average of $|w|^2$ over $B_{\sqrt{2t}}(x) \times (t, 2t)$ is bounded by

$$C(c_1) \mathcal{N}_0(R, T_0; w) t^{-5/2} e^{-|x|^2 / (2t)},$$

which tends to zero. By continuity of w at $(x, 0)$, these averages tend to $|w(x, 0)|^2$; hence $w(x, 0) = 0$. At $x = 0$, (5.7) and continuity give $w(0, 0) = 0$.

We record why the same Gaussian estimate supplies the integral vanishing needed to move the center. Consider any application of Lemma 5.1 centered at x_0 on a ball of radius R_0 , and let y_0 satisfy $0 < d := |y_0 - x_0| < \beta_1 R_0$. Choose $r > 0$ small enough that $B(y_0, 3r)$ lies in $B(x_0, \beta_1 R_0)$, $|y - x_0| \geq d/2$ for $y \in B(y_0, 3r)$, $\beta_2 r^2 < d^2/4$, and $r^2 < \gamma T_0$ for the time length T_0 of this application. For $j \geq 0$, set $\tau_j = 2^{-j-1} r^2$ and $q_j = \frac{1}{2} \sqrt{2\tau_j}$. Since $q_j \leq r/2$, choose a maximal family of pairwise disjoint balls of radius $q_j/2$ with centers $x_{j\ell} \in B(y_0, r)$. These balls lie in $B(y_0, r + q_j/2)$, so comparison of volumes bounds their number by $C(r/q_j)^3 = C r^3 \tau_j^{-3/2}$. Maximality implies that the balls $B_{q_j}(x_{j\ell})$ cover $B(y_0, r)$, and their centers lie in $B(y_0, 2r)$. The corresponding boxes $B_{\sqrt{2\tau_j}}(x_{j\ell}) \times (\tau_j, 2\tau_j)$ cover $B(y_0, r) \times (0, r^2)$ as j varies. Every such box is

eligible for the Gaussian estimate, and its integral is at most $C\mathcal{N}_{x_0}(R_0, T_0; w)e^{-cd^2/\tau_j}$, with an absolute $c > 0$. Therefore

$$\int_0^{r^2} \int_{B_r(y_0)} |w|^2 \leq C\mathcal{N}_{x_0}(R_0, T_0; w)r^3 \sum_{j \geq 0} \tau_j^{-3/2} e^{-cd^2/\tau_j}.$$

For every $p > 0$, $e^{-cd^2/\tau} \leq C(p, d)\tau^p$ for sufficiently small τ . Taking p arbitrarily large shows that the last expression is $O(r^m)$ for every m . Thus w has integral vanishing of infinite order at $(y_0, 0)$. If $y_0 = x_0$, that property is the hypothesis of the application itself. This argument uses the L^2 box estimate only and does not assume pointwise spatial regularity at positive times.

It remains to reach every point of B_R . Define

$$r_0 = \beta_1 R, \quad r_{j+1} = r_j + \beta_1(R - r_j).$$

We have proved both $w(y, 0) = 0$ and integral vanishing at every $|y| < r_0$. Suppose these two conclusions hold for every $|y| < r_j$. Given y with $r_j \leq |y| < r_{j+1}$, choose $s < r_j$ sufficiently close to r_j that $|y| - s < \beta_1(R - s)$, and put $x_0 = sy/|y|$. Then $|x_0| = s < r_j$ and $|y - x_0| < \beta_1(R - |x_0|)$. The restriction of w to $B(x_0, R - |x_0|) \times [0, T_0]$ is continuous there. Its weak derivative fields restrict to this smaller cylinder, all four L^2 norms remain finite, and the differential inequality holds a.e. by restriction. Thus the smaller open cylinder satisfies the regularity and inequality hypotheses of Lemma 5.1; the ball lies in B_R , and integral vanishing at x_0 is the induction hypothesis. Applying the lemma with this center gives the Gaussian estimate at y for all sufficiently small t , since $y \neq x_0$. Dividing its box integral by the box measure gives averages tending to zero, and continuity yields $w(y, 0) = 0$. The covering argument above gives integral vanishing at y , which supplies the induction hypothesis for the next step.

Finally $r_j = R - (1 - \beta_1)^j(R - r_0)$ increases to R . For each $y \in B_R$, some finite j has $|y| < r_j$, proving the conclusion. $\square$

Remark 5.3 (Departure from [4]). The class in Theorem 5.2 uses the space-time weak-derivative $W_2^{2,1}$ convention and a continuous representative up to $t = 0$, both fixed in the statement of the theorem; this makes the pointwise growth assumption and the value $w(x, 0)$ unambiguous. ESS's Lemma 4.3 [4, pp. 230–233] estimates a pointwise value through (4.18). Here the Carleman estimate instead yields an L^2 bound on parabolic boxes, with the cutoff collar controlled by the L^2 norms in (4.1). The initial-time cutoff limit is justified by the dyadic shell sum above, and the final passage from the inner ball to B_R is the explicit radius iteration in the proof.

6 Backward uniqueness on a half-space

Backward uniqueness uses two different decays. The first Carleman estimate turns small Gaussian growth into decay away from the boundary plane. The anisotropic estimate then makes the remaining cutoff errors vanish as its parameter tends to infinity. We work throughout with the space-time weak derivatives defined in Section 2; in particular, all differential inequalities below hold almost everywhere.

Lemma 6.1 (Gaussian averages). *Set $A_0 = 10^{-12}$ and $\beta = 10^{-6}$. For every $c_1 > 0$ there is $\gamma = \gamma(c_1) \in (0, 1/12)$ with the following property. Let $A \in [0, A_0]$, and suppose that w is continuous on $\mathbb{R}_+^3 \times [0, 1)$, has space-time weak derivatives $Dw, D^2w, D_t w \in L^2$ on every bounded subset of Q_+ , and satisfies*

$$\begin{aligned} |D_t w + \Delta w| &\leq c_1(|Dw| + |w|) && \text{a.e. in } Q_+, \\ w(x, 0) &= 0 && \text{for } x \in \mathbb{R}_+^3, \\ |w(x, t)| &\leq e^{A|x|^2} && \text{for } (x, t) \in Q_+. \end{aligned}$$

Writing $\bar{A} = \max\{A, A_0/2\}$, there is $C = C(c_1, A) > 0$ such that, for every $x_3 > 2$ and $0 < t < \gamma$,

$$\frac{1}{t^{5/2}} \int_t^{5t/2} \int_{B_{\sqrt{3t}}(x)} |w(y, s)|^2 dy ds \leq C e^{8\bar{A}|x|^2} e^{-\beta x_3^2/(12t)}. \quad (6.1)$$

Proof. Fix $x_3 > 2$ and $0 < t < \gamma$, and write $h(s) = s e^{(1-s)/3}$. Put $\lambda = \sqrt{3t}$, $\rho = (x_3 - 1)/\lambda$, and $\sigma = 1/6$. The rescaled field

$$v(y, s) = w(x + \lambda y, \lambda^2(s - \sigma))$$

is defined for $y \in B_\rho$ and $\sigma < s < 3$. Its weak derivatives obey

$$|D_s v + \Delta_y v| \leq c_1 \lambda (|D_y v| + |v|), \quad |v(y, s)| \leq e^{2A|x|^2 + 2A\lambda^2|y|^2}.$$

The first inequality follows from the affine change of variables and $\lambda < 1$; the second follows from $|x + \lambda y|^2 \leq 2|x|^2 + 2\lambda^2|y|^2$.

The field v already uses the Carleman time coordinate: its source time is $\lambda^2(s - \sigma)$, where $\sigma = 1/6$. Multiply it by the mollified spatial cutoff, equal to one on $B_{\rho/2}$ and supported in $B_{3\rho/4}$, the final-time cutoff evaluated at $s - \sigma$ (one through $3/2$ and zero after $7/4$), and the initial-time cutoff evaluated at $s - \sigma$ (zero through ε and one from 2ε). Extend their product V_ε by zero. Its spatial transition is contained in $13\rho/20 \leq |y| \leq 3\rho/4$; the late transition is $\sigma + 3/2 \leq s \leq \sigma + 7/4$; and the initial transition is $\sigma + \varepsilon \leq s \leq \sigma + 2\varepsilon$. The weak product rule gives its first spatial, second spatial, and time weak derivatives in L^2 , with compact support in $\mathbb{R}^3 \times (0, 2)$. Thus Lemma 4.4 permits applying Proposition 4.1 to V_ε . Applying the Gaussian Carleman estimate, set

$$I_\varepsilon = \int_0^2 \int_{\mathbb{R}^3} h^{-2a}(s) e^{-|y|^2/(4s)} \left(\frac{a}{s} |V_\varepsilon|^2 + |D_y V_\varepsilon|^2 \right) dy ds.$$

Let $S_\rho = \{13\rho/20 \leq |y| \leq 3\rho/4\}$ and $L_\rho = B_{3\rho/4}$. The localized product-rule estimate has a spatial error on $S_\rho \times (\sigma, \sigma + 7/4)$ involving $|v| + |D_y v|$, a late-time error on $L_\rho \times [\sigma + 3/2, \sigma + 7/4]$ involving $|v|$, and an initial-strip error on $B_\rho \times [\sigma + \varepsilon, \sigma + 2\varepsilon]$ involving $\varepsilon^{-1}|v|$. Applying the Gaussian Carleman estimate to V_ε and squaring this bound gives

$$\begin{aligned} I_\varepsilon &\leq C_G c_1^2 \lambda^2 I_\varepsilon \\ &\quad + C(c_1) \int_{S_\rho \times (\sigma, \sigma + 7/4)} \omega_a (|v|^2 + |D_y v|^2) + C(c_1) \int_{L_\rho \times (\sigma + 3/2, \sigma + 7/4)} \omega_a |v|^2 \\ &\quad + C\varepsilon^{-2} \int_{\sigma + \varepsilon}^{\sigma + 2\varepsilon} \int_{B_\rho} \omega_a |v|^2. \end{aligned} \quad (6.2)$$

Here C_G is absolute and $\omega_a = h(s)^{-2a} e^{-|y|^2/(4s)}$.

The initial-strip error is controlled without extending w by zero. Since $v(y, s) = w(x + \lambda y, \lambda^2(s - \sigma))$, Lemma 4.7 gives

$$\varepsilon^{-2} \int_{\sigma+\varepsilon}^{\sigma+2\varepsilon} \int_{B_\rho} |v(y, s)|^2 dy ds \leq 2\lambda^{-1} \int_0^{2\lambda^2\varepsilon} \int_{B(x, \lambda\rho)} |D_t w(z, r)|^2 dz dr \longrightarrow 0.$$

For fixed a , ω_a is bounded on $B_\rho \times [\sigma, \sigma + 1/6]$, so the weighted initial error also tends to zero. We use this only to choose one sufficiently small ε ; the proof below does not pass the weighted left side to an $\varepsilon \downarrow 0$ limit.

Choose $\gamma(c_1)$ so that

$$\gamma \leq \frac{1}{24}, \quad \gamma \leq \frac{\beta}{3H}, \quad 3C_G c_1^2 \gamma \leq \frac{1}{2}, \quad H = \log h(3/2) > \frac{1}{6}.$$

For $0 < t < \gamma$, set $a = \beta\rho^2/H$. Then $\rho^2 > 1/(3t) > H/\beta$, so $a > 1$ and $\rho > 4$; also $\lambda^2 = 3t \leq 3\gamma$ absorbs the first term in the estimate for I_ε .

For the chosen $a = \beta\rho^2/H$, put $T_\rho = e^{8\bar{A}|x|^2} e^{-2\beta\rho^2}$. The spatial-shell errors lie on $S_\rho = \{|y| \leq 3\rho/4\}$; the late-time mass error lies in $B_{3\rho/4}$ over Carleman times $[\sigma + 3/2, \sigma + 7/4]$, where the unshifted rescaled time $s - \sigma$ lies in $[3/2, 7/4]$. Take $r = (16\sqrt{a})^{-1}$. A finite Besicovitch cover of S_ρ by balls of radius r has spatial overlap at most N , and its doubled balls lie in B_ρ . For Caccioppoli in the rescaled source time, set $a_j = \sigma + jr^2/2$, use inner intervals $(a_j, a_j + r^2)$, and enlarge them to $J_j = (a_j, a_j + 4r^2)$. These cover $(\sigma, \sigma + 7/4)$ up to a null set, remain below 2, and have overlap at most 8; the resulting outer cylinders therefore have overlap at most $8N$. The Gaussian weight varies by a bounded factor on each outer cylinder because $\rho r = \sqrt{H/\beta}/16$ is fixed. Caccioppoli on each inner cylinder, including its initial-time version at $s = \sigma$, controls the shell gradient by outer-cylinder radial mass. The radial mass tail bound, the late-time bound, and the cutoff derivative bounds give shell and late errors bounded by $C(c_1, A)(1+a)(1+\rho)^3 T_\rho$. Choose $0 < \varepsilon < 1/12$ small enough that the weighted initial error in the cutoff estimate above is at most T_ρ ; the initial-strip limit above permits this choice. Absorbing the term proportional to $c_1^2 \lambda^2$ then yields the direct one-cutoff estimate

$$I_\varepsilon \leq C(c_1, A)(1+a)(1+\rho)^3 e^{8\bar{A}|x|^2} e^{-2\beta\rho^2}. \quad (6.3)$$

The spatial cutoff equals one on B_1 , and for this choice of ε the time cutoff equals one on the averaging core $B_1 \times (1/2, 1)$. Since $\omega_a \geq e^{-3/2}$ there, the core mass is directly controlled:

$$D := \int_{1/2}^1 \int_{B_1} |v(y, s)|^2 dy ds \leq e^{3/2} I_\varepsilon.$$

Finally, $a = \beta\rho^2/H$ implies that the polynomial factor is absorbed by the Gaussian tail: $(1+a)(1+\rho)^3 e^{-2\beta\rho^2} \leq C e^{-\beta\rho^2}$. Thus $D \leq C(c_1, A) e^{8\bar{A}|x|^2} e^{-\beta\rho^2}$.

Changing variables back on $B_1 \times (1/2, 1)$ gives the left side of (6.1) up to the absolute factor $3^{5/2}$. Finally, $\rho^2 = (x_3 - 1)^2/(3t) \geq x_3^2/(12t)$ for $x_3 > 2$. $\square$

Remark 6.2 (Departure from [4]). The initial time is not extended by zero. A cutoff above $s = 1/6 + \varepsilon$ and Lemma 4.7 replace that extension; its differentiated error tends to zero by the displayed L^2 estimate. The proof uses Lemma 4.6 for the collar gradients and proves an average estimate, so it does not use the pointwise parabolic regularity bounds or estimate (5.17) of ESS [4, §5, pp. 234–236]. The explicit choice $A_0 = 10^{-12}$, $\beta = 10^{-6}$ also supplies the uniform positive Gaussian exponent needed when the input value A is zero; it satisfies $A_0 < 1/32$, $8A_0 < 1/256$, $16A_0/24 < 1/48$, $16A_0/24 < \beta/48$, $\beta < 1/256$, and $\beta < H/96$, where $H = \log(3/2) - 1/6 > 1/6$.

Lemma 6.3 (Vanishing on a short time interval). *Fix $c_1 > 0$ and let $A \in [0, A_0]$. Suppose w satisfies the continuity, weak-derivative, differential-inequality and zero-trace hypotheses of Lemma 6.1, as well as $|w(x, t)| \leq e^{A|x|^2}$ on Q_+ . There is $\gamma_1 = \gamma_1(c_1) \in (0, \gamma(c_1)/2]$ such that*

$$w(x, t) = 0 \quad \text{for every } x \in \mathbb{R}_+^3 \text{ and } 0 < t < \gamma_1.$$

Proof. Fix $\alpha = 3/4$, and let $c_*(\alpha)$ and $a_0(\alpha)$ be the constants in Proposition 4.2. Put $\lambda = \sqrt{2\gamma_1}$ and, for $y_3 > 0$ and $1/2 < s < 1$, set

$$v(y, s) = w(\lambda y, \lambda^2(s - 1/2)), \quad F(y_3, s) = \frac{(1-s)y_3^{2\alpha}}{s^\alpha}.$$

Then $|D_s v + \Delta_y v| \leq c_1 \lambda (|D_y v| + |v|)$ a.e. and $v(\cdot, s) \rightarrow 0$ locally in L^2 as $s \downarrow 1/2$, by the assumed continuous zero trace. Choose $y_- = 3/\lambda + 1$, $y_+ = 3/\lambda + 3/2$, and $y_0 = 3/\lambda + 2$. Define $B = 2F(y_0, 1/2)$ and $\Phi(y_3, s) = F(y_3, s) - B$. Let ψ_1 be smooth, zero on $(-\infty, y_-]$ and one on $[y_+, \infty)$. Let ψ_2 be smooth, zero on $(-\infty, -3/4]$ and one on $[-2/3, \infty)$. Set $\eta(y_3, s) = \psi_1(y_3)\psi_2(\Phi(y_3, s)/B)$.

The derivatives of η are supported where $\Phi \leq -D/2$, with $D = -2\Phi(y_+, 1/2) > 0$. Indeed, F increases in y_3 and decreases in s , so the support of ψ_1' has $\Phi \leq \Phi(y_+, 1/2) = -D/2$. On the support of ψ_2' , $\Phi \leq -2B/3$. Since $\lambda < 1/2$,

$$\frac{D}{2B} = 1 - \frac{1}{2} \left(\frac{y_+}{y_0} \right)^{3/2} < 1 - \frac{45}{128} < \frac{2}{3},$$

which gives the same bound there. Also $\Phi > 0$ implies $y_3 > y_+$, so $\eta = 1$ on that region.

For $R > 1$, choose a smooth spatial cutoff χ_R equal to one on B_R , zero outside B_{2R} , and satisfying $|D\chi_R| \leq C/R$, $|D^2\chi_R| \leq C/R^2$. For $\varepsilon > 0$, choose ϑ_ε equal to zero for $s \leq 1/2 + \varepsilon$, equal to one for $s \geq 1/2 + 2\varepsilon$, and with $|\vartheta'_\varepsilon| \leq C/\varepsilon$. The field $V_{R,\varepsilon} = \chi_R \eta \vartheta_\varepsilon v$ has compact support in $(\mathbb{R}_+^3 + e_3) \times (0, 1)$: its spatial support stays above $y_3 > 1$, and on the bounded set B_{2R} the condition $\psi_2(\Phi/B) \neq 0$ keeps s a positive distance below 1. It belongs to the Sobolev class of Lemma 4.4 by the weak product rule.

Apply Proposition 4.2 to $V_{R,\varepsilon}$ with parameter $a > \max\{a_0(3/4), 1\}$. The terms containing $D_s v + \Delta_y v$ are bounded by $Cc_1^2 \lambda^2$ times the Carleman left side, plus cutoff errors. Choose γ_1 so that

$$2Cc_*(3/4)c_1^2\gamma_1 \leq \frac{1}{4}, \quad \gamma_1 \leq \frac{\gamma(c_1)}{2},$$

where C is the absolute product-rule constant. These terms are then absorbed. The errors from $D\eta$ are supported in

$$\omega = \{y_3 > y_-, 1/2 < s < 1, \Phi(y_3, s) \leq -D/2\}.$$

Indeed, the ψ_1' term is supported in (y_-, y_+) , while each term from ψ_2' is multiplied by ψ_1 and vanishes for $y_3 \leq y_-$. The preceding cutoff estimates give $\Phi \leq -D/2$ on both supports. After absorption, these errors give

$$\begin{aligned} I_a &:= \int_{1/2}^1 \int_{\mathbb{R}_+^3} e^{-|y'|^2/(4s)} e^{2a\Phi} (a|\eta v|^2 + s|D_y(\eta v)|^2) dy ds \\ &\leq C(c_1) \int_{\omega} (1+y_3)^4 e^{-|y'|^2/(4s)} e^{2a\Phi} (|v|^2 + |D_y v|^2) dy ds + E_{R,a} + E_{\varepsilon,a}. \end{aligned} \quad (6.4)$$

Here $E_{R,a}$ is the error from $D\chi_R, D^2\chi_R$, and $E_{\varepsilon,a}$ is the error from ϑ'_ε . The factor $(1+y_3)^4$ bounds the squares of the first and second derivatives of the fixed cutoff η on $y_3 > 1$.

The lower-time error vanishes as $\varepsilon \downarrow 0$. On the bounded spatial support B_{2R} the Carleman weight is bounded for fixed a, R , and Lemma 4.7 gives

$$E_{\varepsilon,a} \leq C(a, R, \lambda) \varepsilon^{-2} \int_{1/2+\varepsilon}^{1/2+2\varepsilon} \int_{B_{2R}} |v|^2 dy ds \leq C(a, R, \lambda) \int_0^{2\lambda^2\varepsilon} \int_{B_{2\lambda R}} |D_t w|^2 dx dt \longrightarrow 0.$$

We next control the noncompact collar by shifted dyadic cells. For $d = 2^{-k}$, each layer $d/2 < \delta = s - 1/2 < d$ is partitioned into spatial cubes of side $\sqrt{d}/128$ and time intervals of length $d/4096$, with 2048 time labels. At a cell center Y with time $\delta_Y \in [d/2, d]$, the Gaussian average estimate gives an average-box mass bounded by

$$C(c_1, A, \lambda) d^2 e^{8A_0\lambda^2|Y|^2} e^{-\beta Y_3^2/(12d)}.$$

Caccioppoli with $r = \sqrt{\delta_Y}/8$ contributes a factor bounded by $C(c_1)/d$; hence the cell energy has the bound

$$\int_{C_{k,m,\ell}} (|v|^2 + |D_y v|^2) \leq C(c_1, A, \lambda) d \exp\left(16A_0\lambda^2|Y|^2 - \frac{\beta Y_3^2}{48d}\right).$$

After including $(1 + y_3)^4$ and the tangential Gaussian, Gaussian absorption bounds the cell sum by a fixed spatial lattice profile times a dyadic factor $C2^{N(k+1)}e^{-c\beta 2^{k+1}}$ for a fixed integer N . The spatial profile is summable, there are only 2048 time labels on each layer, and the dyadic Gaussian series converges. Consequently, after omitting $e^{2a\Phi}$,

$$J := \int_{\omega} (1 + y_3)^4 e^{-|y'|^2/(4s)} (|v|^2 + |D_y v|^2) dy ds < \infty,$$

with a bound independent of a . The cell-energy power here is d : it comes from the d^2 Gaussian average and the d^{-1} Caccioppoli factor.

For fixed a , the spatial cutoff error tends to zero as $R \rightarrow \infty$: its derivatives are supported in $\{|y| > R\}$, and the weighted integral is finite because the negative quadratic Gaussian dominates the positive normal weight $e^{2ay_3^{2\alpha}}$, with $2\alpha < 2$. For every fixed R , the lower-time error tends to zero as $\varepsilon \downarrow 0$ by Lemma 4.7. Thus, for fixed a , the cutoff estimate has the form $I_a \leq C(c_1)J e^{-Da} + S_R + E_R(\varepsilon)$, where $S_R \rightarrow 0$ as $R \rightarrow \infty$ and $E_R(\varepsilon) \rightarrow 0$ for each fixed R . Applying the two-parameter limit in that order, first $\varepsilon \downarrow 0$ and then $R \rightarrow \infty$, gives

$$I_a \leq C(c_1, A, \lambda) e^{-Da},$$

with a constant independent of a .

For any compact $K \subset \{y_3 > 0, 1/2 < s < 1, \Phi > 0\}$, $\eta = 1$ and Φ has a positive minimum on K . Hence $ae^{2a \min_K \Phi} \int_K |v|^2 \leq I_a \rightarrow 0$ as $a \rightarrow \infty$. Continuity gives $v = 0$ on this open curved region.

It remains to fill the half-space at each fixed time $s_0 \in (1/2, 1)$. Write

$$g(s) = \left(\frac{Bs^\alpha}{1-s} \right)^{1/(2\alpha)},$$

so that $\Phi(y_3, s) > 0$ exactly when $y_3 > g(s)$. Given any target $y = (y', y_3) \in \mathbb{R}_+^3$, choose $H > \max\{2g(s_0), y_3\}$ and let $c = (y', H)$, $R = H - y_3/2$. Then $|y - c| = H - y_3 < R < H$, so $y \in B_R(c) \subset \mathbb{R}_+^3$. By continuity of g , there is $T > 0$ such that $s_0 + T < 1$ and $g(s) < H - R/2$ for $s \in [s_0, s_0 + T]$. Thus v is zero on a space-time neighborhood of (c, s_0) lying above the graph. The

translated field $u(z, t) = v(c + z, s_0 + t)$ satisfies all hypotheses of Theorem 5.2 on $B_R(0) \times (0, T)$: its weak derivatives and differential inequality are inherited; the cylinder has compact closure in the half-space; and its all-orders condition holds because $u = 0$ when $|z| + \sqrt{t} < r_0$ for some $r_0 > 0$. On the rest of the compact cylinder u is bounded, so the condition holds with $C_k = r_0^{-k} \sup |u|$. Theorem 5.2 therefore gives $v(\cdot, s_0) = 0$ on $B_R(c)$, in particular $v(y, s_0) = 0$. As y and s_0 were arbitrary, these balls cover the connected half-space at every $s_0 \in (1/2, 1)$. No time reversal is used. Returning to (x, t) proves the lemma. $\square$

Remark 6.4 (Departure from [4]). The application of Proposition 4.2 to the noncompact cutoff is justified by first inserting χ_R ; its shell terms vanish by the Gaussian-average estimate and Caccioppoli. A cutoff above $s = 1/2 + \varepsilon$, controlled by Lemma 4.7, avoids extending the field by zero at its initial trace. The source invokes Theorem 5.2 to fill the curved zero region in one sentence [4, §5, pp. 236–238]; here the center $c = (y', H)$ and radius $R = H - y_3/2$ are chosen for each target point, and Theorem 5.2 is applied forward from the time s_0 . The modified transition interval for ψ_2 ensures every cutoff derivative lies in $\Phi \leq -D/2$.

Lemma 6.5 (Iteration in time). *Suppose w satisfies the hypotheses of Lemma 6.3 with some $A \in [0, A_0]$. Then $w \equiv 0$ on Q_+ .*

Proof. Set $\gamma_0 = 0$ and $\gamma_{k+1} = \gamma_k + (1 - \gamma_k)\gamma_1$. If $w = 0$ for $0 < t < \gamma_k$, define

$$w^{(k)}(y, s) = w(\sqrt{1 - \gamma_k} y, \gamma_k + (1 - \gamma_k)s), \quad (y, s) \in Q_+.$$

The affine change of variables preserves the space-time weak derivative class and continuity up to $s = 0$. Its initial trace is zero: for $k = 0$ this is the given trace, and for $k \geq 1$ it follows from continuity and $w(\cdot, t) = 0$ for $t < \gamma_k$. Moreover,

$$|D_s w^{(k)} + \Delta_y w^{(k)}| \leq c_1(|D_y w^{(k)}| + |w^{(k)}|), \quad |w^{(k)}(y, s)| \leq e^{A|y|^2},$$

because the time and Laplacian factors are both $1 - \gamma_k$, while the gradient factor is $\sqrt{1 - \gamma_k}$. The local L^2 conditions are preserved by this nonsingular change of variables. Lemma 6.3 applied to $w^{(k)}$ shows $w = 0$ for $0 < t < \gamma_{k+1}$. Induction gives $\gamma_k = 1 - (1 - \gamma_1)^k \uparrow 1$, and every $t < 1$ lies below some γ_k . $\square$

Remark 6.6 (Departure from [4]). The initial trace at each iteration is justified by continuity from the already vanishing time interval; the rescaling factors and the recurrence are displayed rather than left implicit in the proof of Lemma 5.4 [4, §5, p. 238].

Theorem 6.7 (Backward uniqueness on a half-space). *Let $c_1 > 0$, $M \in \mathbb{R}$, and $Q_+ = \mathbb{R}_+^3 \times (0, 1)$. Suppose that w is continuous on $\mathbb{R}_+^3 \times [0, 1)$, has space-time weak derivatives $Dw, D^2w, D_t w$ on Q_+ , and satisfies*

$$\begin{aligned} w(x, 0) &= 0 & (x \in \mathbb{R}_+^3), \\ |D_t w + \Delta w| &\leq c_1(|Dw| + |w|) & \text{a.e. on } Q_+, \\ |w(x, t)| &\leq e^{M|x|^2} & ((x, t) \in Q_+), \end{aligned}$$

with $w, Dw, D^2w, D_t w \in L^2$ on every bounded subset of Q_+ . Then $w \equiv 0$ on Q_+ .

Proof. If $M \leq A_0$, replace M by $A = \max\{M, 0\}$; the growth bound then implies $|w(x, t)| \leq e^{A|x|^2}$ and $A \in [0, A_0]$. Lemma 6.5 proves the conclusion. Suppose next that $M > A_0$, and set

$\lambda^2 = A_0/(2M) < 1/2$. Define $v_0(y, s) = w(\lambda y, \lambda^2 s)$. The weak derivatives transform by the affine chain rule, giving

$$|D_s v_0 + \Delta_y v_0| \leq c_1 (\lambda |D_y v_0| + \lambda^2 |v_0|) \leq c_1 (|D_y v_0| + |v_0|), \quad |v_0(y, s)| \leq e^{(A_0/2)|y|^2}.$$

The zero trace, continuity, and local L^2 conditions are preserved, so Lemma 6.5 gives $v_0 \equiv 0$. Hence $w(x, t) = 0$ for $0 < t < \lambda^2$.

To cover the remaining time, set $t_k = k\lambda^2$ while $t_k < 1$ and $\delta_k = \min\{\lambda^2, 1 - t_k\}$. Once $w(\cdot, t_k) = 0$, define

$$v_k(y, s) = w(\sqrt{\delta_k} y, t_k + \delta_k s), \quad (y, s) \in Q_+.$$

For $k = 0$, the zero trace is the hypothesis; for $k \geq 1$, continuity and the preceding slab conclusion give $w(\cdot, t_k) = 0$. The field v_k is continuous up to $s = 0$ and has the required local weak derivatives by the change of variables. Since $0 < \delta_k \leq \lambda^2 < 1/2$,

$$|D_s v_k + \Delta_y v_k| \leq c_1 (\sqrt{\delta_k} |D_y v_k| + \delta_k |v_k|) \leq c_1 (|D_y v_k| + |v_k|), \quad |v_k(y, s)| \leq e^{M\delta_k |y|^2} \leq e^{(A_0/2)|y|^2}.$$

Thus Lemma 6.5 applies on every slab and yields $w = 0$ for $t_k < t < t_k + \delta_k$. The construction takes at most $\lceil 1/\lambda^2 \rceil$ slabs; the final one has length $\delta_k = 1 - t_k < \lambda^2$ if the endpoint is not reached exactly. They cover $0 < t < 1$, proving the theorem. $\square$

Remark 6.8 (Departure from [4]). The final large-growth iteration in the source asks for another application of the small-growth lemma with exponent M , outside its stated range [4, §5, p. 238]. Here each time slab is spatially and parabolically rescaled by its own length $\delta_k \leq \lambda^2$; this keeps the growth exponent at most $A_0/2$ and the differential-inequality constant at most c_1 . The last partial slab is included explicitly.

Parts II and III. Existence and the associated pressure (proved in the CKN paper)

7 Leray existence and the associated pressure

The six results of Section 3 that concern the existence of suitable Leray–Hopf solutions and their pressures are Theorem 3.5 (existence), Corollary 3.6 (existence with singular set of zero one-dimensional parabolic Hausdorff measure), Theorem 3.7 (associated pressure), and their forced versions Theorem 3.9, Theorem 3.10 and Theorem 3.11. They are stated in Section 3 because the theorems of Parts IV–VI are about Leray–Hopf solutions, but they do not depend on anything in Parts I and IV–VI. They are proved in the last part of the Caffarelli–Kohn–Nirenberg paper [2], where the statements carry the same labels and the proofs occupy the sections on the whole-space pressure (`sec:riesz-pressure`), the Leray-regularized equations (`sec:regularised`), uniform bounds, compactness, the limit (`sec:limit`) and forced Leray solutions (`sec:forced-leray`). They are formalized in the CKN library as `CKN.leray_existence`, `CKN.leray_existence_singularSet`, `CKN.associatedPressure`, `CKN.lerayExistenceForced`, `CKN.lerayExistenceForcedSingularSet` and `CKN.associatedPressureForced`.

The proofs of Parts IV–VI use, besides these statements, the following notation and facts from the CKN paper. The whole-space pressure is defined in the CKN paper as follows, and is used here with the same sign convention. Further facts from the same part ([2, Lemma 11.1] on compactness,

[2, Lemma 11.5] on restriction, the energy interpolation [2, Lemma 8.4], the Riesz duality [2, Lemma 8.2], the regularized solutions of [2, Theorem 9.4] and the limit of [2, Proposition 12.1]) are cited where they are used.

Definition 7.1 (Whole-space pressure). For $1 < r < \infty$, on Schwartz functions let $\mathcal{R}_j$ denote the Fourier multiplier $i\xi_j/|\xi|$. The product $\mathcal{R}_i\mathcal{R}_j$ has a bounded extension to $L^r(\mathbb{R}^3)$. For a jointly measurable tensor field $F = (F_{ij})$ whose slices $F_{ij}(\cdot, t)$ belong to $L^r(\mathbb{R}^3)$ for almost every t , define

$$P[F](\cdot, t) = \sum_{i,j=1}^3 \mathcal{R}_i\mathcal{R}_j F_{ij}(\cdot, t). \quad (7.1)$$

The extensions for different exponents agree on their common domains, so this defines one pressure operator. Indeed, bounded compactly supported inputs belong to $L^r \cap L^s \cap L^2$ for any two finite exponents r, s ; the CKN extensions agree there with the same negative raw operator. Truncating an input in $L^r \cap L^s$ by balls and bounded values gives convergence in both norms, so the two outputs agree as distributions. With this convention

$$-\Delta P[F] = \sum_{i,j=1}^3 \partial_i \partial_j F_{ij} \quad (7.2)$$

in the spatial distribution sense for almost every t . The CKN raw second derivative operator has the opposite sign; the all-exponent bounded extension in the CKN convention represents its negative, which is $\mathcal{R}_i\mathcal{R}_j$ on the common $L^r \cap L^2$ domain.

There is a jointly measurable representative of $P[F]$, and for almost every t ,

$$\|P[F](\cdot, t)\|_{L^r(\mathbb{R}^3)} \leq C_r \sum_{i,j=1}^3 \|F_{ij}(\cdot, t)\|_{L^r(\mathbb{R}^3)}. \quad (7.3)$$

Here and below C_r depends only on r in dimension three. If $F \in L^r(\mathbb{R}^3 \times (0, T))$, then

$$\|P[F]\|_{L^r(\mathbb{R}^3 \times (0, T))} \leq C_r \sum_{i,j=1}^3 \|F_{ij}\|_{L^r(\mathbb{R}^3 \times (0, T))}. \quad (7.4)$$

Part IV. Regularity of $L^\infty L^3$ solutions

8 Local regularity near small-energy points

We use the CKN past cylinder

$$\mathcal{Q}_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0], \quad z_0 = (x_0, t_0),$$

and write $\overline{\mathcal{Q}_r(z_0)} = \overline{B_r(x_0)} \times [t_0 - r^2, t_0]$. The parabolic distance is

$$d_{\text{par}}((x, t), (y, s)) = \max\{|x - y|, |t - s|^{1/2}\}.$$

All vector norms below are Euclidean. Throughout this section, “suitable” means suitable in the sense of Definition 3.1, with force $f = 0$; the spatial gradient is the separate datum Du fixed there.

Lemma 8.1 (Theorem A on a cylinder of radius r). *There are constants $\varepsilon_0, \gamma_0, C_4$ with $0 < \varepsilon_0$, $0 < \gamma_0 \leq 2/3$, and $0 \leq C_4$, supplied by CKN Theorem A at $q = 3$, with the following property. Let $D = \Omega \times I$ be a CKN solution domain and let $(u, Du, p, 0)$ be suitable there with exponent $q = 3$. Suppose $z_0 = (x_0, t_0) \in D$, $r > 0$, $\overline{\mathcal{Q}_r(z_0)} \subset D$, and*

$$r^{-2} \iint_{\mathcal{Q}_r(z_0)} (|u|^3 + |p|^{3/2}) dx dt < \varepsilon_0. \quad (8.1)$$

Then there is a representative $u_{z_0, r}$ of u on $\mathcal{Q}_{r/2}(z_0)$ which extends to its closure and satisfies

$$\begin{aligned} \sup_{\overline{\mathcal{Q}_{r/2}(z_0)}} |u_{z_0, r}| &\leq C_4 r^{-1}, \\ |u_{z_0, r}(z) - u_{z_0, r}(z')| &\leq C_4 r^{-1-\gamma_0} d_{\text{par}}(z, z')^{\gamma_0} \quad (z, z' \in \overline{\mathcal{Q}_{r/2}(z_0)}). \end{aligned}$$

Proof. Set $\Phi(y, s) = (x_0 + ry, t_0 + r^2 s)$ and define $u_r(y, s) = ru(\Phi(y, s))$, $Du_r(y, s) = r^2 Du(\Phi(y, s))$, and $p_r(y, s) = r^2 p(\Phi(y, s))$. The rescaled domain is $D_r = \Phi^{-1}(D)$. The CKN rescaling theorem, together with the equivalence of its suitable and integrable-suitable predicates, makes $(u_r, Du_r, p_r, 0)$ suitable on D_r . The affine map Φ sends the closed unit past cylinder onto $\overline{\mathcal{Q}_r(z_0)}$, so the assumed domain inclusion gives the closed-cylinder hypothesis of Theorem A. The Jacobian of Φ is r^5 , and hence

$$\iint_{\mathcal{Q}_1(0)} |u_r|^3 = r^{-2} \iint_{\mathcal{Q}_r(z_0)} |u|^3, \quad \iint_{\mathcal{Q}_1(0)} |p_r|^{3/2} = r^{-2} \iint_{\mathcal{Q}_r(z_0)} |p|^{3/2}.$$

The strict real-valued bound in (8.1) implies the non-strict extended-integral bound required by Theorem A; the force term vanishes. That theorem gives a representative w on the closed half unit-cylinder with supremum at most C_4 and parabolic Hölder seminorm at most C_4 . Define $u_{z_0, r}(z) = r^{-1} w(\Phi^{-1}(z))$. Since Φ preserves null sets, this is a representative of u on the half-cylinder. Moreover, $d_{\text{par}}(\Phi^{-1}z, \Phi^{-1}z') = r^{-1} d_{\text{par}}(z, z')$. The two stated bounds follow. $\square$

Lemma 8.2 (Top-boundary ε -regularity). *Let $\rho > r > 0$, $z_0 = (x_0, t_0)$, and suppose $(u, Du, p, 0)$ is suitable with exponent $q = 3$ on $B_\rho(x_0) \times (t_0 - \rho^2, t_0)$. If*

$$r^{-2} \iint_{B_r(x_0) \times (t_0 - r^2, t_0)} (|u|^3 + |p|^{3/2}) dx dt < \varepsilon_0/8, \quad (8.2)$$

then, with $V = B_{r/2}(x_0) \times (t_0 - r^2/4, t_0)$, u has a representative on V which extends to the closure and obeys

$$\begin{aligned} \sup_{\overline{V}} |u_{\text{top}}| &\leq 2C_4/r, \\ |u_{\text{top}}(z) - u_{\text{top}}(z')| &\leq 4 \cdot 8^{\gamma_0} C_4 r^{-1-\gamma_0} d_{\text{par}}(z, z')^{\gamma_0} \quad (z, z' \in \overline{V}). \end{aligned}$$

Proof. For $x \in B_{r/2}(x_0)$ and $0 < s < r^2/4$, set $z_{x, s} = (x, t_0 - s)$ and apply Lemma 8.1 with radius $r' = r/2$. The closed full cylinder $\overline{\mathcal{Q}_{r/2}(z_{x, s})}$ lies in $B_r(x_0) \times (t_0 - r^2, t_0)$: spatially, $\overline{B_{r/2}(x)} \subset B_r(x_0)$, and its time interval is $[t_0 - s - r^2/4, t_0 - s]$, strictly between $t_0 - r^2$ and t_0 . Consequently

$$(r/2)^{-2} \iint_{\mathcal{Q}_{r/2}(z_{x, s})} (|u|^3 + |p|^{3/2}) \leq 4r^{-2} \iint_{B_r(x_0) \times (t_0 - r^2, t_0)} (|u|^3 + |p|^{3/2}) < \varepsilon_0/2.$$

The smallness and domain hypotheses of that lemma hold uniformly in x, s . It gives a representative on the closed half-cylinder $\overline{B_{r/4}(x)} \times [t_0 - s - r^2/16, t_0 - s]$, with the displayed bounds.

These open interiors, as x and s vary, cover $V = B_{r/2}(x_0) \times (t_0 - r^2/4, t_0)$. Indeed, for $(y, t) \in V$, put $a = t_0 - t$ and choose $s = a - \min\{a/2, r^2/32\}$. Then $0 < s < r^2/4$ and $0 < a - s < r^2/16$, so (y, t) lies strictly inside the half-cylinder centered at $(y, t_0 - s)$. The local representatives agree on every nonempty overlap of their open interiors: they are continuous there and equal to u almost everywhere there, while a nonempty open overlap has positive spacetime measure. A countable subcover of V shows that the glued representative equals u almost everywhere on V .

The local estimates have the same exponent and the same constants. If $z, z' \in V$ and $d_{\text{par}}(z, z') < r/8$, choose the spatial center to be that of the later point and choose a time center strictly between the later point and t_0 , sufficiently close to the later point. More explicitly, write $a_{\text{late}} < a_{\text{early}}$ for their time deficits from t_0 and take $c = a_{\text{late}} - \min\{a_{\text{late}}/2, r^2/64\} > 0$. Then both deficits lie strictly between c and $c + r^2/16$, because $a_{\text{early}} - a_{\text{late}} < r^2/64$, and both spatial points lie in the radius- $r/4$ ball about the later point. Thus both points lie in one of the half-cylinder interiors, giving the stated local Hölder estimate. For $d_{\text{par}}(z, z') \geq r/8$, the uniform supremum bound gives

$$|u_{\text{top}}(z) - u_{\text{top}}(z')| \leq 4C_4/r \leq 4 \cdot 8^{\gamma_0} C_4 r^{-1-\gamma_0} d_{\text{par}}(z, z')^{\gamma_0}.$$

Hence the glued representative is Hölder on all of V , with a uniform constant. A Hölder function on V extends uniquely with the same bound to its closure, including the face $t = t_0$. $\square$

Set

$$D = B_1(0) \times (-1, 0), \quad \widehat{D} = B_1(0) \times (-1, 0], \quad F_{\text{top}} = B_1(0) \times \{0\}.$$

Throughout, $(u, Du, p, 0)$ is suitable with exponent $q = 3$ on D .

Definition 8.3 (Good point). A point $z_0 = (x_0, t_0) \in \widehat{D}$ is *good* if there is an $r > 0$ such that

$$B_r(x_0) \times (t_0 - r^2, t_0) \subset D, \quad \overline{B_r(x_0)} \times [t_0 - r^2, t_0] \subset \widehat{D},$$

and

$$r^{-2} \iint_{B_r(x_0) \times (t_0 - r^2, t_0)} (|u|^3 + |p|^{3/2}) dx dt < \varepsilon_0/8.$$

Lemma 8.4 (Good points and Hölder gluing). *The good-point set is relatively open in $\widehat{D}$. Let $K \subset \overline{B_{1/2}(0)} \times (-1/4, 0)$ be compact and suppose every point of K is good. Then there are a relatively open neighborhood $N \subset \widehat{D}$ of K and a function $\tilde{u} : N \rightarrow \mathbb{R}^3$ such that $\tilde{u} = u$ almost everywhere on $N \cap D$ and $\tilde{u}|_K$ is parabolically Hölder with exponent γ_0 . In particular, this representative is Hölder on K including its top face $t = 0$, and extends continuously there.*

Moreover, if $z_0 = (x_0, t_0) \in K$ is not good, then for every $R > 0$ such that $B_R(x_0) \times (t_0 - R^2, t_0) \subset D$,

$$R^{-2} \iint_{B_R(x_0) \times (t_0 - R^2, t_0)} (|u|^3 + |p|^{3/2}) dx dt \geq \varepsilon_0/8. \quad (8.3)$$

Proof. Write $G = |u|^3 + |p|^{3/2}$. Fix a good point with witness radius r . As $r' \uparrow r$, the cylinders $B_{r'}(x_0) \times (t_0 - r'^2, t_0)$ increase to the witness cylinder. Monotone convergence and the strict smallness allow us to replace r by some $r' < r$ for which the normalized integral is still below $\varepsilon_0/8$. The resulting cylinder has positive spatial and time margin inside the original witness cylinder.

First note that $G \in L^1_{\text{loc}}(D)$. On each local box $B \times J \Subset D$, clause (S1) of Definition 3.1 gives $u \in L^\infty(J; L^2(B))$ and $u \in L^2(J; H^1(B))$. The local Sobolev inequality gives

$$\|u(\cdot, t)\|_{L^6(B)} \leq C_B (\|Du(\cdot, t)\|_{L^2(B)} + \|u(\cdot, t)\|_{L^2(B)}).$$

Interpolation between $L^2(B)$ and $L^6(B)$ then yields

$$\int_J \|u(\cdot, t)\|_{L^3(B)}^3 dt \leq \left(\operatorname{ess\,sup}_{t \in J} \|u(\cdot, t)\|_{L^2(B)} \right)^{3/2} |J|^{1/4} \left(\int_J \|u(\cdot, t)\|_{L^6(B)}^2 dt \right)^{3/4} < \infty.$$

The pressure clause of (S1) gives $p \in L^{3/2}(B \times J)$, so $G \in L^1(B \times J)$.

For centers sufficiently close to z_0 in the relative topology of $\widehat{D}$, the translated open-top cylinders of radius r' remain admissible in Definition 8.3. If $t_0 = 0$, these centers satisfy $t' \leq 0$; the spatial and lower-time margins from $r' < r$ keep their cylinders inside the original witness cylinder. The function G is integrable there by the witness's strict smallness, so it dominates the translated integrands. If $t_0 < 0$, choose $\eta > 0$ small enough that

$$K_0 = \overline{B}_{r'+\eta}(x_0) \times [t_0 - r'^2 - \eta, t_0 + \eta] \Subset D.$$

Every translated cylinder with center sufficiently close to z_0 is contained in K_0 . In both cases the cylinder indicators converge almost everywhere as the center varies, since their boundaries have measure zero. Dominated convergence, using G on the witness cylinder at the top and $\mathbf{1}_{K_0}G$ at interior points, gives continuity of the translated integral. The strict inequality persists, proving relative openness.

We next construct local representatives around points of K . Fix a good point $z_0 = (x_0, t_0)$ and a witness radius. Shrink it as in the preceding paragraph to a radius $r' < r$, retaining strict smallness. Choose ρ with $r' < \rho < r$. The suitable solution restricts to $B_\rho(x_0) \times (t_0 - \rho^2, t_0)$ by [2, Lemma 11.5], and this cylinder lies in D . Lemma 8.2 gives a representative Hölder on the closed past half-cylinder centered at z_0 . If $t_0 = 0$, its restriction to a relative neighborhood of z_0 in $\widehat{D}$ is the required one-sided representative.

If $t_0 < 0$, use the interior CKN conclusion to obtain a full open neighborhood. At center (x_0, t_0) and radius $r'/2$, the cylinder is contained in the witness cylinder and its normalized integral is less than $\varepsilon_0/2$. Its integral varies continuously under a small time translation. By the local-integrability argument above, $G \in L^1_{\text{loc}}(D)$. Choose a fixed compact set $K_1 \Subset D$ containing the closures of all these cylinders for sufficiently small translations. Their indicators converge almost everywhere, so dominated convergence applies with dominating function $\mathbf{1}_{K_1}G$. Therefore, for some sufficiently small $\sigma > 0$, the CKN cylinder of radius $r'/2$ centered at $(x_0, t_0 + \sigma)$ has normalized integral less than ε_0 and closed full cylinder in D . Choose also $\sigma < r'^2/16$ and $\sigma < -t_0$. Lemma 8.1 then gives a Hölder representative on an open neighborhood of z_0 .

Thus each point of K has a relative neighborhood in $\widehat{D}$ with a local representative equal to u almost everywhere on its intersection with D , Hölder with exponent γ_0 . For any two such neighborhoods with nonempty intersection, the intersection contains a nonempty open subset of D : this is immediate for an interior point of the intersection; if it meets only the top face, both neighborhoods are top neighborhoods and their spatial and past-time interiors overlap below $t = 0$. On that open subset the two representatives agree almost everywhere and are continuous, hence agree pointwise; continuity up to the top gives agreement on the full intersection. Compactness of K gives a finite subcover. The local representatives consequently glue to one function on its relatively open union N , and the finite cover shows it equals u almost everywhere on $N \cap D$.

The finite relative open cover of compact K has a Lebesgue number $\delta > 0$ for the parabolic metric. Let H be the maximum of the local Hölder constants and let M bound the glued function on K ; both are finite. Pairs in K at distance less than δ lie in one patch and satisfy its Hölder estimate. For pairs at distance at least δ , the difference is at most $2M$, which is at most $2M\delta^{-\gamma_0}d_{\text{par}}(z, z')^{\gamma_0}$. This proves the claimed Hölder bound on all of K .

Finally, suppose the normalized integral in (8.3) were strictly less than $\varepsilon_0/8$ for some such R . As $r' \uparrow R$ through smaller radii, the open-top cylinders increase to the cylinder of radius R . Monotone convergence and the strict margin give an $r' < R$ with normalized integral still less than $\varepsilon_0/8$. Since $r' < R$, its closed spatial ball and its half-open time interval lie inside $B_R(x_0) \times (t_0 - R^2, t_0)$, hence inside $\widehat{D}$ as required in Definition 8.3. This makes z_0 good, a contradiction. This proves (8.3). $\square$

Remark 8.5 (Top-face regularity). At the top face $t = 0$, the regularity conclusion used here is a continuous Hölder extension from the past, not CKN’s open-neighborhood `IsRegularPoint` predicate, since $t = 0$ is outside D . Theorem A alone gives that extension only after the shifted-center gluing in Lemma 8.2. This is the one-sided conclusion needed for Hölder continuity on the closure in Theorem 3.12; it does not assert that the boundary point is regular in CKN’s sense.

Lemma 8.6 (Projection of the singular set to time). *Let $I \subset \mathbb{R}$ be an open interval, let $S \subset \mathbb{R}^3 \times I$, and let $\pi_t(x, t) = t$. If the one-dimensional parabolic Hausdorff measure of S is zero, then $\mathcal{L}^1(\pi_t S) = 0$. In particular, if S is the singular set of a CKN-suitable solution on $\mathbb{R}^3 \times I$ with $q = 3$ and zero force, then for almost every $t \in I$ and every $\rho > 0$ there is a $\delta > 0$ such that*

$$\overline{B}_\rho(0) \times (t - \delta, t + \delta) \quad \text{consists entirely of regular points.}$$

Proof. The CKN parabolic Hausdorff measure uses the diameter gauge for d_{par} . If a set in a parabolic cover has diameter d , its time projection has Euclidean diameter at most d^2 , because $|t - s| \leq d_{\text{par}}((x, t), (y, s))^2$. Thus a parabolic cover of S with diameters d_i induces a time cover whose one-half-dimensional diameter cost is at most $\sum_i d_i$. Taking covers with arbitrarily small total cost shows that the Euclidean Hausdorff measure of dimension 1/2 of $\pi_t S$ is zero. To see this implies Lebesgue nullity, enclose each time cover set in an interval of length ℓ_i ; for covers with $\ell_i \leq \eta$, the total interval length is bounded by $\sqrt{\eta} \sum_i \sqrt{\ell_i}$. The latter sum can be made arbitrarily small, proving $\mathcal{L}^1(\pi_t S) = 0$.

For a CKN-suitable solution in the stated class, the CKN version of Theorem C gives zero one-dimensional parabolic Hausdorff measure of its singular set. Also, its regular-point locus is open: if a point has an open neighborhood with a Hölder representative, that same neighborhood and representative witness regularity for each of its points. Hence S is relatively closed in $\mathbb{R}^3 \times I$. Fix $t \in I \setminus \pi_t S$ and $\rho > 0$. The compact slice $\overline{B}_\rho(0) \times \{t\}$ is disjoint from S , and so is covered by finitely many product neighborhoods contained in the regular-point locus. The minimum of their time radii is positive; decrease it if necessary so that $(t - \delta, t + \delta) \subset I$. These neighborhoods cover $\overline{B}_\rho(0) \times (t - \delta, t + \delta)$, as required. $\square$

Remark 8.7. Lemma 8.6 supplies neighborhoods consisting of regular points. By itself it does not supply the higher Sobolev bounds, vorticity differential inequality, or infinite-order vanishing hypotheses of ESS Theorem 4.1; those inputs must be established before that theorem is applied (ESS 2003, printed p. 230, (4.1)–(4.3)).

Lemma 8.8 (Shifted-cylinder regularity). *Let $\Omega \subset \mathbb{R}^3$ be open and let $I \subset \mathbb{R}$ be an open interval. Set $D = \Omega \times I$, and let $z_0 = (x_0, t_0) \in D$ and $R > 0$. Set $z_1 = (x_0, t_0 + R^2/8)$ and suppose $\overline{\mathcal{Q}_{R/2}(z_1)} \subset D$. If u has a representative Hölder on $\overline{\mathcal{Q}_{R/2}(z_1)}$, equal to u almost everywhere on its interior, then z_0 is a regular point in the CKN sense.*

Proof. The closed half-cylinder is

$$\overline{B}_{R/2}(x_0) \times [t_0 - R^2/8, t_0 + R^2/8].$$

Thus z_0 is strictly inside it in both space and time. Choose $0 < \delta < R/\sqrt{8}$. The open parabolic ball $N = \{z : d_{\text{par}}(z, z_0) < \delta\}$ lies inside the open spatial ball and strictly between the two time faces of the half-cylinder, hence $N \subset D$. Restrict the given representative to N . It remains Hölder there and agrees almost everywhere with u there, which is exactly the CKN open-neighborhood definition of a regular point.

In the global application $D = \mathbb{R}^3 \times (0, T)$ and $0 < t_0 < T$. Choose R so that $t_0 - 7R^2/8 > 0$ and $t_0 + R^2/8 < T$. The full cylinder centered at z_1 has time interval $(t_0 - 7R^2/8, t_0 + R^2/8]$, so its closure is in D ; in particular, the half-cylinder hypothesis above holds. This verifies the scale and time-domain conditions used with the shifted center. $\square$

8.1 Vorticity regularity

Vorticity removes the pressure from the momentum equation. When the velocity is bounded, its equation is a heat equation with a bounded zeroth-order multiplier in the divergence-form flux. Local heat estimates and div–curl recovery then give the spatial derivatives needed for unique continuation.

For $z_0 = (x_0, t_0)$, we use $Q_r^-(z_0)$ and its closure as fixed in Section 2. Weak derivatives of the velocity are always carried by the separate field Du of Definition 3.1.

Lemma 8.9 (Weak vorticity equation). *Let $U = \Omega \times I$ with $\Omega \subset \mathbb{R}^3$ open and $I \subset \mathbb{R}$ an open interval. If $(u, Du, p, 0)$ is a suitable weak solution on U , define*

$$\omega_i = \sum_{k,\ell=1}^3 \epsilon_{ik\ell} Du_{\ell k}, \quad C_{ji} = u_j \omega_i - \omega_j u_i.$$

Then, for every $\psi \in C_c^\infty(U; \mathbb{R}^3)$,

$$\iint_U (-\omega_i \partial_t \psi_i - \omega_i \Delta \psi_i) = \iint_U C_{ji} \partial_j \psi_i. \quad (8.4)$$

Here and below repeated indices are summed from 1 to 3. Equivalently, in distributions on U ,

$$\partial_t \omega_i - \Delta \omega_i = - \sum_{j=1}^3 \partial_j (u_j \omega_i - \omega_j u_i), \quad i = 1, 2, 3.$$

The tensor convention in this formula is $C_{ji} = u_j \omega_i - \omega_j u_i$, with divergence $(\operatorname{div} C)_i = \sum_j \partial_j C_{ji}$.

Proof. Condition (S1) of Definition 3.1 gives $u \in L_{\text{loc}}^\infty(I; L_{\text{loc}}^2) \cap L_{\text{loc}}^2(I; H_{\text{loc}}^1)$ and $\omega \in L_{\text{loc}}^2$. The three-dimensional interpolation inequality gives $u \in L_{\text{loc}}^{10/3}(U)$, so $u\omega \in L_{\text{loc}}^{5/4}(U)$ and the right side of (8.4) is integrable.

Use $\varphi = \operatorname{curl} \psi$ in (S3) of Definition 3.1. This is an admissible compactly supported smooth vector field and $\operatorname{div} \varphi = 0$, so the pressure term vanishes. The distributional curl identity $\langle \operatorname{curl} u, \psi \rangle = \langle u, \operatorname{curl} \psi \rangle$ changes the time term into $-\iint \omega \cdot \partial_t \psi$. For the viscous term, the definition of Du and spatial integration by parts give

$$\iint_U Du_{ij} \partial_j (\operatorname{curl} \psi)_i = - \iint_U u_i \Delta (\operatorname{curl} \psi)_i = - \iint_U \omega_i \Delta \psi_i.$$

These integrations have no boundary terms because ψ is compactly supported in U .

It remains to identify the nonlinear term. On every compact subcylinder, spatial mollification of u preserves its zero divergence on a neighborhood of the test support and converges to u in $L_t^2 H_x^1$ and in $L_{t,x}^2$. The corresponding curls converge to ω in $L_{t,x}^2$. Thus the products $u^\varepsilon \otimes u^\varepsilon$ converge in $L_{t,x}^1$ and $u^\varepsilon \omega^\varepsilon$ converge in $L_t^1 L_x^{3/2}$ on the test support. For a smooth divergence-free spatial field v , the product rule and $\operatorname{div}(\operatorname{curl} v) = 0$ give

$$(\operatorname{curl}((v \cdot \nabla)v))_i = \partial_j(v_j(\operatorname{curl} v)_i - (\operatorname{curl} v)_j v_i).$$

Passing to the limit in the distributional pairing proves the same identity for u . Consequently the nonlinear term in (S3), after moving it to the other side, is $\iint C_{ji} \partial_j \psi_i$. This proves (8.4). $\square$

Lemma 8.10 (Localized vorticity energy estimate). *Let $J = (a, \tau)$ be a bounded interval. Suppose $z \in L^2(J \times \mathbb{R}^3; \mathbb{R}^3)$ solves*

$$\partial_t z_i - \Delta z_i = -\partial_j(v_j z_i - z_j v_i) + \partial_j F_{ji} + g_i$$

in distributions, where $v \in L^\infty(J \times \mathbb{R}^3; \mathbb{R}^3)$, $F \in L^2(J \times \mathbb{R}^3; \mathbb{R}^{3 \times 3})$, and $g \in L^2(J; H^{-1}(\mathbb{R}^3; \mathbb{R}^3))$. If the initial trace is $z(a) = z_0 \in L^2(\mathbb{R}^3)$, then

$$z \in C([a, \tau]; L^2(\mathbb{R}^3)) \cap L^2(J; H^1(\mathbb{R}^3))$$

and

$$\begin{aligned} \sup_{t \in [a, \tau]} \|z(t)\|_2^2 + \int_a^\tau \|\nabla z(t)\|_2^2 dt &\leq C \exp(C(1 + \|v\|_\infty^2)(\tau - a)) \\ &\cdot \left(\|z_0\|_2^2 + \|F\|_{L^2(J \times \mathbb{R}^3)}^2 + \|g\|_{L^2(J; H^{-1})}^2 \right). \end{aligned} \quad (8.5)$$

where C is an absolute constant.

Proof. First take smooth data and a smooth solution. Pair the equation with z and integrate over $\mathbb{R}^3$. Integration by parts has no boundary terms, initially by spatial cutoff and then by passage to the limit in H^1 . Since $|v_j z_i - z_j v_i| \leq 2|v||z|$, Young's inequality gives

$$\frac{d}{dt} \|z(t)\|_2^2 + \|\nabla z(t)\|_2^2 \leq C \|v\|_\infty^2 \|z(t)\|_2^2 + C \|F(t)\|_2^2 + C \|g(t)\|_{H^{-1}}^2.$$

The g pairing is estimated in H^{-1}, H^1 duality; the L^2 part of the H^1 norm contributes $C \|g(t)\|_{H^{-1}}^2 + C \|z(t)\|_2^2$, which is included in the Grönwall term. Grönwall's inequality proves (8.5) for smooth solutions.

The trace convention for a distributional solution is the following. Since $z \in L^2(J; L^2)$ and

$$\partial_t z = \Delta z - \operatorname{div}(vz - zv) + \operatorname{div} F + g \in L^2(J; H^{-2}(\mathbb{R}^3)),$$

$z \in W^{1,2}(J; H^{-2})$, so it has a representative in $C([a, \tau]; H^{-2})$. The assertion $z(a) = z_0$ means that this representative has value z_0 at a , under the embedding $L^2 \subset H^{-2}$.

We now prove the assertion for the given distributional solution z directly, without constructing a second solution. By the Riesz representation in H^1 , at almost every time $w = (1 - \Delta)^{-1} g(t) \in H^1(\mathbb{R}^3; \mathbb{R}^3)$ gives $g = g_0 + \operatorname{div} G$ in the sense $g_i = g_{0,i} + \partial_j G_{ji}$, with $g_0 = w$ and $G_{ji} = -\partial_j w_i$. Here $g_0 \in L^2(J \times \mathbb{R}^3; \mathbb{R}^3)$, $G \in L^2(J \times \mathbb{R}^3; \mathbb{R}^{3 \times 3})$ and $\|g_0\|_{L^2(J \times \mathbb{R}^3)}^2 + \|G\|_{L^2(J \times \mathbb{R}^3)}^2 = \|g\|_{L^2(J; H^{-1})}^2$;

the argument below uses only such a representation. Since $z \in L^2(J \times \mathbb{R}^3)$ and $|v_j z_i - z_j v_i| \leq \|v\|_\infty(|z_i| + |z_j|)$, the flux

$$H_{ji} = -(v_j z_i - z_j v_i) + F_{ji} + G_{ji}$$

belongs to $L^2(J \times \mathbb{R}^3)$, with $\|H\|_2^2 \leq 36\|v\|_\infty^2 \|z\|_2^2 + 3\|F\|_2^2 + 3\|G\|_2^2$ on every time slab, and each component solves the linear equation

$$\partial_t z_i - \Delta z_i = \partial_j H_{ji} + g_{0,i}$$

in distributions on $J \times \mathbb{R}^3$, with all data in $L^2(J \times \mathbb{R}^3)$.

Let $k \in C_c^\infty(\mathbb{R}^3)$, and write $k*$ for convolution in the space variable. Fix $x \in \mathbb{R}^3$ and test the equation with $k(x - \cdot)\theta$, $\theta \in C_c^\infty(J)$. This shows that $t \mapsto (k * z_i(t))(x)$ has the distributional derivative

$$S_{k,i}(x, t) = ((\Delta k) * z_i(t) + (\partial_j k) * H_{ji}(t) + k * g_{0,i}(t))(x)$$

on J , which belongs to $L^2(J)$. By the trace convention, applied to the smooth compactly supported test $k(x - \cdot)$, the continuous representative of this function has the value $(k * z_{0,i})(x)$ at a . Hence, for almost every $t \in J$,

$$(k * z_i(t))(x) = Z_{k,i}(x, t) := (k * z_{0,i})(x) + \int_a^t S_{k,i}(x, s) ds,$$

and $t \mapsto Z_{k,i}(x, t)$ is absolutely continuous on $[a, \tau]$. Differentiate $Z_{k,i}(x, t)^2$ in t , integrate over (a, t) and over $x \in \mathbb{R}^3$ (Fubini), and integrate by parts in x . The kernels are smooth with compact support and all fields are in L^2 , so no boundary terms occur, and $(\partial_j k) * H_{ji} = \partial_j(k * H_{ji})$ and $(\Delta k) * z_i = \Delta Z_{k,i}$ for almost every t . Together with $2|\langle \nabla Z_k, k * H \rangle| \leq \|\nabla Z_k\|_2^2 + \|k * H\|_2^2$ and $2|\langle Z_k, k * g_0 \rangle| \leq \|Z_k\|_2^2 + \|k * g_0\|_2^2$, this gives for every $t \in [a, \tau]$

$$\|Z_k(t)\|_2^2 + \int_a^t \|\nabla Z_k\|_2^2 \leq \|k * z_0\|_2^2 + \int_a^t (\|k * H\|_2^2 + \|k * z\|_2^2 + \|k * g_0\|_2^2).$$

Let ρ_ε be a standard spatial mollifier and apply this inequality with $k = \rho_\varepsilon - \rho_\delta$. Since $\|\rho_\varepsilon * h - h\|_{L^2} \rightarrow 0$ for every h in $L^2(\mathbb{R}^3)$ and in $L^2(J \times \mathbb{R}^3)$, the right side tends to zero as $\varepsilon, \delta \rightarrow 0$. Hence the curves Z_{ρ_ε} are Cauchy in $C([a, \tau]; L^2(\mathbb{R}^3))$, and $\nabla Z_{\rho_\varepsilon}$ is Cauchy in $L^2(J \times \mathbb{R}^3)$. The limit curve is continuous in L^2 , equals $z(t)$ for almost every t , and has the value z_0 at $t = a$; the limit of the gradients is the weak gradient of z . This proves $z \in C([a, \tau]; L^2(\mathbb{R}^3)) \cap L^2(J; H^1(\mathbb{R}^3))$. Passing to the limit in the inequality with $k = \rho_\varepsilon$ gives it for z without the kernel. With the bound on $\|H\|_2^2$, the quantities $e(t) = \|z(t)\|_2^2$ and $d(t) = \int_a^t \|\nabla z\|_2^2$ satisfy

$$e(t) + d(t) \leq \|z_0\|_2^2 + 3\|F\|_2^2 + 3\|G\|_2^2 + \|g_0\|_2^2 + (36\|v\|_\infty^2 + 1) \int_a^t e(s) ds,$$

where the norms of F , G and g_0 are taken over $J \times \mathbb{R}^3$. Grönwall's inequality then gives (8.5) with $C = 36$. $\square$

Lemma 8.11 (Local heat gain). *Let $B_r \Subset B_R \subset \mathbb{R}^3$ be concentric balls, $a < s < b$, and let $m = 1$ or $m = 2$. If $z \in L^2((a, b); H^m(B_R))$ and*

$$\partial_t z - \Delta z = G \quad \text{in } B_R \times (a, b), \quad G \in L^2((a, b); H^{m-1}(B_R)),$$

then, for each $a < s < b$,

$$z \in C([s, b]; H^m(B_r)) \cap L^2((s, b); H^{m+1}(B_r))$$

and

$$\|z\|_{C([s,b];H^m(B_r))} + \|z\|_{L^2((s,b);H^{m+1}(B_r))} \leq C \left(\|z\|_{L^2((a,b);H^m(B_R))} + \|G\|_{L^2((a,b);H^{m-1}(B_R))} \right), \quad (8.6)$$

where C depends only on m , the radii and their gap, and $s - a, b - a$.

Proof. Choose a smooth spatial cutoff supported in B_R and equal to 1 on B_r , and a smooth time cutoff that vanishes near a and equals 1 on $[s, b]$. Multiplying by their product η gives, after extension by zero in space,

$$\partial_t(\eta z) - \Delta(\eta z) = \eta G + (\partial_t \eta - \Delta \eta)z - 2\nabla \eta \cdot \nabla z.$$

The right side belongs to $L^2((a, b); H^{m-1}(\mathbb{R}^3))$: the last term uses $z \in L^2 H^m$, and all cutoff derivatives are bounded. The field ηz vanishes for t near a . For a smooth zero-initial solution w of $\partial_t w - \Delta w = H$ with compact support in space, the energies of the spatial derivatives $\partial^\beta w$, $|\beta| \leq m$, give

$$\sup_{t \in [a, b]} \|w(t)\|_{H^m}^2 + \int_a^b \|w(t)\|_{H^{m+1}}^2 dt \leq C(m, b - a) \int_a^b \|H(t)\|_{H^{m-1}}^2 dt.$$

At integer order we work with the derivative energies $\sum_{|\beta| \leq m} \|\partial^\beta w(t)\|_2^2$ in place of the Bessel norm: the Fourier weights $(1 + |\xi|^2)^m$ and $\sum_{|\beta| \leq m} \xi^{2\beta}$ are comparable, so these energies define a norm equivalent to $\|w(t)\|_{H^m}^2$ with constants depending only on m . Each $\partial^\beta w$ satisfies its own energy identity, obtained by pairing its equation with $\partial^\beta w$. For $\beta = \beta' k$ we write $\partial^\beta H = \partial_k \partial^{\beta'} H$ and move ∂_k onto $\partial^\beta w$, so that only $m - 1$ derivatives of H enter; Young's inequality absorbs half of $\|\nabla \partial^\beta w\|_2^2$. The remaining term $\|\partial^\beta w\|_2^2$ is handled by Grönwall's inequality, which produces a factor depending on $b - a$ (of size e^{b-a}); for long intervals this dependence cannot be dropped, so the constant is $C(m, b - a)$.

The same argument applies to $w = \eta z$, with the energy estimate of Lemma 8.10 in place of the smooth computation. Write $H = \eta G + (\partial_t \eta - \Delta \eta)z - 2\nabla \eta \cdot \nabla z$. Testing with derivatives of test functions shows that each $\partial^\beta w$, $|\beta| \leq m$, solves $\partial_t \partial^\beta w - \Delta \partial^\beta w = \partial^\beta H$ in distributions on $\mathbb{R}^3 \times (a, b)$. For $|\beta| \leq m - 1$ this source belongs to L^2 ; for $\beta = \beta' k$ it is the divergence $\partial_k(\partial^{\beta'} H)$ of the L^2 field $\partial^{\beta'} H$. Since $\partial^\beta w$ vanishes for t near a , its pairings with spatial test functions have absolutely continuous versions with value 0 at a , so its initial trace is zero. Lemma 8.10 with $v = 0$ therefore gives $\partial^\beta w \in C([a, b]; L^2(\mathbb{R}^3)) \cap L^2((a, b); H^1(\mathbb{R}^3))$, with

$$\sup_{t \in [a, b]} \|\partial^\beta w(t)\|_2^2 + \int_a^b \|\nabla \partial^\beta w\|_2^2 dt \leq C \int_a^b (\|H\|_{H^{m-1}}^2 + \|\partial^\beta w\|_2^2) dt,$$

where C depends on $b - a$. The weak-derivative relations between the continuous curves of $\partial^\beta w$ and $\partial^{\beta_j} w$ hold at almost every time, and by continuity at every time. Summing over $|\beta| \leq m$, using that $\eta = 1$ on $B_r \times [s, b]$ and bounding the right side by $\|z\|_{L^2((a,b);H^m(B_R))}^2 + \|G\|_{L^2((a,b);H^{m-1}(B_R))}^2$ gives $z \in C([s, b]; H^m(B_r)) \cap L^2((s, b); H^{m+1}(B_r))$ and (8.6). There is no terminal boundary term to discard: the estimate includes the value at b in its supremum.

The bootstrap variant at the end of the proof of Theorem 8.14 uses only the following first-order form of this gain, which needs no prior information on ∇w and gives bounds up to the top time. Let $\rho_\varepsilon \geq 0$ be a smooth space-time mollifier with integral one, supported in $B_\varepsilon \times (-\varepsilon, \varepsilon)$. For a field w on $B_R \times (a, b)$ define its backward mollification

$$w_\varepsilon(x, t) = \iint \rho_\varepsilon(y, \sigma) w(x - y, t - 5\varepsilon - \sigma) dy d\sigma.$$

It uses only values of w at times in $(t-6\varepsilon, t-4\varepsilon)$. Hence it is smooth on $B_{R-\varepsilon} \times (a+6\varepsilon, b]$, including the top time, and it commutes with weak derivatives. Suppose $w, F_1, F_2, F_3 \in L^2(B_R \times (a, b))$ and $\partial_t w - \Delta w = \operatorname{div} F$ in distributions. Then $\partial_t w_\varepsilon - \Delta w_\varepsilon = \operatorname{div} F_\varepsilon$ pointwise. Fix $r < R' < R$ and $a < a' < s < b$. Choose a cutoff η supported in $B_{R'}$ in space, vanishing near a' in time, and equal to 1 on $B_r \times [s, b]$. Pair the equation with $\eta^2 w_\varepsilon$ on $B_{R'} \times (a', t)$ and absorb the cross terms by Young's inequality. For small ε this gives

$$\sup_{s < t < b} \|w_\varepsilon(t)\|_{L^2(B_r)}^2 + \int_s^b \|\nabla w_\varepsilon(t)\|_{L^2(B_r)}^2 dt \leq C(\|w_\varepsilon\|_{L^2(B_{R'} \times (a', b))}^2 + \|F_\varepsilon\|_{L^2(B_{R'} \times (a', b))}^2),$$

where C depends only on the gaps. As $\varepsilon \rightarrow 0$, $w_\varepsilon \rightarrow w$ and $F_\varepsilon \rightarrow F$ in $L^2(B_{R'} \times (a', b))$. Thus ∇w_ε is bounded in $L^2(B_r \times (s, b))$. Every weak limit is the weak gradient of w there, so

$$\|\nabla w\|_{L^2(B_r \times (s, b))} \leq C(\|w\|_{L^2(B_R \times (a, b))} + \|F\|_{L^2(B_R \times (a, b))}).$$

The same estimate for $w_\varepsilon - w_\delta$ shows that $w_\varepsilon(t)$ is Cauchy in $L^2(B_r)$, uniformly in $t \in (s, b)$, as $\varepsilon, \delta \rightarrow 0$. $\square$

Lemma 8.12 (Local div–curl recovery). *Let $B_r \Subset B_R \subset \mathbb{R}^3$ be concentric balls, let $m \in \{0, 1, 2\}$, and suppose $v \in H^m(B_R; \mathbb{R}^3)$, $\operatorname{div} v = 0$, and $\operatorname{curl} v = \zeta \in H^m(B_R; \mathbb{R}^3)$. Then*

$$\|v\|_{H^{m+1}(B_r)} \leq C(\|\zeta\|_{H^m(B_R)} + \|v\|_{H^m(B_R)}), \quad (8.7)$$

where C depends only on m, r, R . The same estimate holds after taking an L^∞ or L^2 norm in time.

Proof. For a compactly supported smooth vector field W on $\mathbb{R}^3$, two integrations by parts give

$$\int_{\mathbb{R}^3} |\nabla W|^2 = \int_{\mathbb{R}^3} (\operatorname{div} W)^2 + \int_{\mathbb{R}^3} |\operatorname{curl} W|^2,$$

the integrated form of $|\xi|^2 |\widehat{W}(\xi)|^2 = |\xi \cdot \widehat{W}(\xi)|^2 + |\xi \times \widehat{W}(\xi)|^2$. Spatial derivatives commute with div and curl . Applying the identity to $W = \partial^\alpha V$ for every derivative ∂^α of order $|\alpha| \leq m$ and adding $\|V\|_{H^m}^2$ proves

$$\|V\|_{H^{m+1}(\mathbb{R}^3)} \leq C_m(\|\operatorname{div} V\|_{H^m} + \|\operatorname{curl} V\|_{H^m} + \|V\|_{H^m})$$

for compactly supported smooth V . Now let $V \in H^m(\mathbb{R}^3; \mathbb{R}^3)$ have compact support, and let $\operatorname{div} V$ and $\operatorname{curl} V$, taken in distributions, belong to H^m . For a standard mollifier ρ_ε the fields $V_\varepsilon = \rho_\varepsilon * V$ are smooth with compact support, $\operatorname{div} V_\varepsilon = \rho_\varepsilon * \operatorname{div} V$ and $\operatorname{curl} V_\varepsilon = \rho_\varepsilon * \operatorname{curl} V$. Since $\rho_\varepsilon * f \rightarrow f$ in H^m for $f \in H^m$, the smooth estimate applied to $V_\varepsilon - V_\delta$ shows that V_ε is Cauchy in $H^{m+1}(\mathbb{R}^3)$. The limits of the derivatives of order $m+1$ are the weak derivatives of V , so $V \in H^{m+1}(\mathbb{R}^3)$ and the estimate holds for V . Choose a cutoff χ supported in B_R and equal to 1 on B_r , with an intermediate concentric ball for its derivatives. For $V = \chi v$, extended by zero, testing the equations for v with $\chi\varphi$ gives in distributions on $\mathbb{R}^3$

$$\operatorname{div} V = \nabla \chi \cdot v, \quad \operatorname{curl} V = \chi \zeta + \nabla \chi \times v.$$

By the product rule these expressions and V are bounded in $H^m(\mathbb{R}^3)$ by $C(\|\zeta\|_{H^m(B_R)} + \|v\|_{H^m(B_R)})$. The estimate for compactly supported fields and $V = v$ on B_r give (8.7). Integrating the pointwise-in-time estimate, or taking its essential supremum, gives the time-dependent versions.

The bootstrap variant at the end of the proof of Theorem 8.14 uses the case $m = 0$ in time-integrated form. In components, the identity above reads

$$\int |\nabla V|^2 = \int (\operatorname{div} V)^2 + \frac{1}{2} \sum_{i,k=1}^3 \int (\partial_i V_k - \partial_k V_i)^2$$

for compactly supported smooth V . With the cutoff above, for smooth v and $r < R'$ it bounds $\|\nabla v\|_{L^2(B_r)}$ by the $L^2(B_{R'})$ norms of $\operatorname{div} v$, of the antisymmetric derivatives $\partial_i v_k - \partial_k v_i$, and of v . Now let $Y \in L^2(B_R \times (a, b); \mathbb{R}^3)$ satisfy $\operatorname{div} Y = 0$ and $\partial_i Y_k - \partial_k Y_i \in L^2(B_R \times (a, b))$ in distributions. Apply the smooth bound at each time to the backward mollifications Y_ε from the proof of Lemma 8.11. They are divergence free, and their antisymmetric derivatives are the mollified ones. Integrate over (s, b) with $a < s < b$. The gradients are then bounded in $L^2(B_r \times (s, b))$, and a weak limit gives

$$\|\nabla Y\|_{L^2(B_r \times (s, b))} \leq C \left(\sum_{i,k=1}^3 \|\partial_i Y_k - \partial_k Y_i\|_{L^2(B_R \times (a, b))} + \|Y\|_{L^2(B_R \times (a, b))} \right).$$

No time-continuous representative of Y is needed. $\square$

Lemma 8.13 (Products at the two bootstrap levels). *For compactly supported $v \in H^2(\mathbb{R}^3; \mathbb{R}^3) \cap L^\infty$, and for $v \in H^3(\mathbb{R}^3; \mathbb{R}^3)$, respectively,*

$$\|v \otimes v\|_{H^2} \leq C \|v\|_\infty \|v\|_{H^2}, \quad (8.8)$$

$$\|v \otimes v\|_{H^3} \leq C \|v\|_{H^2} \|v\|_{H^3}. \quad (8.9)$$

The corresponding local estimates hold after inserting a cutoff; their constants also depend on the distance between the two balls.

Proof. The derivatives of order zero and one in (8.8) are bounded by the product rule and $\|v\|_\infty$. For the second derivatives, the only additional term is bounded by $C \|\nabla v\|_4^2$. Integration by parts for compactly supported smooth v gives

$$\begin{aligned} \int_{\mathbb{R}^3} |\nabla v|^4 &= - \sum_{a,j=1}^3 \int_{\mathbb{R}^3} v_a \partial_j (|\nabla v|^2 \partial_j v_a) \\ &\leq C \|v\|_\infty \int_{\mathbb{R}^3} |\nabla v|^2 |\nabla^2 v| \\ &\leq C \|v\|_\infty \|\nabla v\|_4^2 \|\nabla^2 v\|_2. \end{aligned}$$

If $\|\nabla v\|_4 \neq 0$, division yields $\|\nabla v\|_4^2 \leq C \|v\|_\infty \|\nabla^2 v\|_2$; the zero case is immediate. Approximation proves the first estimate. For (8.9), the same argument applied to two compactly supported smooth scalar functions f, g gives the bilinear form $\|fg\|_{H^2} \leq C(\|f\|_\infty \|g\|_{H^2} + \|g\|_\infty \|f\|_{H^2})$: the mixed term $\nabla f \otimes \nabla g$ is bounded by $\|\nabla f\|_4 \|\nabla g\|_4$ and the inequality above. With $H^2 \hookrightarrow L^\infty$ this shows that $H^2(\mathbb{R}^3)$ is an algebra, $\|fg\|_{H^2} \leq C \|f\|_{H^2} \|g\|_{H^2}$. The derivatives of order at most three of $v_i v_j$ are those of order at most two of $v_i v_j$ and of $\partial_k(v_i v_j) = \partial_k v_i v_j + v_i \partial_k v_j$. Hence

$$\|v_i v_j\|_{H^3} \leq C \left(\|v\|_{H^2}^2 + \sum_{k=1}^3 \|\partial_k v\|_{H^2} \|v\|_{H^2} \right) \leq C \|v\|_{H^2} \|v\|_{H^3},$$

since $\|\partial_k v\|_{H^2} \leq \|v\|_{H^3}$ and $\|v\|_{H^2} \leq \|v\|_{H^3}$. Smooth approximation and cutoffs give the stated local forms.

The bootstrap variant at the end of the proof of Theorem 8.14 uses these bounds only in the following local, time-integrated forms. For smooth f with $|f| \leq M$ on B_R and $r < R$, the integration by parts above, with weight η^4 for a cutoff η equal to one on B_r , and Young's inequality give

$$\int_{B_r} |\nabla f|^4 \leq CM^2 \int_{B_R} (|\nabla f|^2 + |\nabla^2 f|^2).$$

Apply this at each time to the components of the backward mollifications u_ε of u ; these are also bounded by M . Integrate in time and pass to the limit by Fatou's lemma. If $|u| \leq M$ and $Du, D^2u \in L^2$ on a box, then $Du \in L^4$ on a smaller box, and $\|Du\|_4^4 \leq CM^2(\|Du\|_2^2 + \|D^2u\|_2^2)$. At the next level, fix a time. The embedding $H^2 \hookrightarrow L^\infty$ for $\nabla u_\varepsilon(t)$ and the smooth case of Lemma 8.12 bound $\sup_{B_r} |\nabla u_\varepsilon(t)|^2$ by the squared L^2 norms of $u_\varepsilon(t)$, $\omega_\varepsilon(t)$, $\nabla \omega_\varepsilon(t)$ and $\nabla^2 \omega_\varepsilon(t)$ on a larger ball. Also, by the same div–curl estimate and the uniform-in-time bounds of the first-order heat gain in the proof of Lemma 8.11 for ω and $\nabla \omega$, the sum $\|D^2 u_\varepsilon(t)\|_2 + \|\nabla \omega_\varepsilon(t)\|_2$ on that ball is bounded uniformly in t . Integrating in time and applying Fatou's lemma give

$$\iint |Du|^2 (|D^2 u|^2 + |\nabla \omega|^2) < \infty$$

on a smaller box, with a bound by the data. These are the products in the first and second spatial derivatives of the flux $u_j \omega_i - \omega_j u_i$. $\square$

Theorem 8.14 (Vorticity regularity near a bounded-velocity point). *Let $z_0 = (x_0, t_0)$, and let $U = \Omega \times I$ be an open neighborhood of $\overline{Q_1^-(z_0)}$. Suppose $q > 5/2$, $M \geq 0$, $(u, Du, p, 0)$ is suitable on U , and $|u| \leq M$ almost everywhere on $Q_1^-(z_0)$. Set*

$$E = \left(\iint_{Q_1^-(z_0)} |Du|^2 dx dt \right)^{1/2}.$$

Then there is a constant $C = C(M, E)$ such that, on $Q_{1/2}^-(z_0)$,

$$\operatorname{ess\,sup}_{Q_{1/2}^-(z_0)} (|u| + |Du|) \leq C, \quad (8.10)$$

$$\|\omega\|_{W_2^{2,1}(Q_{1/2}^-(z_0))} + \sup_{t \in [t_0 - 1/4, t_0]} \|\tilde{\omega}(\cdot, t)\|_{H^2(B_{1/2}(x_0))} + \|\tilde{\omega}\|_{C(\overline{Q_{1/2}^-(z_0)})} \leq C, \quad (8.11)$$

where $\omega = \operatorname{curl} u$ is the weak vorticity and $\tilde{\omega}$ is a representative continuous on the closed past half-cylinder. In addition,

$$|\partial_t \omega - \Delta \omega| \leq C(|\omega| + |\nabla \omega|) \quad \text{almost everywhere on } Q_{1/2}^-(z_0). \quad (8.12)$$

The bounds for u and Du are essential bounds on the open past half-cylinder, uniform as $t \uparrow t_0$; no time-continuous representative of u or Du is asserted.

Proof. The suitable-solution energy clause gives $E < \infty$ and $u \in L_t^2 H_x^1$ on compact subcylinders. The weak vorticity equation is (8.4). We use nested radii

$$1 > \frac{7}{8} > \frac{3}{4} > \frac{5}{8} > \frac{9}{16} > \frac{17}{32} > \frac{33}{64} > \frac{65}{128} > \frac{1}{2}.$$

All cutoff constants below depend only on these fixed gaps. We denote by $C(M, E)$ a finite bound depending only on the displayed parameters and the fixed radii; its value may increase from line to line.

Initial energy level. On Q_1^- , $\|\omega\|_{L^2} \leq CE$. Let $J = (t_0 - 1, t_0)$, and choose a spatial cutoff $\varphi \in C_c^\infty(B_1(x_0))$ equal to one on $\overline{B_{7/8}(x_0)}$ and a time cutoff θ that vanishes near $t_0 - 1$ and equals one on $[t_0 - (7/8)^2, t_0]$. Set $\chi(x, t) = \varphi(x)\theta(t)$. Define $v = u$ on $B_1(x_0) \times J$ and extend it by zero to $J \times \mathbb{R}^3$; extend $z = \chi\omega$ by zero outside $B_1(x_0)$. Then $\|v\|_\infty \leq M$ and $v_j z_i - z_j v_i = \chi C_{ji}$. Multiplying the weak equation by χ gives

$$\partial_t(\chi\omega_i) - \Delta(\chi\omega_i) = -\partial_j(\chi C_{ji} + 2\omega_i \partial_j \chi) + \partial_j \chi C_{ji} + (\partial_t \chi + \Delta \chi)\omega_i.$$

Every term on the right is in $L_t^2 H_x^{-1}$; the two non-divergence terms are in $L_{t,x}^2$, and the divergence flux is in $L_{t,x}^2$ because $|C| \leq 2M|\omega|$. The cutoff field has zero initial trace. Lemma 8.10, with the zero-extended multiplier v , therefore gives

$$\omega \in C([t_0 - (7/8)^2, t_0]; L^2(B_{7/8})) \cap L^2((t_0 - (7/8)^2, t_0); H^1(B_{7/8})), \quad \|\omega\|_{L^\infty L^2 \cap L^2 H^1} \leq C(M, E). \quad (8.13)$$

No spatial boundary term occurs because the cutoff is compactly supported in B_1 ; the energy estimate includes the terminal time t_0 .

Velocity through order two. For almost every time, $\operatorname{div} u = 0$ by (S2) and $\operatorname{curl} u = \omega$ by definition. Lemma 8.12 with $m = 0$, together with $\|u(t)\|_{L^2(B_{7/8})} \leq CM$, gives

$$\begin{aligned} u &\in L^\infty((t_0 - (3/4)^2, t_0); H^1(B_{3/4})) \\ &\quad \cap L^2((t_0 - (3/4)^2, t_0); H^2(B_{3/4})), \\ \|u\|_{L^\infty H^1 \cap L^2 H^2} &\leq C(M, E). \end{aligned}$$

Choose a spatial cutoff supported in $B_{3/4}$ and equal to one on $B_{5/8}$. Applying (8.8) to the cutoff velocity gives $u \otimes u \in L_t^2 H_x^2(B_{5/8})$, with norm at most $C(M, E)$. Since $\operatorname{div} u = 0$, the weak vorticity equation is also

$$\partial_t \omega_i - \Delta \omega_i = -\epsilon_{ik\ell} \partial_k \partial_j (u_\ell u_j).$$

This identity follows by taking the curl of the weak momentum equation; it is valid distributionally because $u \otimes u \in L_{\text{loc}}^1$ and its localized derivatives just obtained are in $L_t^2 H_x^2(B_{5/8})$. Its right side belongs to $L_t^2 L_x^2(B_{5/8})$. Apply Lemma 8.11 with $m = 1$, outer ball $B_{5/8}$, inner ball $B_{9/16}$, initial time $t_0 - (3/4)^2$, and terminal time t_0 . The prior $L_t^2 H_x^1$ bound follows from (8.13). We obtain

$$\begin{aligned} \omega &\in C([t_0 - (9/16)^2, t_0]; H^1(B_{9/16})) \\ &\quad \cap L^2((t_0 - (9/16)^2, t_0); H^2(B_{9/16})), \\ \|\omega\|_{CH^1 \cap L^2 H^2} &\leq C(M, E). \end{aligned}$$

A first application of Lemma 8.12 with $m = 1$ from $B_{9/16}$ to $B_{17/32}$ uses the $L_t^\infty H_x^1$ bound on ω and gives $u \in L_t^\infty H_x^2(B_{17/32})$. A separate application with $m = 2$ on the same pair of balls uses $\omega \in L_t^2 H_x^2(B_{9/16})$ from the heat gain and $u \in L_t^2 H_x^2(B_{9/16})$ from the earlier $m = 0$ step; it gives $u \in L_t^2 H_x^3(B_{17/32})$. Combining these outputs yields

$$\begin{aligned} u &\in L^\infty((t_0 - (9/16)^2, t_0); H^2(B_{17/32})) \\ &\quad \cap L^2((t_0 - (9/16)^2, t_0); H^3(B_{17/32})), \\ \|u\|_{L^\infty H^2 \cap L^2 H^3} &\leq C(M, E). \end{aligned}$$

One more vorticity level. A cutoff of u supported in $B_{17/32}$ and equal to one on $B_{33/64}$ is bounded in $L_t^\infty H_x^2$ and $L_t^2 H_x^3$. Estimate (8.9) therefore gives $u \otimes u \in L_t^2 H_x^3(B_{33/64})$, with norm bounded by $C(M, E)$. The right side of the last vorticity equation is now in $L_t^2 H_x^1(B_{33/64})$. The prior estimate gives $\omega \in L_t^2 H_x^2$ there, so Lemma 8.11 with $m = 2$, outer ball $B_{33/64}$, inner ball $B_{65/128}$, initial time $t_0 - (17/32)^2$ and terminal time t_0 yields

$$\begin{aligned} \omega &\in C([t_0 - (65/128)^2, t_0]; H^2(B_{65/128})) \\ &\cap L^2((t_0 - (65/128)^2, t_0); H^3(B_{65/128})), \\ \|\omega\|_{CH^2 \cap L^2 H^3} &\leq C(M, E). \end{aligned}$$

Applying Lemma 8.12 with $m = 2$ from $B_{65/128}$ to $B_{1/2}$ gives

$$u \in L^\infty((t_0 - 1/4, t_0); H^3(B_{1/2})), \quad \|u\|_{L^\infty H^3} \leq C(M, E).$$

The Sobolev embedding $H^3(B_{1/2}) \hookrightarrow W^{1,\infty}(B_{1/2})$ proves the gradient bound in (8.10). The same level gives $\|\tilde{\omega}(t)\|_{H^2(B_{1/2})} \leq C(M, E)$ for every t in the closed time interval. The continuous embedding $H^2(B_{65/128}) \hookrightarrow C(\overline{B_{1/2}})$ and continuity in time with values in H^2 supply the representative $\tilde{\omega} \in C(\overline{Q_{1/2}^-})$.

Finally, on $Q_{1/2}^-$ the vorticity equation gives

$$\partial_t \omega = \Delta \omega - \partial_j (u_j \omega_i - \omega_j u_i).$$

The level-two bounds put $\omega, \nabla \omega, \nabla^2 \omega$ in L^2 and the right side in L^2 . Indeed, $\Delta \omega \in L^2$; expanding the flux derivative is legitimate since u and Du are essentially bounded, $\operatorname{div} u = \operatorname{div} \omega = 0$, and it is bounded pointwise by $|u| |\nabla \omega| + |Du| |\omega|$. Thus $\omega \in W_2^{2,1}(Q_{1/2}^-)$ and

$$|\partial_t \omega - \Delta \omega| \leq |u| |\nabla \omega| + |Du| |\omega| \leq C(M, E)(|\omega| + |\nabla \omega|)$$

a.e. This proves (8.11)–(8.12).

A variant of the bootstrap. The same conclusions also follow from L^2 space-time bounds alone, on nested boxes $B_\rho(x_0) \times (t_0 - \sigma, t_0)$ with the same top time, without Lemma 8.10. All smoothing uses the backward mollifications from the proof of Lemma 8.11, which are smooth up to t_0 . Each box below is slightly smaller than the previous one, and every bound depends only on M, E and the fixed gaps. Write $C_{ji} = u_j \omega_i - \omega_j u_i$ as in Lemma 8.9. Then $|C| \leq 2M|\omega|$ and $|\omega| \leq \sqrt{2}|Du|$.

The vorticity solves $\partial_t \omega_i - \Delta \omega_i = -\partial_j C_{ji}$ with $\omega, C \in L^2$, so the first-order heat gain gives $\nabla \omega \in L^2$. For each j , the field $\partial_j u$ is divergence free, and its antisymmetric derivatives are given by $\partial_j \omega$. The integrated div–curl bound therefore gives $D^2 u \in L^2$, and the L^4 bound gives $Du \in L^4$. Differentiating the weak equation shows that $\partial_m \omega$ solves the heat equation with flux $\partial_m C$. Since $|\partial_m C| \leq 2|Du| |\omega| + 2M |\nabla \omega|$, this flux is in L^2 , and the first-order heat gain gives $\nabla^2 \omega \in L^2$. By the integrability of $|Du|^2(|D^2 u|^2 + |\nabla \omega|^2)$, the flux $\partial_k \partial_m C$ in the equation for $\partial_k \partial_m \omega$ is also in L^2 .

Each of $\omega, \nabla \omega$ and $\nabla^2 \omega$ therefore solves a heat equation with L^2 flux. Their backward mollifications are Cauchy in L^2 of a ball, uniformly in time. By $H^2 \hookrightarrow L^\infty$ on a slightly larger ball, the mollifications of ω are uniformly Cauchy on $\overline{Q_{1/2}^-(z_0)}$. The limit is the continuous representative $\tilde{\omega}$, and the same uniform bounds give $\sup_t \|\tilde{\omega}(t)\|_{H^2(B_{1/2})} + \|\tilde{\omega}\|_{C(\overline{Q_{1/2}^-})} \leq C(M, E)$. Next, fix a time. The embedding $H^2 \hookrightarrow L^\infty$ for $\nabla u_\varepsilon(t)$ and the smooth div–curl estimate at that time bound $|\nabla u_\varepsilon|$ on $Q_{1/2}^-$ by $C(M, E)$. These use $|u_\varepsilon| \leq M$ and the uniform-in-time bounds of the mollified $\omega, \nabla \omega$ and $\nabla^2 \omega$. Since $\nabla u_\varepsilon \rightarrow Du$ in L^2 , a subsequence converges almost everywhere, which

gives (8.10). Finally, $\partial_t \omega_i = \Delta \omega_i - \partial_j C_{ji}$ has right side in L^2 . Hence $\omega \in W_2^{2,1}(Q_{1/2}^-)$ with the stated bound, and the pointwise flux estimate above gives (8.12). This variant does not use time continuity of u or Du . $\square$

Remark 8.15 (Departure from [4]). ESS Lemma 2.2 and Remark 2.3 [4, Lemma 2.2, Remark 2.3, pp. 216, 220–221] refer the higher spatial orders to a smooth-solution estimate and linear Stokes regularity; that citation does not by itself justify the passage from a suitable weak solution to the regularity used here. We give the finite weak-to-regular bootstrap needed for this theorem, following the vorticity and div–curl route of Lemarié-Rieusset [6, Theorems 13.1–13.2, pp. 397–403]. The proof retains the dependence on the local gradient energy E needed to start the vorticity energy estimate, and establishes joint continuity only for ω , which is the trace used below.

Remark 8.16 (Dependence of the bounds). Every constant C in the theorem depends only on (M, E) and the fixed radii in the proof. To make the dependence on $E = \|Du\|_{L^2(Q_1^-)}$ uniform in a family, take a larger cylinder $\mathcal{Q} = B_R(x_0) \times (t_0 - T_-, t_0 + T_+) \Subset U$ containing $\overline{Q_1^-(z_0)}$ in its interior, and choose $\psi \in C_c^\infty(\mathcal{Q})$ with $\psi \geq 0$ and $\psi = 1$ on $\overline{Q_1^-(z_0)}$. Set $M_{\mathcal{Q}} = \text{ess sup}_{\mathcal{Q}} |u|$. With $f = 0$, the local energy inequality (S4) and Hölder’s inequality give

$$E^2 \leq \frac{1}{2} \left[M_{\mathcal{Q}}^2 (\|\partial_t \psi\|_\infty + \|\Delta \psi\|_\infty) |\mathcal{Q}| + \|\nabla \psi\|_\infty (M_{\mathcal{Q}}^3 |\mathcal{Q}| + 2M_{\mathcal{Q}} |\mathcal{Q}|^{1/3} \|p\|_{L^{3/2}(\mathcal{Q})}) \right].$$

Thus fixed larger-cylinder geometry and uniform bounds for $\sup_{\mathcal{Q}} |u|$ and $\|p\|_{L^{3/2}(\mathcal{Q})}$ give a uniform E , and hence a uniform theorem constant when M is also uniform. Finiteness of E for each suitable solution follows from (S1) in Definition 3.1. The argument proves continuity up to the top face for ω ; it does not assert joint continuity of u or Du in time.

9 Stability of suitable weak solutions and the blow-up limit

The hypotheses of Theorem 3.12 give energy bounds and an $L_t^\infty L_x^3$ bound, but they do not include the local energy inequality required by Definition 3.1. We first recover an energy identity and a representative continuous under spatial $L^{3/2}$ tests. We then fix a pressure decomposition on the original cylinder; its harmonic part has enough interior control to vanish under blow-up scaling.

9.1 Energy equality and weak continuity

Lemma 9.1 (Local energy equality). *Suppose that u , Du , and p satisfy the hypotheses of Theorem 3.12, and set*

$$M := \text{ess sup}_{t \in (-1, 0)} \|u(\cdot, t)\|_{L^3(B_1)} < \infty. \quad (9.1)$$

For every $r \in (0, 1)$, $u \in L^4(B_r \times (-1, 0))$ and, for every $\psi \in C_c^\infty(B_r \times (-1, 0))$,

$$2 \iint_{B_r \times (-1, 0)} |Du|^2 \psi \, dx \, dt = \iint_{B_r \times (-1, 0)} \left[|u|^2 (\partial_t \psi + \Delta \psi) + (|u|^2 + 2p) u \cdot \nabla \psi \right] dx \, dt. \quad (9.2)$$

Consequently $(u, Du, p, 0)$ is a suitable weak solution on $B_r \times (-1, 0)$, in the sense of Definition 3.1 with $q = 3$.

Proof. For almost every $t \in (-1, 0)$, each component of $u(\cdot, t)$ belongs to $H^1(B_r)$ with weak gradient given by the corresponding row of $Du(\cdot, t)$. The ball Sobolev–Poincaré inequality applied

after subtracting the component means, followed by Hölder's inequality for those means, gives the scale-explicit estimate

$$\|u(\cdot, t)\|_{L^6(B_r)} \leq C \left(\|Du(\cdot, t)\|_{L^2(B_r)} + r^{-1} \|u(\cdot, t)\|_{L^2(B_r)} \right), \quad (9.3)$$

where C is an absolute constant in dimension three for three components (CKN library). Interpolation between L^3 and L^6 gives

$$\int_{B_r} |u(x, t)|^4 dx \leq \|u(\cdot, t)\|_{L^3(B_r)}^2 \|u(\cdot, t)\|_{L^6(B_r)}^2 \leq CM^2 \left(\|Du(\cdot, t)\|_{L^2(B_r)}^2 + r^{-2} \|u(\cdot, t)\|_{L^2(B_r)}^2 \right). \quad (9.4)$$

Integrating in time and using the energy bound in Theorem 3.12 proves $u \in L^4(B_r \times (-1, 0))$. In particular $F := u \otimes u$ belongs to L^2 on this cylinder.

Fix a compact subset of $B_r \times (-1, 0)$ containing the support of ψ in its interior, and choose a nonnegative smooth space-time mollifier ρ of integral one, with $\rho_\epsilon(z) = \epsilon^{-4} \rho(z/\epsilon)$. For sufficiently small ϵ , convolution is taken only where its translated support remains in $B_1 \times (-1, 0)$. Write $u_\epsilon = u * \rho_\epsilon$, $F_\epsilon = F * \rho_\epsilon$, and $p_\epsilon = p * \rho_\epsilon$ there. Spatial divergence commutes with this convolution, so $\nabla \cdot u_\epsilon = 0$. The local L^s convergence of mollification for finite s follows from translation continuity in L^s : $\|g * \rho_\epsilon - g\|_{L^s} \leq \int \rho_\epsilon(h) \|g(\cdot - h) - g\|_{L^s} dh \rightarrow 0$ after restricting to a slightly larger compact subset.

Convolving the weak momentum identity (S3) gives

$$\partial_t u_\epsilon + \nabla \cdot F_\epsilon - \Delta u_\epsilon + \nabla p_\epsilon = 0. \quad (9.5)$$

Set $R_\epsilon = F_\epsilon - u_\epsilon \otimes u_\epsilon$. Testing (9.5), rewritten with $u_\epsilon \otimes u_\epsilon$ on the left and $-\nabla \cdot R_\epsilon$ on the right, against $2\psi u_\epsilon$ and integrating by parts yields

$$\begin{aligned} 2 \iint |Du_\epsilon|^2 \psi &= \iint [|u_\epsilon|^2 (\partial_t \psi + \Delta \psi) + (|u_\epsilon|^2 + 2p_\epsilon) u_\epsilon \cdot \nabla \psi] \\ &\quad + 2 \iint R_\epsilon : Du_\epsilon \psi + 2 \iint R_{\epsilon, ij} (u_\epsilon)_i \partial_j \psi. \end{aligned} \quad (9.6)$$

All these integrations by parts have no boundary terms because ψ is compactly supported in the region where the convolutions are defined. The pressure term reduces to $p_\epsilon u_\epsilon \cdot \nabla \psi$ because $\nabla \cdot u_\epsilon = 0$.

The approximate-identity convergence gives $u_\epsilon \rightarrow u$ in L^4 , $Du_\epsilon \rightarrow Du$ in L^2 , and $F_\epsilon \rightarrow F$ in L^2 on the support of ψ . Also $u_\epsilon \otimes u_\epsilon \rightarrow u \otimes u$ in L^2 , so $R_\epsilon \rightarrow 0$ in L^2 . Thus the two error integrals in (9.6) tend to zero by Cauchy–Schwarz. Since $p_\epsilon \rightarrow p$ in $L^{3/2}$ and $u_\epsilon \rightarrow u$ in L^3 on this finite-measure set, $p_\epsilon u_\epsilon \rightarrow pu$ in L^1 . The remaining terms converge in L^1 by the stated L^4 and L^2 convergences. Passing to the limit in (9.6) proves (9.2).

The energy, pressure, measurability, and slice weak-gradient clauses of Definition 3.1 on $B_r \times (-1, 0)$ follow by restricting the hypotheses of Theorem 3.12 to compactly contained boxes. Conditions (S2) and (S3) restrict to tests supported there, and (S4) follows from (9.2) for nonnegative tests. This proves suitability with $q = 3$ and $f = 0$. $\square$

Lemma 9.2 (Weakly continuous L^3 representative). *Under the hypotheses of Theorem 3.12, there is a map*

$$\bar{u}: [-(3/4)^2, 0] \longrightarrow L^3(B_{3/4}; \mathbb{R}^3) \quad (9.7)$$

that agrees with $u(\cdot, t)$ for almost every t and satisfies, for every $w \in L^{3/2}(B_{3/4}; \mathbb{R}^3)$, continuity of

$$t \mapsto \int_{B_{3/4}} \bar{u}(x, t) \cdot w(x) dx, \quad (9.8)$$

together with

$$\sup_{-(3/4)^2 \leq t \leq 0} \|\bar{u}(\cdot, t)\|_{L^3(B_{3/4})} \leq M, \quad (9.9)$$

where M is defined in (9.1).

Proof. Fix $\varphi \in C_c^\infty(B_{3/4}; \mathbb{R}^3)$. Testing (S3) with $\eta(t)\varphi(x)$ shows that the distributional derivative of $t \mapsto \int u(x, t) \cdot \varphi(x) dx$ is the locally integrable function

$$\int_{B_{3/4}} [u_i u_j \partial_j \varphi_i - Du_{ij} \partial_j \varphi_i + p \nabla \cdot \varphi](x, t) dx. \quad (9.10)$$

The first term has absolute value at most $M^2 \|\nabla \varphi\|_{L^3(B_{3/4})}$ for almost every t . For $s < t$, Cauchy–Schwarz in space-time bounds the increment of the second term by $\|Du\|_{L^2(Q)} \|\nabla \varphi\|_{L^2(B_{3/4})} |t-s|^{1/2}$. Hölder’s inequality bounds the integrated pressure term by $\|p\|_{L^{3/2}(Q)} \|\nabla \cdot \varphi\|_{L^3(B_{3/4})} |t-s|^{1/3}$. Therefore the scalar pairing has an absolutely continuous representative, denoted by ℓ_φ , on $[-(3/4)^2, 0]$, and this representative obeys

$$\begin{aligned} |\ell_\varphi(t) - \ell_\varphi(s)| &\leq M^2 \|\nabla \varphi\|_{L^3(B_{3/4})} |t-s| + \|Du\|_{L^2(Q)} \|\nabla \varphi\|_{L^2(B_{3/4})} |t-s|^{1/2} \\ &\quad + \|p\|_{L^{3/2}(Q)} \|\nabla \cdot \varphi\|_{L^3(B_{3/4})} |t-s|^{1/3}. \end{aligned} \quad (9.11)$$

The value at $t = 0$ is the limit from below; the lower endpoint is interior to the original time interval and the same absolute-continuity argument applies there.

Choose a full-measure set of times on which $u(\cdot, t) \in L^3(B_{3/4})$ with norm at most M and on which its pairings agree with the absolutely continuous representatives for a countable dense family of smooth tests. At each time in this set, Hölder’s inequality bounds the pairing by $M \|w\|_{L^{3/2}}$. Density extends this bound to every $w \in L^{3/2}(B_{3/4}; \mathbb{R}^3)$, and the duality $(L^{3/2})^* = L^3$ represents the time-slice pairing by an element of L^3 with norm at most M .

For any t in the closed interval, choose good times converging to t . Since $B_{3/4}$ has finite measure, Hölder’s inequality makes these slices uniformly bounded in $L^2(B_{3/4})$. Their pairings against the countable L^2 -dense family of smooth tests converge by (9.11); these limits determine an L^2 element $\bar{u}(\cdot, t)$ and weak L^2 convergence of the slices to it. Passing the slice bound to this limit gives $|\int_{B_{3/4}} \bar{u}(\cdot, t) \cdot w| \leq M \|w\|_{L^{3/2}}$ for every L^2 test w . Applying the L^3 norming estimate to $w_N = \min(\|\bar{u}(\cdot, t)\|, N) \bar{u}(\cdot, t) \in L^2$ and letting $N \rightarrow \infty$ gives $\bar{u}(\cdot, t) \in L^3$ with norm at most M . The smooth-test pairings then extend by density to every $L^{3/2}$ test; on the full-measure set they agree with those of u , so this representative agrees with u almost everywhere and satisfies (9.9). $\square$

9.2 The pressure split

For the pressure estimates below we use the whole-space pressure operator of Definition 7.1, with the sign convention fixed by (S3). The input tensor is extended by zero outside B_1 before applying that operator.

Lemma 9.3 (Pressure split and rescaling). *Suppose that u , Du , and p satisfy the hypotheses of Theorem 3.12, and let M be as in (9.1). For almost every $t \in (-1, 0)$ define*

$$F_{ij}(x, t) := \mathbf{1}_{B_1}(x) u_i(x, t) u_j(x, t), \quad p_1(\cdot, t) := P[F](\cdot, t), \quad p_2 := p - p_1 \quad \text{on } B_1 \times (-1, 0). \quad (9.12)$$

Then $p_1 \in L^\infty((-1, 0); L^{3/2}(\mathbb{R}^3))$, $p_2 \in L^{3/2}(B_1 \times (-1, 0))$, and for almost every t the slice $p_2(\cdot, t)$ is weakly harmonic in B_1 . There is an absolute constant C such that

$$\begin{aligned} \operatorname{ess\,sup}_{t \in (-1, 0)} \|p_1(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} &\leq CM^2, \\ \int_{-1}^0 \|p_2(\cdot, t)\|_{L^\infty(B_{3/4})}^{3/2} dt &\leq C \left(\|p\|_{L^{3/2}(Q)}^{3/2} + M^3 \right). \end{aligned} \quad (9.13)$$

Extend u by zero outside Q , extend p_1 by zero in time, and extend p_2 by zero outside $B_1 \times (-1, 0)$; denote these extensions by tildes. Given $z_0 = (x_0, t_0) \in \overline{B_{1/2}} \times [-1/4, 0]$ and $R > 0$, set

$$\begin{aligned} v^R(x, t) &:= R\tilde{u}(x_0 + Rx, t_0 + R^2t), \\ p_1^R(x, t) &:= R^2\tilde{p}_1(x_0 + Rx, t_0 + R^2t), \\ p_2^R(x, t) &:= R^2\tilde{p}_2(x_0 + Rx, t_0 + R^2t). \end{aligned} \quad (9.14)$$

For almost every $t \in \mathbb{R}$ these satisfy

$$\begin{aligned} \int_{\mathbb{R}^3} |v^R(x, t)|^3 dx &= \int_{\mathbb{R}^3} |\tilde{u}(x, t_0 + R^2t)|^3 dx, \\ \int_{\mathbb{R}^3} |p_1^R(x, t)|^{3/2} dx &= \int_{\mathbb{R}^3} |\tilde{p}_1(x, t_0 + R^2t)|^{3/2} dx. \end{aligned} \quad (9.15)$$

For every bounded measurable $\Omega \subset \mathbb{R}^3$, all sufficiently small R also satisfy $x_0 + R\Omega \subset B_{3/4}$ and

$$\int_{\mathbb{R}} \|p_2^R(\cdot, t)\|_{L^\infty(\Omega)}^{3/2} dt = R \int_{\mathbb{R}} \|\tilde{p}_2(\cdot, s)\|_{L^\infty(x_0 + R\Omega)}^{3/2} ds \leq CR \left(\|p\|_{L^{3/2}(Q)}^{3/2} + M^3 \right). \quad (9.16)$$

In particular, $p_2^R \rightarrow 0$ in $L^{3/2}(\mathbb{R}; L^\infty(\Omega))$ as $R \downarrow 0$.

Proof. For almost every t , the zero-extended tensor $F(\cdot, t)$ belongs to $L^{3/2}(\mathbb{R}^3)$ and

$$\sum_{i,j=1}^3 \|F_{ij}(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq 9 \|u(\cdot, t)\|_{L^3(B_1)}^2. \quad (9.17)$$

The slice estimate and joint measurability in Definition 7.1 give the first bound in (9.13). Since $p \in L^{3/2}(Q)$ and p_1 is in that space on $B_1 \times (-1, 0)$, their difference p_2 also belongs to $L^{3/2}$ there.

For $\zeta \in C_c^\infty(Q)$, use $\nabla \zeta$ as the vector test in (S3). The time term vanishes by (S2) applied to $\partial_t \zeta$. In the viscous term, integration by parts in x_j gives $-\iint u_i \partial_i \Delta \zeta$, which vanishes by (S2) applied to $\Delta \zeta$. These integrations have no boundary terms because ζ is compactly supported in Q . The remaining terms yield

$$\iint_Q p \Delta \zeta dx dt = - \sum_{i,j=1}^3 \iint_Q u_i u_j \partial_i \partial_j \zeta dx dt. \quad (9.18)$$

By Definition 7.1, $-\Delta p_1 = \sum_{i,j} \partial_i \partial_j F_{ij}$ in spatial distributions. Since $F = u \otimes u$ on B_1 , subtracting this identity from (9.18) gives $\Delta_x p_2 = 0$ in distributions on $B_1 \times (-1, 0)$. To obtain the slice assertion, test with products $\zeta(x)\eta(t)$, first for a countable dense family of $\zeta \in C_c^\infty(B_1)$; Fubini's theorem gives a common full-measure set of times on which the harmonic identity holds for that family. The $L^{3/2}$ slice bound for p_2 extends the identity to every spatial test function, so $p_2(\cdot, t)$ is weakly harmonic for almost every t .

The CKN weakly harmonic interior estimate gives, for almost every t ,

$$\|p_2(\cdot, t)\|_{L^\infty(B_{3/4})} \leq C_{17} \|p_2(\cdot, t)\|_{L^{3/2}(B_1)}, \quad (9.19)$$

where C_{17} is absolute in dimension three (CKN library). The triangle inequality and the first estimate in (9.13) give

$$\begin{aligned} \int_{-1}^0 \|p_2(\cdot, t)\|_{L^\infty(B_{3/4})}^{3/2} dt &\leq C \int_{-1}^0 \int_{B_1} \left(|p(x, t)|^{3/2} + |p_1(x, t)|^{3/2} \right) dx dt \\ &\leq C \left(\|p\|_{L^{3/2}(Q)}^{3/2} + M^3 \right), \end{aligned}$$

which proves the second estimate in (9.13).

The first scaling identity in (9.15) follows by changing variables $y = x_0 + Rx$ and using $|R\tilde{u}|^3 dx = |\tilde{u}|^3 dy$. The second follows from the same change of variables and $|R^2 \tilde{p}_1|^{3/2} dx = |\tilde{p}_1|^{3/2} dy$. For the last identity, the same spatial change and then $s = t_0 + R^2 t$ give the factor $R^3 R^{-2} = R$ in (9.16). Because $x_0 \in \overline{B_{1/2}}$ and Ω is bounded, $x_0 + R\Omega \subset B_{3/4}$ for all sufficiently small R , with the threshold depending only on $\sup_{x \in \Omega} |x|$. The second estimate of (9.13) then proves (9.16); its right side tends to zero with R . $\square$

Remark 9.4 (Departure from [4]). ESS define their local p_1 by the weak Dirichlet problem in (3.11)–(3.12) on printed pp. 222–223 of [4], and take p_2 to be the resulting harmonic remainder; here p_1 is the whole-space Riesz pressure of the zero extension of $u \otimes u$. The new p_2 is harmonic on B_1 , and the needed interior bound and rescaled vanishing are proved directly from this fixed decomposition and the assumed $L^{3/2}(Q)$ pressure norm. Thus the proof does not transfer (3.19) on printed p. 223 of [4] to a pressure remainder redefined after rescaling. ESS derive the time continuity in (3.8) through the stronger estimate (3.10) on printed p. 222; here the pairing modulus follows directly from (S3), the energy bound, and the $L^{3/2}$ pressure bound. In the local energy equality, the full pressure remains in $L^{3/2}$: pairing it with the mollified velocity converges in L^1 by the $L^{3/2}$ – L^3 Hölder inequality, so no L^2 hypothesis on p is imposed.

9.3 The blow-up limit

The compactness argument needs uniform control on each fixed region of the rescaled past. The local energy inequality supplies the gradient bound; the pressure split gives the pressure bound and, after blow-up, identifies its limit with the whole-space pressure of the limiting velocity.

We use the open past cylinder $Q_r^-(z_0)$ fixed in Section 2. The open top is useful when $t_0 = 0$, since all rescaled times then remain in the original time interval.

Proposition 9.5 (Blow-up limit). *Suppose $u_{\text{orig}}, Du_{\text{orig}}, p_{\text{orig}}$ satisfy the hypotheses of Theorem 3.12 on $D = B_1 \times (-1, 0)$. Let*

$$K = \overline{B_{1/2}} \times [-1/4, 0],$$

and let $z_0 = (x_0, t_0) \in K$ be a bad point in the sense of Definition 8.3, with its top-face interpretation. Let $R_k \rightarrow 0$, and discard finitely many terms so that $Q_{R_k}^-(z_0) \subset D$ for every k .

For almost every $t \in (-1, 0)$, let

$$F(x, t) = \mathbf{1}_{B_1}(x) u_{\text{orig}}(x, t) \otimes u_{\text{orig}}(x, t), \quad p_1(\cdot, t) = P[F(\cdot, t)], \quad p_2 = p_{\text{orig}} - p_1 \quad \text{on } D,$$

as in Lemma 9.3. Extend $\tilde{u}_{\text{orig}} = \mathbf{1}_D u_{\text{orig}}$, $\widetilde{Du}_{\text{orig}} = \mathbf{1}_D Du_{\text{orig}}$, and $\tilde{p}_{\text{orig}} = \mathbf{1}_D p_{\text{orig}}$ by zero to $\mathbb{R}^3 \times \mathbb{R}$. Extend p_1 by zero in time only, retaining its whole-space spatial values, and extend p_2 by zero outside D ; call these extensions $\tilde{p}_1, \tilde{p}_2$. Define

$$\begin{aligned} v^k(x, t) &= R_k \tilde{u}_{\text{orig}}(x_0 + R_k x, t_0 + R_k^2 t), & Dv^k(x, t) &= R_k^2 \widetilde{Du}_{\text{orig}}(x_0 + R_k x, t_0 + R_k^2 t), \\ p^k(x, t) &= R_k^2 \tilde{p}_{\text{orig}}(x_0 + R_k x, t_0 + R_k^2 t), & p_1^k(x, t) &= R_k^2 \tilde{p}_1(x_0 + R_k x, t_0 + R_k^2 t), \\ & & p_2^k(x, t) &= R_k^2 \tilde{p}_2(x_0 + R_k x, t_0 + R_k^2 t). \end{aligned}$$

Let $D_k = \{(x, t) : (x_0 + R_k x, t_0 + R_k^2 t) \in D\}$. Then $p_1^k = P[v^k \otimes v^k]$, and $p^k = p_1^k + p_2^k$ on D_k . For time-slice assertions, use the $L^3(B_{3/4})$ equivalence class $\bar{u}(\cdot, \tau)$ from Lemma 9.2 whenever $\tau = t_0 + R_k^2 t \in [-(3/4)^2, 0]$ and the spatial region under consideration rescales into $B_{3/4}$. The weak-continuity lemma identifies its smooth-test pairings with the absolutely continuous representatives from the momentum equation. This choice agrees with the zero-extended spacetime field almost everywhere and applies throughout each fixed bounded cylinder $B_R \times [a, 0]$ for all sufficiently large k ; in particular it specifies the terminal slice at $t = 0$.

There is a subsequence, still denoted k , and fields u, Du, q on $\mathbb{R}^3 \times (-\infty, 0)$ with the following properties.

- (a) For every bounded cylinder $C = B_R \times (a, 0)$, with $R < \infty$ and $a < 0$, a constant depending only on C and the norms in Theorem 3.12 satisfies

$$\iint_C (|p^k|^{3/2} + |Dv^k|^2) dx dt \leq C_C$$

for all sufficiently large k .

- (b) On each such cylinder, v^k has the pairing modulus required in [2, Lemma 11.1]: for every compact $K_0 \subset \mathbb{R}^3$, every finite interval $J = [a, 0]$ with $a < 0$, every $w \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ supported in K_0 , and all $s, t \in J$,

$$\begin{aligned} \left| \int_{\mathbb{R}^3} (v^k(x, t) - v^k(x, s)) \cdot w(x) dx \right| &\leq C_{K_0, J} \left(|t - s| (\|\Delta w\|_{L^2(\mathbb{R}^3)} + \|\nabla w\|_{L^\infty(\mathbb{R}^3)}) \right. \\ &\quad \left. + |t - s|^{1/3} \|\operatorname{div} w\|_{L^\infty(\mathbb{R}^3)} \right), \end{aligned}$$

where the constants are independent of k, s, t, w for all sufficiently large k . For $J \subset (-\infty, 0)$ this is the hypothesis of [2, Lemma 11.1]; the same estimate extends to $t = 0$ using the trace just defined. Also the $L^{10/3}(C)$ norms are uniformly bounded for all sufficiently large k on every bounded cylinder $C = B_R \times (a, 0)$.

- (c) The subsequence satisfies

$$v^k \longrightarrow u \quad \text{strongly in } L^3(C)$$

for every bounded cylinder $C = B_R \times (a, 0)$, and $v^k(\cdot, t) \rightharpoonup u(\cdot, t)$ weakly in $L_{\text{loc}}^2(\mathbb{R}^3)$ for every $t < 0$. Also $Dv^k \rightharpoonup Du$ weakly in $L^2(C)$ for every such C , where Du is the weak spatial

gradient of u . Moreover,

$$\begin{aligned} u &\in L^\infty((-\infty, 0); L^3(\mathbb{R}^3)), \\ p_1^k &\longrightarrow q \quad \text{strongly in } L^{3/2}(C), \\ p_2^k &\longrightarrow 0 \quad \text{strongly in } L^{3/2}(C), \\ p^k &\longrightarrow q \quad \text{strongly in } L^{3/2}(C). \end{aligned}$$

(d) The limit $(u, Du, q, 0)$ is a suitable weak solution with exponent 3 on every $B_\rho \times (a, 0)$, where $\rho < \infty$ and $a < 0$. The limiting pressure is the whole-space associated pressure

$$q = P[u \otimes u] \quad \text{on } \mathbb{R}^3 \times (-\infty, 0).$$

(e) There is a terminal slice $u(\cdot, 0) \in L^2_{\text{loc}}(\mathbb{R}^3)$ such that $v^k(\cdot, 0) \rightharpoonup u(\cdot, 0)$ weakly in $L^2_{\text{loc}}(\mathbb{R}^3)$, and

$$\int_{B_1(x)} |u(y, 0)|^2 dy = 0 \quad \text{for every } x \in \mathbb{R}^3.$$

(f) For every k ,

$$\iint_{Q_1^-(0)} (|v^k|^3 + |p^k|^{3/2}) dx dt \geq \varepsilon_0/8.$$

If $u = 0$ almost everywhere on $\mathbb{R}^3 \times (-T, 0)$ for some $T > 1$, then

$$\lim_{k \rightarrow \infty} \iint_{Q_1^-(0)} (|v^k|^3 + |p^k|^{3/2}) dx dt = 0.$$

Proof. Put $M = \text{ess sup}_{t \in (-1, 0)} \|u_{\text{orig}}(\cdot, t)\|_{L^3(B_1)}$. Lemma 9.1 makes $(u_{\text{orig}}, Du_{\text{orig}}, p_{\text{orig}}, 0)$ suitable on $B_{3/4} \times (-1, 0)$, while Lemma 9.2 supplies the all-time local L^3 representative up to $t = 0$. For each fixed bounded cylinder in the rescaled variables, its inverse image lies in $B_{3/4} \times (-1, 0)$ for all sufficiently large k , except for its top face, which has measure zero. Since $t_0 + R_k^2 t \leq t_0 \leq 0$ whenever $t \leq 0$, these rescalings use only past times.

Local energy and pressure bounds. The velocity scaling and Lemma 9.3 give the bound M for the global spatial L^3 norm of $v^k(\cdot, t)$ for almost every rescaled time. For almost every such t , both $p_1^k(\cdot, t)$ and $P[v^k(\cdot, t) \otimes v^k(\cdot, t)]$ belong to $L^{3/2}(\mathbb{R}^3)$ and solve the Poisson equation with right-hand side $\sum_{i,j=1}^3 \partial_i \partial_j (v_i^k v_j^k)$. Their difference is harmonic, hence smooth, and belongs to $L^{3/2}(\mathbb{R}^3)$. The mean-value property gives $|h(x)| \leq |B_R|^{-2/3} \|h\|_{L^{3/2}(\mathbb{R}^3)}$ for this difference h and every $R > 0$; letting $R \rightarrow \infty$ shows that $h = 0$. Hence $p_1^k = P[v^k \otimes v^k]$ on almost every slice, and the $L^{3/2}$ boundedness of P gives

$$\|p_1^k(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq CM^2.$$

For every bounded spatial set Ω ,

$$\int_{\mathbb{R}} \|p_2^k(\cdot, t)\|_{L^\infty(\Omega)}^{3/2} dt \leq CR_k (\|p_{\text{orig}}\|_{L^{3/2}(D)}^{3/2} + M^3).$$

Here and below C may change from line to line and depends only on the displayed bounded sets and the fixed data norms. In particular, p^k is uniformly bounded in $L^{3/2}(C)$ on each bounded

cylinder C . For a fixed bounded spatial set Ω , Hölder's inequality and the velocity scaling give, uniformly for t in any fixed interval $[a, 0]$,

$$\int_{\Omega} |v^k(x, t)|^2 dx \leq R_k^{-1} \int_{x_0 + R_k \Omega} |\bar{u}(y, t_0 + R_k^2 t)|^2 dy \leq |\Omega|^{1/3} M^2. \quad (9.20)$$

This holds for all sufficiently large k , since $x_0 + R_k \Omega \subset B_{3/4}$ and the time arguments lie in $[-(3/4)^2, 0]$.

Fix $C = B_R \times (a, 0)$. Choose a nonnegative spatial cutoff $\zeta \in C_c^\infty(B_{R+1})$ that equals one on B_R . For $b < 0$, choose time cutoffs $\eta_b \in C_c^\infty((a-1, 0))$, $0 \leq \eta_b \leq 1$, equal to one on $[a, b]$, with $\|\eta_b'\|_{L^1} \leq 2$, uniformly as $b \uparrow 0$. Apply the local energy inequality to $\psi(x, t) = \zeta(x)\eta_b(t)$ for the rescaled suitable solution. The terms with $\nabla \zeta$ are bounded by the local $L_t^\infty L_x^3$ bound and the local $L^{3/2}$ pressure bound; the term with η_b' is bounded by $2 \sup_t \int_{\text{supp } \zeta} |v^k|^2$. The term with $\Delta \zeta$ has the same slice L^2 bound. These bounds do not depend on b or k . Thus $\iint_{B_R \times (a, b)} |Dv^k|^2 \leq C_C$; letting $b \uparrow 0$ proves (a), including its top-face endpoint.

The weak momentum identity (S3), tested with a spatial field w and a scalar time test, gives the absolutely continuous pairing identity

$$\int_{\mathbb{R}^3} (v^k(x, t) - v^k(x, s)) \cdot w(x) dx = \int_s^t \int_{\mathbb{R}^3} (v^k \cdot \Delta w + (v^k \otimes v^k) : \nabla w + p^k \operatorname{div} w) dx d\tau.$$

The integration by parts in the diffusion term has no spatial boundary contribution because w is compactly supported. The first term is bounded by $C|t-s|\|\Delta w\|_2$ using the local slice L^2 bound. The second is bounded by $C|t-s|\|\nabla w\|_\infty$ using the $L_t^\infty L_x^3$ bound. For the last term, the local $L^{3/2}$ pressure bound and Hölder's inequality give

$$\int_s^t \int_{\text{supp } w} |p^k| |\operatorname{div} w| \leq C_{K_0, J} |t-s|^{1/3} \|\operatorname{div} w\|_\infty.$$

The identity first holds for $s, t < 0$. The trace from Lemma 9.2 and the integrability of the right side let $t \uparrow 0$, proving the same modulus on $[a, 0]$.

For the local $L^{10/3}$ bound, choose a spatial cutoff η supported in a bounded neighborhood of B_R and equal to one there. The three-dimensional Gagliardo–Nirenberg inequality applied to $\eta v^k(\cdot, t)$, extended by zero, gives

$$\|\eta v^k(\cdot, t)\|_{10/3}^{10/3} \leq C \|\eta v^k(\cdot, t)\|_2^{4/3} \|\nabla(\eta v^k)(\cdot, t)\|_2^2.$$

The local slice bound and (a), integrated over $(a, 0)$, give the asserted uniform $L^{10/3}$ bound.

For each integer $m \geq 1$, choose $N(m)$ so that the local energy, pairing-modulus, and $L^{10/3}$ bounds above hold on $\bar{B}_{m+1} \times [-(m+1), 0]$ for every $k \geq N(m)$. Choose a strictly increasing sequence $\nu(j)$ with $\nu(j) \geq \max_{m \leq j} N(m)$. Let χ_j be a smooth spatial cutoff equal to one on $\bar{B}_j$ and supported in B_{j+1} , and let θ_j be a smooth time ramp vanishing before $-(j+1)$ and equal to one on $[-j, 0]$. Apply [2, Lemma 11.1] once on $\mathbb{R}^3 \times (-\infty, 0)$ to the cutoff sequence $\hat{v}^j = \theta_j \chi_j v^{\nu(j)}$. The preceding bounds and the cutoff derivative estimates give the required uniform hypotheses on each compact cylinder; the finitely many early indices at each stage are controlled by their individual moduli. Since $\hat{v}^j = v^{\nu(j)}$ on any fixed cylinder for all sufficiently large j , the resulting all-slice weak L_{loc}^2 convergence, local strong L^2 convergence, and local weak-gradient convergence transfer to the original rescaled fields. [2, Corollary 11.3] then gives strong L^3 convergence on every

compact subcylinder below $t = 0$. Fix $R < \infty$ and $a < 0$. For sufficiently small $\delta > 0$, the uniform $L^{10/3}$ bound and Hölder's inequality give

$$\|v^k\|_{L^3(B_R \times (a, a+\delta))}^3 \leq (|B_R|\delta)^{1/10} \|v^k\|_{L^{10/3}(B_R \times (a, a+\delta))}^3,$$

and the identical estimate holds on $B_R \times (-\delta, 0)$ and for u by Fatou's lemma. Both time-boundary strips therefore have L^3 norms tending to zero uniformly in k . On $B_R \times (a + \delta, -\delta)$ the strong convergence follows from [2, Corollary 11.3], since $\overline{B_R} \times [a + \delta, -\delta]$ is compactly contained in a slightly larger space-time cylinder. Combining the interior convergence with the two strip estimates proves strong L^3 convergence on $B_R \times (a, 0)$. The local weak-gradient convergence upgrades to weak convergence in $L^2(B_R \times (a, 0))$: (a) gives a uniform L^2 bound on that cylinder, and compactly supported tests in its interior are dense in L^2 , so the local weak limit determines the weak limit on the whole cylinder. The distributional gradient identity passes to the limit by the strong local L^2 convergence and identifies Du as the weak spatial gradient of u .

By strong local L^3 convergence, take a diagonal subsequence of the zero-extended fields that converges almost everywhere on every bounded space-time cylinder. Fubini's theorem gives a full-measure set of times on which this subsequence converges almost everywhere in space; intersect it with the common full-measure set on which every slice $v^k(\cdot, t)$ has global L^3 norm at most M . For each such time, Fatou's lemma on each ball gives the same bound for $u(\cdot, t)$ there. Monotone convergence over an increasing sequence of balls yields $\|u(\cdot, t)\|_{L^3(\mathbb{R}^3)} \leq M$, and hence $u \in L^\infty((-\infty, 0); L^3)$.

The pressure limit. Set $F_k = v^k \otimes v^k$ and $F = u \otimes u$. The just-proved strong L^3 convergence gives $F_k \rightarrow F$ strongly in $L^{3/2}(B_L \times (a, 0))$ for each fixed $L < \infty$ and $a < 0$. Fix $R < L/2$. The near-field pressure satisfies

$$\|P[\mathbf{1}_{B_L}(F_k - F)]\|_{L^{3/2}(B_R \times (a, 0))} \leq C\|F_k - F\|_{L^{3/2}(B_L \times (a, 0))} \rightarrow 0$$

by the $L^{3/2}$ bound for the double Riesz transforms in Definition 7.1. Away from the support, their kernel is the second derivative of the Newtonian kernel and obeys $|K_{ij}(x - y)| \leq C|x - y|^{-3}$. Thus, for $|x| \leq R$, $L \geq 2R$, and almost every t ,

$$\begin{aligned} |P[\mathbf{1}_{\mathbb{R}^3 \setminus B_L} F_k](x, t)| &\leq C \int_{|y| > L} |x - y|^{-3} |F_k(y, t)| dy \\ &\leq C \| |x - \cdot|^{-3} \|_{L^3(\mathbb{R}^3 \setminus B_L)} \|F_k(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq CL^{-2} M^2. \end{aligned}$$

The same estimate holds for F . Taking the $L^{3/2}$ norm on $B_R \times (a, 0)$, the far-field contribution is at most $C|B_R|^{2/3}|a|^{2/3}L^{-2}M^2$. First let $k \rightarrow \infty$ at fixed L , then let $L \rightarrow \infty$. This proves $p_1^k = P[F_k] \rightarrow P[F]$ strongly in $L^{3/2}(B_R \times (a, 0))$. Lemma 9.3 gives $p_2^k \rightarrow 0$ in $L_t^{3/2}L_x^\infty$ on bounded spatial sets, so $p_2^k \rightarrow 0$ in $L^{3/2}(B_R \times (a, 0))$ as well. Since $p^k = p_1^k + p_2^k$ on every fixed cylinder for all sufficiently large k , the pressure limit is $q = P[u \otimes u]$.

Suitability of the limit. Fix $\rho < \infty$ and $a < 0$. On every open box compactly contained in $B_\rho \times (a, 0)$, its inverse image lies in $B_{3/4} \times (-1, 0)$ for all sufficiently large k . Restriction by [2, Lemma 11.5] and the Navier–Stokes rescaling $v^k = R_k u_{\text{orig}}$, $Dv^k = R_k^2 Du_{\text{orig}}$, $p^k = R_k^2 p_{\text{orig}}$ make the tail of the sequence suitable there with exponent 3 and zero force. On each such box, (a) and the slice bound (9.20) give the uniform essential-supremum L^2 and space-time gradient bounds; (b)–(c) give strong L^3 , weak L^2 gradients, and strong local $L^{3/2}$ pressure convergence. Apply [2, Theorem 11.4] on each box to the corresponding tail. These conclusions for all compactly contained boxes prove suitability of $(u, Du, q, 0)$ on $B_\rho \times (a, 0)$.

The terminal slice. For a fixed compact spatial set and a smooth compactly supported test w , the modulus in (b) is uniform up to $t = 0$. Since the pairings converge for every $t < 0$, it follows that the sequence $\int v^k(x, 0) \cdot w(x) dx$ is Cauchy: for $\delta > 0$, compare each endpoint pairing with its value at $-\delta$, use (b) twice, pass $k, \ell \rightarrow \infty$ at $-\delta$, and then let $\delta \downarrow 0$. The local L^2 bounds make the limiting functionals bounded on L^2 ; density and the Riesz representation theorem give a compatible slice $u(\cdot, 0) \in L^2_{\text{loc}}$ and weak L^2_{loc} convergence of $v^k(\cdot, 0)$ to it. The same modulus shows that this is the weak trace of u from times below zero.

Fix $x \in \mathbb{R}^3$. Since $t_0 \in [-1/4, 0]$, Lemma 9.2 gives $\bar{u}(\cdot, t_0) \in L^3(B_{3/4})$. For all sufficiently large k , $B(x_0 + R_k x, R_k) \subset B_{3/4}$, and change of variables followed by Hölder's inequality gives

$$\begin{aligned} \int_{B_1(x)} |v^k(y, 0)|^2 dy &= R_k^{-1} \int_{B(x_0 + R_k x, R_k)} |\bar{u}(y, t_0)|^2 dy \\ &\leq |B_1|^{1/3} \left(\int_{B(x_0 + R_k x, R_k)} |\bar{u}(y, t_0)|^3 dy \right)^{2/3} \\ &\longrightarrow 0. \end{aligned}$$

The final limit follows from absolute continuity of the integral of the fixed $L^3(B_{3/4})$ function on sets whose measure tends to zero. Weak convergence and lower semicontinuity of the $L^2(B_1(x))$ norm prove the zero terminal-slice assertion in (e), for every center x .

Finally, by Lemma 8.4 and equation (8.3), the bad-point lower bound applies at every admissible open-top radius. At radius R_k it gives, for every k ,

$$R_k^{-2} \iint_{Q_{R_k}^-(z_0)} (|u_{\text{orig}}|^3 + |p_{\text{orig}}|^{3/2}) dx dt \geq \varepsilon_0/8.$$

The Navier–Stokes scaling changes this exactly into the lower bound in (f). If $u = 0$ on $\mathbb{R}^3 \times (-T, 0)$ with $T > 1$, then $q = P[u \otimes u] = 0$ there. Strong $L^3(Q_1^-)$ convergence gives $\int_{Q_1^-} |v^k|^3 \rightarrow 0$; the pressure convergence in (c) gives $\int_{Q_1^-} |p^k|^{3/2} \rightarrow 0$. This proves the final assertion. $\square$

Remark 9.6 (Departure from [4]). The rescaling and the roles of the local energy and pressure bounds are those of (3.13)–(3.20), printed pp. 223–225, of [4]. The compactness argument gives strong local space-time L^2 , strong L^3 , and weak convergence of every velocity slice; it does not use the strong $C([a, b]; L^2_{\text{loc}})$ convergence asserted in (3.23), printed p. 225, of [4]. The endpoint conclusion replacing (3.34), printed p. 226, follows from the uniform pairing modulus, weak slice convergence at $t = 0$, and lower semicontinuity. The pressure split is the fixed whole-space Riesz split of Lemma 9.3; its harmonic remainder decays by that lemma's scaling estimate, proved for this split, rather than by transferring (3.19) to a newly defined pressure remainder.

10 Proof of the local theorem

Lemma 10.1 (Flatness from a zero neighborhood). *Let $R', T', r_0, M > 0$, $x_* \in \mathbb{R}^3$, and let $w: B_{R'}(x_*) \times [0, T'] \rightarrow \mathbb{R}^3$ be continuous. Suppose $|w| \leq M$ on this cylinder and*

$$w = 0 \quad \text{on } B_{r_0}(x_*) \times (0, r_0^2),$$

where $r_0 \leq R'$ and $r_0^2 \leq T'$. Then, for every integer $k \geq 1$,

$$|w(x, s)| \leq M r_0^{-k} (|x - x_*| + \sqrt{s})^k \quad \text{for } (x, s) \in B_{R'}(x_*) \times (0, T'). \quad (10.1)$$

For $k = 0$ the same conclusion holds with the factor on the right interpreted as M .

Proof. Put $d = |x - x_*| + \sqrt{s}$. If $d < r_0$, then $x \in B_{r_0}(x_*)$ and $0 < s < r_0^2$, so $w(x, s) = 0$. If $d \geq r_0$, then $M r_0^{-k} d^k \geq M \geq |w(x, s)|$. This proves the assertion, including the case $k = 0$. $\square$

Lemma 10.2 (Exterior vorticity at the terminal face). *Let $T_2 > 2$ and $R_2 > 0$. Suppose $(u, Du, q, 0)$ is suitable with exponent 3 on $\mathbb{R}^3 \times (-4T_2, 0)$, $u \in L^\infty((-4T_2, 0); L^3(\mathbb{R}^3))$, and its weakly continuous local trace satisfies $u(\cdot, 0) = 0$. Suppose there are constants $M, E < \infty$ such that for every $z_1 = (x_1, t_1)$ with $|x_1| > R_2$ and $t_1 \in (-2, 0)$, $(u, Du, q, 0)$ is suitable on an open neighborhood of $Q_1^-(z_1)$, $|u| \leq M$ on $Q_1^-(z_1)$, and*

$$\left(\iint_{Q_1^-(z_1)} |Du|^2 dx dt \right)^{1/2} \leq E.$$

Then $\omega = \operatorname{curl} u$ has a representative continuous on $\{|x| > R_2\} \times (-2, 0]$. On every bounded spatial set whose closure lies in $\{|x| > R_2\}$, this representative belongs to $W_2^{2,1}$ up to $t = 0$, satisfies

$$|\partial_t \omega - \Delta \omega| \leq C(M, E)(|\omega| + |\nabla \omega|) \quad \text{a.e. on } \{|x| > R_2\} \times (-2, 0), \quad (10.2)$$

and has $\omega(\cdot, 0) = 0$ there. In addition, its representative satisfies

$$|\omega(x, t)| \leq C_\omega(M, E) \quad (|x| > R_2, -2 < t \leq 0),$$

with a constant independent of the exterior point. The coefficient in the inequality depends only on M, E ; the local $W_2^{2,1}$ norm may also depend on the bounded set.

Proof. Apply Theorem 8.14 to each indicated unit past cylinder. Its bounds depend only on the common M and E . In particular, on $B_{1/2}(x_1) \times (t_1 - 1/4, t_1)$ they give a common bound for ω in $W_2^{2,1}$, for ω in $L_t^\infty H_x^2$, and for the continuous representative on the closed half-cylinder. Representatives from two cylinders agree on overlaps: they are continuous and represent the same weak curl almost everywhere, so if they differed at a point of an open overlap they would differ on a set of positive measure. They therefore glue to a continuous representative on the exterior open region below $t = 0$. The common continuous bound on each half-cylinder also gives a bound independent of the exterior point. Indeed, for $|x| > R_2$ and $-2 < t < 0$, choose t_1 with $t < t_1 < \min\{t + 1/4, 0\}$. Then $(x, t) \in B_{1/2}(x) \times (t_1 - 1/4, t_1)$, so the half-cylinder estimate gives $|\omega(x, t)| \leq C_0(M, E)$. Continuity up to $t = 0$ gives the same bound there.

Fix a ball B whose closure is contained in $\{|x| > R_2\}$. Cover $\bar{B}$ by finitely many balls $B_{1/2}(x_j)$ with $|x_j| > R_2$. For the initial time portion, the interval $(-2, -7/4)$ is contained in the half-cylinder time interval centered at $-7/4$. The compact interval $[-7/4, -1/4]$ is covered by finitely many intervals $(s_\ell - 1/4, s_\ell)$ with centers $s_\ell \in (-2, 0)$, since this family is an open cover of that compact interval. For the terminal portion set $\tau_n = -1/(n+4)$ for $n \geq 1$ and $I_n = (\tau_n - 1/4, \tau_n)$. Then $\tau_n \uparrow 0$ and, for every fixed $t \in (-1/4, 0)$, $\mathbf{1}_{I_n}(t) \rightarrow 1$. For each spatial center x_j and each of the finitely many initial time centers, and for each (x_j, τ_n) , the half-cylinder estimates give with one constant $C = C(M, E)$

$$\operatorname{ess\,sup}_{t \in I} \|\omega(\cdot, t)\|_{H^2(B \cap B_{1/2}(x_j))} + \|\partial_t \omega\|_{L^2((B \cap B_{1/2}(x_j)) \times I)} \leq C,$$

where I denotes the corresponding time interval. They also bound each spatial derivative through order two in L^2 on that patch and interval. The finite covers give the asserted bounds on

the initial portion. On the terminal portion, Fatou's lemma applied to the indicators $\mathbf{1}_{I_n}$ and to the squares of $\partial_t \omega$ and of each spatial derivative through order two gives finite L^2 norms on $B \times (-1/4, 0)$. Since the I_n cover this terminal portion and the H^2 bounds are uniform, countability also gives $\text{ess sup}_{-1/4 < t < 0} \|\omega(t)\|_{H^2(B)} \leq C_B$. Combining the two portions shows that on $B \times (-2, 0)$ these derivatives have finite L^2 norms, that $\text{ess sup}_{-2 < t < 0} \|\omega(t)\|_{H^2(B)} \leq C_B$, and that $\partial_t \omega \in L^2(B \times (-2, 0))$. Thus $\omega \in W^{1,2}((-2, 0); L^2(B))$. The fundamental theorem for Bochner–Sobolev functions gives

$$\|\omega(t) - \omega(s)\|_{L^2(B)} \leq C_B |t - s|^{1/2} \quad (-2 < s < t < 0).$$

Interpolate between $L^2(B)$ and $H^2(B)$ at order $7/4$ and use $H^{7/4}(B) \hookrightarrow L^\infty(B)$. The uniform H^2 bound then gives

$$\|\omega(t) - \omega(s)\|_{L^\infty(B)} \leq C_B |t - s|^{1/16}.$$

Consequently ω has a unique continuous extension to $t = 0$ on B . The spatial embedding $H^2(B) \hookrightarrow C^{0,\alpha}(\overline{B})$ for every $\alpha < 1/2$ shows that this extension is jointly continuous in space and time. The local $W_2^{2,1}$ bounds and the inequality (10.2) pass to the union by countable covering; the endpoint is a null time slice for the inequality.

It remains to identify the trace. For every smooth compactly supported vector test φ in B and almost every $t < 0$, the definition of weak curl gives

$$\int_B \omega(x, t) \cdot \varphi(x) \, dx = \int_B u(x, t) \cdot \text{curl } \varphi(x) \, dx.$$

The left side converges to the pairing with the continuous extension at $t = 0$. By the weakly continuous L_{loc}^3 trace and $u(\cdot, 0) = 0$, the right side tends to zero. Hence the extension has zero curl-trace as a distribution on B . Since it is continuous in space, it vanishes at every point of B at $t = 0$. The ball was arbitrary. $\square$

Lemma 10.3 (Small pressure mass near a regular point). *Let $(u, Du, q, 0)$ be suitable with exponent 3 on an open set containing $\overline{Q_{r_0}^-(z_0)}$, and suppose $q = P[u \otimes u]$ there. If u has a parabolically Hölder representative on a neighborhood of z_0 , then*

$$\lim_{r \downarrow 0} r^{-2} \iint_{Q_r^-(z_0)} (|u|^3 + |q|^{3/2}) \, dx \, dt = 0. \quad (10.3)$$

Proof. Shrink r_0 so that the closure of the resulting cylinder is inside the Hölder neighborhood. On this compact cylinder, $|u|$ and the spatial Hölder seminorm of u are bounded by a constant H . The velocity term in (10.3) is at most $CH^3 r^5$, so after division by r^2 it tends to zero.

Choose $\chi \in C_c^\infty(B_{r_0}(x_0))$ equal to one on $B_{r_0/2}(x_0)$ and put $F = \chi u \otimes u$, extended by zero in space. For almost every time, $F(\cdot, t)$ is uniformly compactly supported and uniformly C^α for some $\alpha > 0$. The kernel formula for the double Riesz transforms defining P , its spherical cancellation, and $|K(x - y)| \leq C|x - y|^{-3}$ give

$$\|P[F(\cdot, t)]\|_{L^\infty(B_{r_0/2}(x_0))} \leq C(H, r_0, \chi).$$

Indeed, in the principal-value integral subtract $F(x, t)$ near x ; the remaining near-field integrand is bounded by $CH|x - y|^{\alpha-3}$, which is integrable in dimension three, and the far-field integral is bounded because F has fixed compact support.

Set $h = q - P[F]$ on $B_{r_0/2}(x_0)$. The pressure identity from (S3) and the definition of P imply $\Delta_x h = 0$ there for almost every time. Also $h \in L^{3/2}$ on a slightly larger compact cylinder. The harmonic interior estimate therefore gives a function $g \in L^1$ of time such that

$$\|h(\cdot, t)\|_{L^\infty(B_{r_0/4}(x_0))}^{3/2} \leq g(t) \quad \text{for a.e. } t.$$

For $r < r_0/4$, the pressure mass of $P[F]$ on $Q_r^-(z_0)$ is at most Cr^5 . The pressure mass of h is at most

$$Cr^3 \int_{t_0-r^2}^{t_0} g(t) dt = o(r^3),$$

by absolute continuity of the integral of g . After division by r^2 both terms tend to zero. The inequality $|a + b|^{3/2} \leq C(|a|^{3/2} + |b|^{3/2})$ completes the proof. $\square$

Lemma 10.4 (Liouville theorem for a curl-free L^3 field). *If $v \in L^3(\mathbb{R}^3; \mathbb{R}^3)$ and $\operatorname{div} v = 0$, $\operatorname{curl} v = 0$ in distributions, then $v = 0$ almost everywhere.*

Proof. The vector identity $\Delta v = \nabla(\operatorname{div} v) - \operatorname{curl}(\operatorname{curl} v)$, valid in distributions, shows that every component of v is distributionally harmonic. For completeness, mollify one component on each relatively compact ball. The mollifications are smooth harmonic functions on smaller balls. A smooth harmonic function equals its convolution with any radial smooth unit-mass kernel supported in such a ball, by the mean-value property. Differentiating this identity bounds every derivative on a compact subball by a constant times the L^1 norm on a slightly larger ball. The mollifications have uniform such bounds because they converge to v_i in L^1_{loc} . Arzelà–Ascoli, applied successively to derivatives and then diagonally on an exhaustion, gives a smooth harmonic representative equal to v_i almost everywhere. This proves the needed form of Weyl’s lemma.

For this representative, the mean-value identity and Hölder’s inequality give, for every $x \in \mathbb{R}^3$ and $R > 0$,

$$|v_i(x)| \leq \frac{1}{|B_R|} \int_{B_R(x)} |v_i(y)| dy \leq |B_R|^{-1/3} \|v_i\|_{L^3(\mathbb{R}^3)}.$$

Since $|B_R|^{-1/3} = |B_1|^{-1/3} R^{-1}$, letting $R \rightarrow \infty$ shows $v_i(x) = 0$. This holds for each component. $\square$

Proof of Theorem 3.12. We argue by contradiction. By Lemma 9.1, the hypotheses of Theorem 3.12 make $(u, Du, p, 0)$ suitable with exponent 3 on $B_1 \times (-1, 0)$: every compactly supported test in this open cylinder is supported in some $B_r \times (-1, 0)$, $r < 1$, where that lemma gives the local energy inequality. Let

$$K = \overline{B_{1/2}} \times [-1/4, 0].$$

If u has no Hölder representative on K , the contrapositive of Lemma 8.4 (with its amended top-face definition of good points) gives $z_0 = (x_0, t_0) \in K$ that is not good. Choose any sequence $R_k \downarrow 0$ small enough that the open-top cylinders $Q_{R_k}^-(z_0)$ lie in $B_1 \times (-1, 0)$. Proposition 9.5, using (8.3), supplies a subsequential blow-up limit $(U, DU, q, 0)$ on $\mathbb{R}^3 \times (-\infty, 0)$. Clause (c) of that proposition gives $U \in L^\infty((-\infty, 0); L^3(\mathbb{R}^3))$; clause (d) gives suitability with exponent 3 on every bounded cylinder in the open past and $q = P[U \otimes U]$; clause (e) gives a terminal slice $U(\cdot, 0) = 0$ in L^2_{loc} ; and clause (f) gives the lower bound

$$\iint_{Q_1^-(0)} (|v^k|^3 + |p^k|^{3/2}) dx dt \geq \varepsilon_0/8 \quad \text{for every } k,$$

with convergence of these integrals to zero if $U = 0$ on $\mathbb{R}^3 \times (-T, 0)$ for any $T > 1$. Here v^k, p^k are the rescalings in Proposition 9.5; its top-face cylinders are open at $t = 0$, so no future values are used.

Exterior regularity. Fix $T_2 > 2$. The bounds $U \in L^\infty L^3$ and $q = P[U \otimes U]$, together with the $L^{3/2}$ boundedness of the double Riesz transforms, imply $U \in L^3(\mathbb{R}^3 \times (-4T_2, 0))$ and $q \in L^{3/2}(\mathbb{R}^3 \times (-4T_2, 0))$. Hence

$$\iint_{(\mathbb{R}^3 \setminus B_A) \times (-4T_2, 0)} (|U|^3 + |q|^{3/2}) dx dt \longrightarrow 0 \quad (A \rightarrow \infty).$$

Let ε_A be the threshold in Lemma 8.1, and set $r_e = 1/4$. Choose A so large that this tail is less than $\varepsilon_A r_e^2$. For every center $z = (x, t)$ with $|x| > A + 1$ and $-2T_2 - 1 < t < 0$, choose a top time $t' > t$ with $t' < 0$ and $t' - t < r_e^2/8$. The closed full cylinder $\overline{Q_{r_e}^-(x, t')}$ is contained in the open solution domain and in the region used for the tail estimate: its spatial ball lies outside B_A , and its time interval lies in $(-4T_2, 0)$ because $T_2 > 2$. Its normalized integral is less than ε_A . Theorem 8.1 gives a representative bounded by C_4/r_e on its half-cylinder, which contains (x, t) . This proves an essential bound $|U| \leq M$ on $\{|x| > A + 1\} \times (-2T_2 - 1, 0)$, with $M = C_4/r_e$.

Set $R_2 = A + 3$ and $R_3 = R_2 + 2$. For every $z_1 = (x_1, t_1)$ with $|x_1| > R_3$ and $t_1 \in (-2T_2, 0)$, the cylinder $Q_1^-(z_1)$ lies in the bounded-velocity region just obtained and in $\mathbb{R}^3 \times (-4T_2, 0)$. Its local gradient energy has a bound independent of x_1, t_1 . To see this, apply the local energy inequality with a fixed spatial cutoff equal to one on $B_1(x_1)$ and supported in $B_2(x_1)$, and with a time cutoff that rises from zero to one on $(t_1 - 2, t_1 - 1)$, equals one on $[t_1 - 1, t_1]$, and falls to zero on $(t_1, t_1 + \delta_1)$, where $\delta_1 = \min\{-t_1/2, 1/8\}$. This interval lies before 0, and the total variation of the cutoff is 2, uniformly in t_1 . Its support lies in $(-4T_2, 0)$ because $t_1 > -2T_2$ and $T_2 > 2$. The terms with U^2 are bounded by the global $L_t^\infty L_x^3$ bound on balls of fixed volume; the terms with U^3 and qU are bounded by the global $L_t^\infty L_x^3$ and $L_t^\infty L_x^{3/2}$ bounds and Hölder's inequality. The local energy inequality with this test bounds the gradient integral on the whole open-top cylinder by a constant uniform in x_1, t_1 . The integrals over $B_1(x_1) \times (t_1 - 1, s)$ increase to this full integral as $s \uparrow t_1$, so monotone convergence includes the top endpoint. Thus

$$\left(\iint_{Q_1^-(z_1)} |DU|^2 \right)^{1/2} \leq E$$

for a constant E independent of the exterior center. For each such z_1 , the closure of $Q_1^-(z_1)$ is contained in an open cylinder compactly contained in $\mathbb{R}^3 \times (-4T_2, 0)$; suitability restricts to that neighborhood. All hypotheses of Theorem 8.14 therefore hold with the same M, E .

We first verify the trace hypothesis of Lemma 10.2. For every smooth compactly supported test field ψ , clause (b) of Proposition 9.5 gives the rescaled fields a uniform pairing modulus on intervals $[a, 0]$. By clause (c), their pairings converge at every negative time to those of U . By clause (e), their pairings at time zero converge to those of the terminal slice $U(\cdot, 0)$, which is zero in L_{loc}^2 . Passing to the limit in the modulus therefore gives

$$\left| \int U(x, t) \cdot \psi(x) dx - \int U(x, 0) \cdot \psi(x) dx \right| \leq C_\psi (|t| + |t|^{1/3}) \longrightarrow 0 \quad (t \uparrow 0).$$

Thus U has the required weakly continuous local trace and that trace is zero. The uniform $L_t^\infty L_x^3$ bound in clause (c) and density of smooth compactly supported tests in $L^{3/2}$ extend this pairing continuity to weak continuity in local L^3 . Lemma 10.2 now gives a continuous representative of $\omega = \text{curl } U$ on $\{|x| > R_3\} \times (-2, 0]$, locally in $W_2^{2,1}$ up to the top, with a uniform inequality

$$|\partial_t \omega - \Delta \omega| \leq c_{\text{ext}}(|\omega| + |\nabla \omega|)$$

and terminal value $\omega(\cdot, 0) = 0$.

Backward uniqueness outside a ball. Fix a unit vector e and choose $O \in SO(3)$ with $Oe_3 = e$. For $(y, s) \in \mathbb{R}_+^3 \times [0, 1)$ set

$$x = R_3 e + \sqrt{2} O y, \quad t = -2s, \quad w(y, s) = A_\omega^{-1} \omega(R_3 e + \sqrt{2} O y, -2s),$$

where $A_\omega = \max\{1, \|\omega\|_{L^\infty(\{|x| > R_3\} \times (-2, 0))}\}$. Then $x \cdot e = R_3 + \sqrt{2} y_3 > R_3$ and the map sends $\{x \cdot e > R_3\} \times (-2, 0]$ onto $\mathbb{R}_+^3 \times [0, 1)$. The weak chain rule gives

$$(\partial_s + \Delta_y)w = -2A_\omega^{-1}(\partial_t \omega - \Delta_x \omega)(R_3 e + \sqrt{2} O y, -2s), \quad |\nabla_x \omega| / A_\omega = 2^{-1/2} |\nabla_y w|.$$

Thus $|\partial_s w + \Delta_y w| \leq c_1(|w| + |\nabla_y w|)$ with $c_1 = 2c_{\text{ext}}$, after increasing it if necessary. The zero terminal trace gives $w(y, 0) = 0$; the exterior bound gives $|w| \leq 1 \leq e^0$; and the local $W_2^{2,1}$ estimates give all required L^2 bounds on bounded subsets. Theorem 6.7 yields $w = 0$ on $\mathbb{R}_+^3 \times (0, 1)$. As e was arbitrary, choosing $e = x/|x|$ shows

$$\omega = 0 \quad \text{on } (\mathbb{R}^3 \setminus \overline{B_{R_3}}) \times (-2, 0). \quad (10.4)$$

Regular time slices and unique continuation. By Proposition 9.5(d), the limit is suitable on every bounded subcylinder of $\mathbb{R}^3 \times (-2, 0)$. Every compactly supported test in the full open domain $\mathbb{R}^3 \times (-2, 0)$ is supported in one such bounded cylinder, so these local suitability statements give the global CKN suitability predicate. Clause (c) gives $U \in L^\infty((-2, 0); L^3(\mathbb{R}^3))$, and clause (d) gives the global pressure identity $q = P[U \otimes U]$. Boundedness of the double Riesz transforms on $L^{3/2}(\mathbb{R}^3)$ therefore gives $q \in L^\infty((-2, 0); L^{3/2}(\mathbb{R}^3))$, the required global pressure integrability; the local gradient and energy conditions are those in (d). Thus the limit lies in the exponent-three CKN class on the full open domain. CKN Theorem C therefore says that the singular set S there has zero one-dimensional parabolic Hausdorff measure. By Lemma 8.6, for almost every $t_* \in (-2, 0)$ there is an interval I about t_* such that $\overline{B_{4R_3}} \times I$ consists of regular points. Shrink I so that its closure is compactly contained in $(-2, 0)$.

We record the small-scale input at each regular point. If z is regular, choose $r_0 > 0$ so that $\overline{Q_{r_0}^-(z)}$ lies in its regularity neighborhood and in the open solution domain. By Proposition 9.5(d), the limit is suitable with exponent 3 there; by clauses (c)–(d), its pressure is the global associated pressure $q = P[U \otimes U]$. Lemma 10.3 applies directly and gives

$$r^{-2} \iint_{Q_r^-(z)} (|U|^3 + |q|^{3/2}) \longrightarrow 0.$$

For small r , Lemma 8.1 also gives a Hölder representative and an explicit bound on a smaller cylinder. Since the closed cylinder is inside the open domain, Theorem 8.14 applies there after parabolic rescaling. Cover the compact cylinder $\overline{B_{4R_3}} \times \bar{I}$ by finitely many of these smaller regular neighborhoods. The resulting vorticity representatives glue; on every smaller cylinder they satisfy $W_2^{2,1}$ bounds, are continuous, and obey

$$|\partial_t \omega - \Delta \omega| \leq c_I(|\omega| + |\nabla \omega|). \quad (10.5)$$

The constants and the bound $|\omega| \leq M_I$ are uniform on this compact cylinder.

Define

$$V = \{x \in B_{4R_3} : \text{for some } r > 0, \omega = 0 \text{ on } B_r(x) \times I\}.$$

This set is relatively open. It is nonempty by (10.4), since the exterior zero region intersects B_{4R_3} . We show it is relatively closed. Let $x \in \overline{V} \cap B_{4R_3}$. Choose $R' > 0$ with $\overline{B_{2R'}(x)} \subset B_{4R_3}$, and

choose $x_* \in V$ so close to x that $|x - x_*| < R'/2$. By the definition of V , there is $r > 0$ such that $\omega = 0$ on $B_r(x_*) \times I$. Fix any $t_* \in I$. Choose $T' > 0$ with $T' < t_* - \inf I$, and choose $r_0 > 0$ so small that $r_0 < \min\{r, R'\}$ and $r_0^2 < T'$. On $B_{R'}(x_*) \times (t_* - T', t_*]$, the vorticity is continuous, belongs to $W_2^{2,1}$ below the top, is bounded by M_I , and satisfies (10.5). Reverse time by setting $w(y, s) = \omega(x_* + y, t_* - s)$ for $y \in B_{R'}$ and $0 \leq s < T'$. Then

$$|\partial_s w + \Delta_y w| \leq c_I(|w| + |\nabla_y w|), \quad w = 0 \text{ on } B_{r_0}(0) \times (0, r_0^2).$$

Lemma 10.1 gives the pointwise infinite-order bound required by Theorem 5.2, with positive constants $C_k = \max\{1, M_I r_0^{-k}\}$; its $W_2^{2,1}$ hypotheses follow from the local vorticity bounds. Theorem 5.2 implies $\omega(\cdot, t_*) = 0$ on $B_{R'}(x_*)$, which contains a neighborhood of x . This argument applies to every $t_* \in I$, with the same R' and the same spatial neighborhood of x . Hence $x \in V$. Connectedness of B_{4R_3} gives $V = B_{4R_3}$, and therefore $\omega(\cdot, t_*) = 0$ throughout B_{4R_3} for every $t_* \in I$. Together with (10.4), this proves $\omega(\cdot, t_*) = 0$ on $\mathbb{R}^3$ for every time t_* in the full-measure set supplied by Theorem C and Lemma 8.6.

For almost every such t , $U(\cdot, t) \in L^3(\mathbb{R}^3)$, $\operatorname{div} U(\cdot, t) = 0$, and $\operatorname{curl} U(\cdot, t) = \omega(\cdot, t) = 0$ in spatial distributions. Lemma 10.4 gives $U(\cdot, t) = 0$ for almost every $t \in (-2, 0)$, hence $U = 0$ almost everywhere on $\mathbb{R}^3 \times (-2, 0)$.

This contradicts Proposition 9.5(f): the bad-point lower bound (8.3) gives a lower bound $\varepsilon_0/8$ for every rescaled integral on $Q_1^-(0)$, whereas the last clause of (f), with $T = 2 > 1$, says those same integrals tend to zero. The contradiction proves that u has a parabolically Hölder representative on K , which is the conclusion of Theorem 3.12. $\square$

Remark 10.5 (Departure from [4]). The proof follows the blow-up, exterior backward-uniqueness, and interior unique-continuation structure of Theorem 1.4 in [4, pp. 222–229]. The endpoint good-point criterion is stated on open-top past cylinders, and the pressure split is the whole-space Riesz split of Lemma 9.3; the endpoint lower bound and pressure convergence are supplied by Proposition 9.5. We give the terminal vorticity extension and the infinite-order vanishing step explicitly, and use CKN Theorem C for the regular-time projection. These choices make the top-face trace and both applications of unique continuation explicit in the regularity classes used here.

11 Proof of the global theorem

The pressure needed for the local theorem is the canonical whole-space pressure of the velocity tensor. Its mixed-norm bound follows directly from the slice-wise Riesz estimate.

Lemma 11.1 (Pressure bound under the critical velocity hypothesis). *Suppose (u, Du) satisfies the hypotheses of Theorem 3.13. Let $p_* = P[u \otimes u]$ be the pressure witness selected in the proof of Theorem 3.7. Then $p_* \in L^{5/3}(\mathbb{R}^3 \times (0, T))$, the identity (S3) of Definition 3.1 holds with $p = p_*$ and $f = 0$, and $p_* \in L^\infty((0, T); L^{3/2}(\mathbb{R}^3))$. There is a constant C depending only on the dimension and the $L^{3/2}$ double-Riesz-transform bound such that*

$$\operatorname{ess\,sup}_{t \in (0, T)} \|p_*(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq C \left(\operatorname{ess\,sup}_{t \in (0, T)} \|u(\cdot, t)\|_{L^3(\mathbb{R}^3)} \right)^2.$$

Proof. The proof of Theorem 3.7 takes $F_{ij} = u_i u_j$ and sets $p_* = P[F]$. [2, Lemma 8.4] gives $F \in L^{5/3}(\mathbb{R}^3 \times (0, T))$; the proof there uses [2, Lemma 8.2] to verify the gradient part of (S3). Thus

the existential theorem's chosen witness is exactly $P[u \otimes u]$, with the stated global integrability and (S3) identity.

For almost every t , the hypothesis of Theorem 3.13 gives $u(\cdot, t) \in L^3(\mathbb{R}^3)$. The slice estimate in Definition 7.1, applied with $r = 3/2$, yields

$$\|p_*(\cdot, t)\|_{L^{3/2}} \leq C_{3/2} \sum_{i,j=1}^3 \|u_i(\cdot, t)u_j(\cdot, t)\|_{L^{3/2}} \leq 9C_{3/2} \|u(\cdot, t)\|_{L^3}^2.$$

Taking the essential supremum proves the estimate. The pressure representative in Definition 7.1 is jointly measurable, so it also has the measurability required below. $\square$

Proof of Theorem 3.13. Fix $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times (0, T)$. Choose $R > 0$ so small that

$$R^2 < \min \left\{ \frac{8t_0}{7}, 8(T - t_0) \right\}, \quad t_1 = t_0 + \frac{R^2}{8}.$$

Then $0 < t_0 - 7R^2/8 < t_1 < T$. Set $I_R = (t_0 - 7R^2/8, t_1)$ and define, on $Q = B_1 \times (-1, 0)$,

$$\begin{aligned} \tilde{u}(y, s) &= Ru(x_0 + Ry, t_1 + R^2s), \\ \widetilde{Du}_{ij}(y, s) &= R^2 Du_{ij}(x_0 + Ry, t_1 + R^2s), \\ \tilde{p}(y, s) &= R^2 p_*(x_0 + Ry, t_1 + R^2s). \end{aligned}$$

The affine map $(y, s) \mapsto (x_0 + Ry, t_1 + R^2s)$ sends Q onto $B_R(x_0) \times I_R$, whose closure is contained in $\mathbb{R}^3 \times (0, T)$. Composition with this map preserves measurability.

The change of variables gives

$$\begin{aligned} \operatorname{ess\,sup}_{s \in (-1, 0)} \int_{B_1} |\tilde{u}(y, s)|^2 dy &\leq R^{-1} \operatorname{ess\,sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty, \\ \iint_Q |\tilde{u}|^2 dy ds &= R^{-3} \iint_{B_R(x_0) \times I_R} |u|^2 dx dt < \infty, \\ \iint_Q |\widetilde{Du}|^2 dy ds &= R^{-1} \iint_{B_R(x_0) \times I_R} |Du|^2 dx dt < \infty, \\ \iint_Q |\tilde{p}|^{3/2} dy ds &= R^{-2} \iint_{B_R(x_0) \times I_R} |p_*|^{3/2} dx dt < \infty. \end{aligned}$$

The first three right-hand sides are finite by (LH1) of Definition 3.3 and the last is finite by Lemma 11.1. Also,

$$\operatorname{ess\,sup}_{s \in (-1, 0)} \int_{B_1} |\tilde{u}(y, s)|^3 dy \leq \operatorname{ess\,sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty.$$

For almost every $s \in (-1, 0)$, the physical time $t = t_1 + R^2s$ is a slice on which $Du(\cdot, t)$ is the weak spatial gradient of $u(\cdot, t)$. If $\psi \in C_c^\infty(B_1)$ and $\psi_R(x) = \psi((x - x_0)/R)$, that weak-gradient identity and the change of variables give, componentwise,

$$\int_{B_1} \tilde{u}_i(y, s) \partial_j \psi(y) dy = -R^{-1} \int_{B_R(x_0)} Du_{ij}(x, t) \psi_R(x) dx = - \int_{B_1} \widetilde{Du}_{ij}(y, s) \psi(y) dy.$$

Thus $\widetilde{Du}$ is the weak gradient on almost every time slice.

It remains to check (S2) and (S3), including the test scaling. For $\eta \in C_c^\infty(Q)$ scalar and $\varphi \in C_c^\infty(Q; \mathbb{R}^3)$, put

$$\eta_R(x, t) = \eta\left(\frac{x - x_0}{R}, \frac{t - t_1}{R^2}\right), \quad \varphi_R(x, t) = \varphi\left(\frac{x - x_0}{R}, \frac{t - t_1}{R^2}\right).$$

These are smooth compactly supported tests in $\mathbb{R}^3 \times (0, T)$. Since $dy ds = R^{-5} dx dt$ and $\nabla_y \eta = R \nabla_x \eta_R$ under this substitution, (LH1) gives

$$\iint_Q \tilde{u} \cdot \nabla_y \eta dy ds = R^{-3} \iint_{B_R(x_0) \times I_R} u \cdot \nabla_x \eta_R dx dt = 0,$$

which is (S2). For the vector test, $\partial_t \varphi_R = R^{-2} \partial_s \varphi$ and $\partial_{x_j}(\varphi_R)_i = R^{-1} \partial_{y_j} \varphi_i$; all four terms of (S3) consequently have the same scaling. In full,

$$\begin{aligned} & \iint_Q \left[-\tilde{u} \cdot \partial_s \varphi - \tilde{u}_i \tilde{u}_j \partial_{y_j} \varphi_i + \tilde{D} u_{ij} \partial_{y_j} \varphi_i - \tilde{p} \operatorname{div}_y \varphi \right] dy ds \\ &= R^{-2} \iint_{B_R(x_0) \times I_R} \left[-u \cdot \partial_t \varphi_R - u_i u_j \partial_{x_j} (\varphi_R)_i + D u_{ij} \partial_{x_j} (\varphi_R)_i - p_* \operatorname{div}_x \varphi_R \right] dx dt = 0, \end{aligned}$$

by (S3) for p_* from Lemma 11.1. This verifies every hypothesis of Theorem 3.12; its statement requires (S2) and (S3), and does not require the local energy inequality (S4).

The local theorem supplies a parabolically Hölder representative of $\tilde{u}$ on $\overline{B_{1/2} \times (-1/4, 0)}$. The point z_0 has scaled coordinates $(0, -1/8)$. In physical variables the closed half-cylinder is

$$\overline{B_{R/2}(x_0)} \times [t_0 - R^2/8, t_1].$$

It is contained in $\mathbb{R}^3 \times (0, T)$, and the representative equals u almost everywhere on its interior. Thus the hypotheses of Lemma 8.8 hold. The open cylinder $B_{1/4} \times (-3/16, -1/16)$ contains that point and has closure inside the half-cylinder above. Pulling its representative back by the inverse scaling therefore gives a Hölder representative of u on the open cylinder N below, since the parabolic metric scales by the factor R under this map:

$$N = B_{R/4}(x_0) \times (t_0 - R^2/16, t_0 + R^2/16) \subset \mathbb{R}^3 \times (0, T).$$

The conclusion of Lemma 8.8 is the open-neighborhood condition in the definition of a regular point. Since z_0 was arbitrary, every point of $\mathbb{R}^3 \times (0, T)$ is regular. $\square$

Remark 11.2 (Departure from [4]). The pressure estimate (3.4) of [4], printed p. 221, is obtained there from the pressure equation after invoking coercive Stokes estimates. Here the proof of Theorem 3.7 selects the canonical pressure $P[u \otimes u]$, and the slice-wise double-Riesz bound gives (3.4) directly from the $L_t^\infty L_x^3$ hypothesis. The weak continuity asserted in (3.1) is not needed for this regularity conclusion. To obtain the open-neighborhood condition in the definition of a regular point, the local theorem is applied at the later center $(x_0, t_0 + R^2/8)$; this puts z_0 strictly inside its Hölder half-cylinder, as required by the shifted-cylinder argument of Lemma 8.8. We use the coherent backward cylinder $B_1 \times (-1, 0)$ in Theorem 3.12: the printed statement of ESS Theorem 1.4 on printed p. 214 says $B_1 \times (0, 1)$, although its displayed hypotheses and conclusion use negative times. This proof establishes the regularity conclusion stated as Theorem 3.13.

Part V. L^5 bounds and uniqueness

A finite $L_t^\infty L_x^3$ bound controls the solution at positive times, but it does not itself give space-time integrability near the initial time. We first recover an L^3 trace and construct a solution from that trace whose L^5 norm is small on a short interval. A comparison argument transfers that estimate to the given weak solution. Away from time zero, the local regularity theorem and the pressure tail give a global velocity bound, which turns the finite energy into the remaining L^5 estimate.

12 The trace and heat flow

Write $Q_\tau = \mathbb{R}^3 \times (0, \tau)$ and let $S(t) = e^{t\Delta}$ be the heat semigroup with the Fourier convention fixed in [2, Section 9].

Lemma 12.1 (The initial trace is in L^3). *Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$, and suppose*

$$M^3 := \operatorname{ess\,sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty.$$

Then $a \in L^3(\mathbb{R}^3; \mathbb{R}^3)$ and $\|a\|_{L^3} \leq M$.

Proof. The set of times at which $\|u(t)\|_3 \leq M$ has full measure. Choose $t_n \downarrow 0$ in this set. The strong L^2 initial trace in Definition 3.3 gives $u(t_n) \rightarrow a$ in L^2 , hence in measure on $\mathbb{R}^3$. Passing to a subsequence, $u(t_{n_k}) \rightarrow a$ almost everywhere. Fatou’s lemma gives $\int_{\mathbb{R}^3} |a|^3 dx \leq \liminf_{k \rightarrow \infty} \int_{\mathbb{R}^3} |u(x, t_{n_k})|^3 dx \leq M^3$, so $a \in L^3$ and $\|a\|_{L^3} \leq M$. Its membership in J is already part of the Leray–Hopf hypothesis. $\square$

Lemma 12.2 (Critical heat estimate). *For $b \in L^3(\mathbb{R}^3; \mathbb{R}^3)$ and $\tau > 0$,*

$$\|S(\cdot)b\|_{L^5(Q_\tau)} \leq C\|b\|_{L^3}, \quad (12.1)$$

where C depends only on the dimension. If in addition $b \in L^2$, then

$$\|S(\cdot)b\|_{L^4(Q_\tau)} \leq C(\|b\|_{L^2} + \|b\|_{L^3}). \quad (12.2)$$

The constants do not depend on τ .

Proof. First take $b \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$. The Gaussian heat kernel gives a smooth solution $h = S(t)b$ with enough decay to integrate by parts in space. Testing $\partial_t h - \Delta h = 0$ against $|h|h$ yields

$$\frac{1}{3} \frac{d}{dt} \|h(t)\|_3^3 + c \int_{\mathbb{R}^3} |h| |\nabla h|^2 dx + c \int_{\mathbb{R}^3} |\nabla(|h|^{3/2})|^2 dx \leq 0,$$

with a numerical $c > 0$. At zeros of h , this identity follows first with $|h|$ replaced by $(|h|^2 + \eta^2)^{1/2}$ and then by letting $\eta \downarrow 0$; the nonnegative terms pass by Fatou’s lemma. There is no boundary contribution, since one may first insert a cutoff equal to one on B_R and use the Gaussian decay as $R \rightarrow \infty$. With $g = |h|^{3/2}$, integration in time gives

$$\operatorname{ess\,sup}_{0 < t < \tau} \|g(t)\|_2^2 + \int_{Q_\tau} |\nabla g|^2 dx dt \leq C\|b\|_3^3.$$

The multiplicative Sobolev inequality proved in the argument for [2, Lemma 8.4] gives

$$\int_{Q_\tau} |g|^{10/3} dx dt \leq C \left(\operatorname{ess\,sup}_{0 < t < \tau} \|g(t)\|_2 \right)^{4/3} \int_{Q_\tau} |\nabla g|^2 dx dt \leq C \|b\|_3^5,$$

which is (12.1). Approximation in L^3 extends the estimate to all $b \in L^3$.

For the second estimate, the heat energy identity and L^2 contraction give

$$\operatorname{ess\,sup}_{0 < t < \tau} \|h(t)\|_2 + \|\nabla h\|_{L^2(Q_\tau)} \leq C \|b\|_2.$$

The spatial inequality in [2, Lemma 8.4], applied at almost every time, therefore gives $\|h\|_{L^{10/3}(Q_\tau)} \leq C \|b\|_2$. Interpolating this bound with (12.1) yields

$$\|h\|_{L^4(Q_\tau)} \leq \|h\|_{L^{10/3}(Q_\tau)}^{1/2} \|h\|_{L^5(Q_\tau)}^{1/2} \leq C \|b\|_2^{1/2} \|b\|_3^{1/2} \leq C (\|b\|_2 + \|b\|_3),$$

which proves (12.2). The interpolation inequality is on the product measure space Q_τ and its constant is one. $\square$

Lemma 12.3 (Small heat orbit). *If $a \in L^2(\mathbb{R}^3; \mathbb{R}^3) \cap L^3(\mathbb{R}^3; \mathbb{R}^3)$, then*

$$\|S(\cdot)a\|_{L^5(Q_\tau)} + \|S(\cdot)a\|_{L^4(Q_\tau)} \longrightarrow 0 \quad (\tau \downarrow 0).$$

Proof. Choose $a_\rho \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ with $a_\rho \rightarrow a$ in both L^2 and L^3 . Such a sequence is obtained by first cutting off outside large balls and then convolving with a smooth approximate identity; truncation and translation continuity give convergence in each of the two norms. By Lemma 12.2,

$$\|S(\cdot)(a - a_\rho)\|_{L^5(Q_\tau)} + \|S(\cdot)(a - a_\rho)\|_{L^4(Q_\tau)} \leq C (\|a - a_\rho\|_2 + \|a - a_\rho\|_3)$$

uniformly in τ . For fixed ρ , the Gaussian kernel and Young's convolution inequality give, for $t > 0$,

$$\|S(t)a_\rho\|_5 \leq Ct^{-3/40} \|a_\rho\|_4, \quad \|S(t)a_\rho\|_4 \leq Ct^{-1/8} \|a_\rho\|_3.$$

Integrating the fifth and fourth powers in time gives

$$\|S(\cdot)a_\rho\|_{L^5(Q_\tau)} + \|S(\cdot)a_\rho\|_{L^4(Q_\tau)} \leq C\tau^{1/8} (\|a_\rho\|_4 + \|a_\rho\|_3).$$

Given $\epsilon > 0$, choose ρ so that the approximation error is below $\epsilon/2$, then choose τ so that the smooth-data term is below $\epsilon/2$. This proves the limit. The pointwise heat bounds are the cases $(s, s_1) = (5, 4)$ and $(4, 3)$ of (7.3) in [4, printed p. 243]; their time integrals above are computed directly, without using the general mixed-norm assertion (7.4). $\square$

13 The forced Stokes response

The fixed point needs an estimate that controls both the critical L^5 norm and an L^4 norm used to keep the quadratic forcing in L^2 . The pressure in this linear equation must be included in the endpoint energy test; it is controlled by the same Riesz bounds as the force.

Lemma 13.1 (Zero-data Stokes estimate). *Let $\tau > 0$ and let $F \in L^{5/2}(Q_\tau; \mathbb{R}^{3 \times 3}) \cap L^2(Q_\tau; \mathbb{R}^{3 \times 3})$. There is a unique energy-class pair (z, q) with zero initial datum satisfying, in distributions,*

$$\partial_t z - \Delta z = \operatorname{div} F - \nabla q, \quad \operatorname{div} z = 0, \quad z(0) = 0,$$

where $(\operatorname{div} F)_j = \sum_i \partial_i F_{ij}$ and $q = -P[F]$. It has

$$z \in C([0, \tau]; L^2) \cap L^2(0, \tau; H^1) \cap C([0, \tau]; L^3) \cap L^5(Q_\tau) \cap L^4(Q_\tau), \quad q \in L^2(Q_\tau) \cap L^{5/2}(Q_\tau),$$

and $\partial_t z \in L^2(0, \tau; V')$, where $V = \{\phi \in H^1(\mathbb{R}^3; \mathbb{R}^3) : \operatorname{div} \phi = 0\}$ and V' is its dual on the divergence-free L^2 space. Moreover

$$\|z\|_{L^\infty(0, \tau; L^3)} + \|z\|_{L^5(Q_\tau)} \leq C\|F\|_{L^{5/2}(Q_\tau)}, \quad (13.1)$$

$$\|z\|_{L^4(Q_\tau)} \leq C(\|F\|_{L^{5/2}(Q_\tau)} + \|F\|_{L^2(Q_\tau)}), \quad (13.2)$$

$$\|z\|_{L^\infty(0, \tau; L^2)} + \|\nabla z\|_{L^2(Q_\tau)} \leq C\|F\|_{L^2(Q_\tau)}. \quad (13.3)$$

The constants depend only on dimension and the pressure-operator bounds at exponents 2 and 5/2, and are independent of τ .

Proof. For smooth tensor data, the Fourier formula in [2, (168)] gives the divergence-free evolution $z(t) = \int_0^t K(t-s)F(s)ds$. The pressure is $q = -P[F]$: the identity $-\Delta P[F] = \operatorname{div} \operatorname{div} F$ gives $\Delta q = \operatorname{div} \operatorname{div} F$, which is the divergence of the displayed Stokes equation. The L^2 energy estimate follows by pairing the projected equation with z in the divergence-free energy space, where the Leray projection is orthogonal and the pressure is absent. Fourier truncation justifies this pairing for smooth data. Cauchy–Schwarz and Young’s inequality give (13.3). The equation then gives the asserted $L^2(0, \tau; V')$ bound for $\partial_t z$.

For the critical estimate, test instead with $|z|z$. First take F smooth and compactly supported in space and time. The Riesz kernel gives $q(x, t) = O(|x|^{-3})$ and $\nabla q(x, t) = O(|x|^{-4})$ outside a fixed ball; the heat representation then gives $z(x, t) = O(|x|^{-4})$ there on a finite time interval. These bounds make all cutoff errors tend to zero as the spatial cutoff radius tends to infinity. A smooth regularization at $z = 0$ justifies the chain rule. The diffusion term controls $|z||\nabla z|^2$ and $|\nabla(|z|^{3/2})|^2$. Integration by parts in the force and pressure terms gives

$$\frac{1}{3} \frac{d}{dt} \|z(t)\|_3^3 + c \int_{\mathbb{R}^3} |z| |\nabla z|^2 + c \int_{\mathbb{R}^3} |\nabla(|z|^{3/2})|^2 \leq C \int_{\mathbb{R}^3} (|F|^2 + |q|^2) |z| dx,$$

with numerical constants $c, C > 0$. Indeed, $|\nabla(|z|z)| \leq 2|z||\nabla z|$ and $|\operatorname{div}(|z|z)| \leq 2|z||\nabla z|$; Cauchy–Schwarz absorbs half of the weighted diffusion term and leaves the right side displayed. Cutoff errors vanish for smooth approximants by their spatial decay. Definition 7.1 gives $\|q\|_{L^{5/2}(Q_\tau)} \leq C\|F\|_{L^{5/2}(Q_\tau)}$.

Let $Z = \|z\|_{L^5(Q_\tau)}$, $A = \operatorname{ess\,sup}_{0 < t < \tau} \|z(t)\|_3^3$, and $D = \int_{Q_\tau} |\nabla(|z|^{3/2})|^2$. The zero initial trace and the last differential inequality imply

$$A + D \leq C(\|F\|_{5/2}^2 + \|q\|_{5/2}^2)Z \leq C\|F\|_{5/2}^2 Z.$$

The multiplicative inequality used in Lemma 12.2, now with $g = |z|^{3/2}$, gives

$$Z^5 \leq CA^{2/3}D \leq C(A + D)^{5/3} \leq C\|F\|_{5/2}^{10/3} Z^{5/3}.$$

If $Z = 0$ the desired estimate holds; otherwise division and taking the power $3/10$ give $Z \leq C\|F\|_{5/2}$. The same differential inequality now bounds $A^{1/3}$ by $C\|F\|_{5/2}$, proving (13.1). These estimates first hold for smooth data and solutions.

To pass to general F , choose smooth tensor fields converging to F in both L^2 and $L^{5/2}$. The L^2 estimate applied to differences makes the responses Cauchy in $C_t L_x^2$ and $L_t^2 H_x^1$; the L^5 and L^4 estimates make them Cauchy in the corresponding space-time norms. Applying the $L_t^\infty L_x^3$ estimate to differences and using continuity in L^3 for smooth responses makes them Cauchy in $C_t L_x^3$, so the limit has zero strong L^3 trace. The pressure sequence converges in $L^2 \cap L^{5/2}$ by Definition 7.1. Passing to the limit in the equation against compactly supported tests proves the distributional equation; the bounds pass by completeness and weak lower semicontinuity. The energy-space solution is unique: the difference solves the homogeneous heat equation with zero data, and its L^2 energy identity forces it to vanish. The L^4 bound follows by interpolating the spatial energy estimate, which gives $\|z\|_{10/3} \leq C\|F\|_2$, with (13.1); then $\|z\|_4 \leq C\|F\|_2^{1/2}\|F\|_{5/2}^{1/2} \leq C(\|F\|_2 + \|F\|_{5/2})$. This proves the lemma. Its endpoint estimate is the special $L^{5/2} \rightarrow L^5$ Stokes argument in Theorem 7.3 of [4, printed pp. 244–246], with the pressure term and approximation limit written out here. $\square$

Remark 13.2 (Departure from [4]). The proof of Theorem 7.3 in [4, printed pp. 244–246] supplies the same critical estimate after testing with $|z|z$. Here the zero-data estimate is isolated as a linear response, the pressure is fixed as $-P[F]$ under Definition 7.1, and the difference estimates needed for completion are recorded explicitly. No general mixed-norm estimate from Lemma 7.1 is used.

14 A solution near the initial time

For $U, V \in L^5(Q_\tau) \cap L^4(Q_\tau)$, let $\mathcal{B}(U, V)$ be the zero-data Stokes response of the tensor $-U \otimes V$. In this definition we use the bilinear extension of Lemma 13.1; it is well-defined because $U \otimes V \in L^{5/2} \cap L^2$ by Hölder's inequality. Write $X_\tau = L^5(Q_\tau) \cap L^4(Q_\tau)$ with norm $\|U\|_{X_\tau} = \|U\|_5 + \|U\|_4$. The lemma gives a constant K , independent of τ , such that

$$\|\mathcal{B}(U, V)\|_{X_\tau} \leq K\|U\|_{X_\tau}\|V\|_{X_\tau}, \quad \|\mathcal{B}(U, U) - \mathcal{B}(V, V)\|_{X_\tau} \leq K(\|U\|_{X_\tau} + \|V\|_{X_\tau})\|U - V\|_{X_\tau}. \quad (14.1)$$

Proposition 14.1 (Short-time solution from an L^3 trace). *For every $a \in L^2(\mathbb{R}^3; \mathbb{R}^3) \cap L^3(\mathbb{R}^3; \mathbb{R}^3) \cap J$, there is $\tau > 0$ and a Leray–Hopf solution (U, DU) on $\mathbb{R}^3 \times [0, \tau]$ with datum a such that $U \in L^5(Q_\tau) \cap L^4(Q_\tau)$. Moreover $U(t) \rightarrow a$ strongly in L^3 as $t \downarrow 0$, and the associated pressure $P[U \otimes U]$ belongs to $L^2(Q_\tau) \cap L^{5/2}(Q_\tau)$.*

Proof. Let $h(t) = S(t)a$. Lemma 12.3 permits a choice of $\tau > 0$ with $H := \|h\|_{X_\tau} < 1/(8K)$. If $H = 0$, take $U = 0$; otherwise the map $\Phi(U) = h + \mathcal{B}(U, U)$ maps the closed ball of radius $2H$ in X_τ into itself and contracts it. In fact, (14.1) gives $\|\Phi(U) - h\|_{X_\tau} \leq 4KH^2 \leq H/2$ on that ball and Lipschitz constant at most $4KH < 1/2$. The contraction theorem gives a unique fixed point U in that ball. The heat flow and every Stokes response are divergence free, so $\operatorname{div} U = 0$ in distributions.

Set $U_0 = h$ and $U_{n+1} = h + \mathcal{B}(U_n, U_n)$. The contraction estimate gives convergence to U in X_τ . Since $U_n \rightarrow U$ in L^4 , the tensors $U_n \otimes U_n$ converge to $U \otimes U$ in L^2 ; the L^5 convergence also gives convergence in $L^{5/2}$. Thus the forcing tensors converge in both data spaces of Lemma 13.1. The associated Stokes responses converge in $C_t L_x^2 \cap L_t^2 H_x^1$, so $U \in C([0, \tau]; L^2) \cap L^2(0, \tau; H^1)$ with $U(0) = a$ strongly in L^2 . Their $L_t^\infty L_x^3$ difference estimate gives convergence in $C_t L_x^3$ as well. The heat flow is strongly continuous in L^3 , and the nonlinear response tends to zero in L^3 at

time zero; hence $U(t) \rightarrow a$ strongly in L^3 . The pressure convergence from Lemma 13.1 gives $P[U \otimes U] \in L^2 \cap L^{5/2}$, and passing to the limit in the mild equation gives

$$\partial_t U - \Delta U + \operatorname{div}(U \otimes U) + \nabla P[U \otimes U] = 0$$

in distributions. This is the unregularized Navier–Stokes equation; the Fourier multiplier supplies only the linear Stokes response, and the tensor in the fixed point is the actual $U \otimes U$.

It remains to verify the Leray–Hopf clauses, including the energy inequality. The response energy estimate and $U \otimes U \in L^2$ give $\partial_t U \in L^2(0, \tau; V')$. The Hilbert-space energy identity for $L^2(0, \tau; H^1)$ fields with time derivative in $L^2(0, \tau; V')$ follows by time convolution: on each compact subinterval convolve in time, pair the equation with the convolved field, and pass in the $L^2 H^1$ and $L^2 V'$ norms; continuity of U in L^2 supplies the endpoint values. The nonlinear pairing is integrable since $U \otimes U \in L^2$ and $DU \in L^2$. Its spatial integral vanishes: for a cutoff χ_R equal to one on B_R , supported in B_{2R} , and with $|\nabla \chi_R| \leq C/R$, integration by parts gives a convection error supported on the annulus $A_R = B_{2R} \setminus B_R$ and bounded by $CR^{-1} \int_{Q_\tau \cap A_R} |U|^3$. Hölder's inequality and $|Q_\tau \cap A_R| \leq C\tau R^3$ bound this by $C\tau^{1/4} R^{-1/4} \|U\|_{L^4(Q_\tau)}^3$, which tends to zero. The diffusion cutoff error is bounded by $CR^{-1} \|U\|_{L^2(Q_\tau)} \|DU\|_{L^2(Q_\tau)}$ and also tends to zero by the energy bounds. There is no pressure boundary term in this divergence-free pairing. Thus $\int (U \otimes U) : \nabla U = 0$. We obtain the energy equality

$$\frac{1}{2} \|U(t)\|_2^2 + \int_0^t \|DU(s)\|_2^2 ds = \frac{1}{2} \|a\|_2^2 \quad (0 \leq t \leq \tau),$$

first for almost every t and then for every t by L^2 continuity and absolute continuity of the dissipation integral. This implies (LH1), (LH2), and the stronger equality version of (LH4); the distributional equation gives (LH3), and the strong L^2 trace gives (LH5). The pressure term does not enter the energy pairing because we work in the divergence-free space V . This proves the proposition. $\square$

15 Weak–strong comparison

The short-time solution is a Leray–Hopf solution, so it can be compared with the given weak solution using only the latter's energy inequality. The comparison identity is obtained by mollifying the time slices of both solutions in space and removing the mollification; the L^5 integrability of one solution makes every nonlinear term integrable down to time zero, and (LH4) is used for the other solution only from zero.

Lemma 15.1 (Weak–strong uniqueness for L^5 solutions). *Let $T > 0$, and let (u, Du) and (v, Dv) be Leray–Hopf solutions on $\mathbb{R}^3 \times [0, T]$ with the same datum in J . If $u \in L^5(\mathbb{R}^3 \times (0, T))$, then $u = v$ almost everywhere on $\mathbb{R}^3 \times (0, T)$.*

Proof. By Theorem 3.7, each of u and v has an associated pressure in $L^{5/3}(Q_T)$ with which it satisfies the momentum equation against compactly supported smooth vector tests; the pairing of its slices with a fixed test field is continuous on $[0, T]$ and equals the pairing with a at $t = 0$. The spatial and temporal exponents of u are both 5, and $2/5 + 3/5 = 1$. The only nontrivial nonlinear integrability check is

$$\begin{aligned} \int_0^T \|v(t)\|_{10/3} \|Dv(t)\|_2 \|u(t)\|_5 dt &\leq C \int_0^T \|v(t)\|_2^{2/5} \|Dv(t)\|_2^{8/5} \|u(t)\|_5 dt \\ &\leq \epsilon \|Dv\|_{L^2(Q_T)}^2 + C_\epsilon \|v\|_{L_t^\infty L_x^2}^2 \|u\|_{L_t^5 L_x^5}^5 < \infty, \end{aligned}$$

where the first inequality is spatial interpolation between L^2 and L^6 , and the second is Young's inequality with exponents $5/4$ and 5 . It makes the convection terms $u \cdot (v \cdot \nabla)v$ and $v \cdot (u \cdot \nabla)u$ integrable on Q_T , down to time zero. The gradient pairing is integrable by the energy bounds, and the pressures pair integrably with both velocities, which lie in $L^{5/2}(Q_T)$ by interpolation.

We derive the cross-testing identity without any regularity of u beyond L^5 . Let $u_n(\cdot, t)$ and $v_n(\cdot, t)$ be the spatial mollifications of the slices at radius $1/(n+1)$, and let χ_n be a smooth cutoff equal to one on B_{n+1} with $|\nabla \chi_n| \leq C/(n+1)$. At a fixed point y , testing the momentum equation of u with the kernel $\rho_n(y - \cdot)e_k$ and a time cutoff shows that $t \mapsto u_n(y, t)_k$ is the primitive in time of an integrable tested flux, and likewise for v_n . The product rule for primitives then expresses $\int \chi_n v_n(t) \cdot u_n(t) dy - \int \chi_n v_n(0) \cdot u_n(0) dy$ as a space-time integral of mollified velocities, gradients, convection terms and pressures. The mollified slices are divergence free, so the pressure contributions only pair the mollified pressures with $\nabla \chi_n$. As $n \rightarrow \infty$, spatial mollification converges in the specific space-time L^p classes used for these terms, and the product estimates above and the energy bounds give convergence of the cross integrals; the terms carrying $\nabla \chi_n$ vanish by the cutoff derivative bound and the same integrability estimates. The initial pairing converges to $\|a\|_2^2$. For almost every $t \in (0, T)$ this gives

$$\int_{\mathbb{R}^3} v(t) \cdot u(t) dx - \|a\|_2^2 = - \int_0^t \int_{\mathbb{R}^3} (u \cdot (v \cdot \nabla)v + \nabla u : \nabla v) - \int_0^t \int_{\mathbb{R}^3} (v \cdot (u \cdot \nabla)u + \nabla v : \nabla u) .$$

No test by an arbitrary weak solution against itself is used: the identity pairs mollified slices, and the limits use only the integrability above. With $v = u$, the transport term $\int u \cdot (u \cdot \nabla)u dx$ vanishes for almost every time by the divergence-free transport identity: the Leray–Hopf slice bounds and $u \in L^5(Q_T)$ provide the integrability needed to justify it by cutoff and mollification. Thus the identity is the energy equality for u at almost every time. We use only (LH4) for v on $[0, t]$. Combining these gives, for almost every $t \in (0, T)$,

$$\frac{1}{2} \|w(t)\|_2^2 + \int_0^t \|Dw(s)\|_2^2 ds \leq \int_0^t \int_{\mathbb{R}^3} u \cdot (w \cdot \nabla)w dx ds, \quad w = v - u .$$

The transport identities $\int b \cdot \nabla(f \cdot g) dx = 0$ for divergence-free b , applied at almost every time with the integrability above, reduce the combination to this form. This is the zero-start energy comparison of [10, (8.12)], printed p. 174, where it is derived from Serrin regularity and their testing lemma (Theorem 8.17 and Lemma 8.18, printed pp. 173–174); the mollified identity above replaces both. The weak solution contributes only its energy inequality from zero.

For almost every s , interpolation and Hölder's inequality give

$$\begin{aligned} \left| \int_{\mathbb{R}^3} u \cdot (w \cdot \nabla)w dx \right| &\leq \|u(s)\|_5 \|w(s)\|_{10/3} \|Dw(s)\|_2 \\ &\leq C \|u(s)\|_5 \|w(s)\|_2^{2/5} \|Dw(s)\|_2^{8/5} \\ &\leq \frac{1}{2} \|Dw(s)\|_2^2 + C \|u(s)\|_5^5 \|w(s)\|_2^2 . \end{aligned}$$

The coefficient $s \mapsto \|u(s)\|_5^5$ is integrable on $(0, T)$. Substitution and Gronwall's inequality yield

$$\|w(t)\|_2^2 \leq \|w(0)\|_2^2 \exp \left(C \int_0^t \|u(s)\|_5^5 ds \right) = 0 .$$

The strong L^2 initial traces give $w(0) = 0$. Gronwall therefore proves $w(t) = 0$ in L^2 for almost every t , hence $u = v$ almost everywhere on the open cylinder. This proof does not use energy equality for v , and it does not use the positive-time regularity of u . $\square$

16 Positive times and the global conclusion

We now prove the proposed successor of Theorem 3.13. The only estimate still needed away from $t = 0$ is a velocity bound uniform in space. Local regularity gives such a bound on bounded spatial sets; at spatial infinity, the tails of the velocity and its canonical pressure make the rescaled local energy arbitrarily small at a fixed radius.

Lemma 16.1 (A global bound on positive-time slabs). *Under the hypotheses of Theorem 3.13, put $M^3 = \operatorname{ess\,sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx$. For every $0 < \delta < T$ there is*

$$B_\delta < \infty$$

depending on $\delta, T, M, \|a\|_2$, the absolute constants in Lemma 8.2, the pressure bound in Lemma 11.1, and the given solution through a finite collection of local regularity neighborhoods, such that

$$\operatorname{ess\,sup}_{(x, t) \in \mathbb{R}^3 \times (\delta, T)} |u(x, t)| \leq B_\delta.$$

Proof. Let $p_* = P[u \otimes u]$ be the canonical pressure selected by Theorem 3.7. Lemma 11.1 gives the momentum identity (S3), the pressure equation $\Delta p_* = -\operatorname{div} \operatorname{div}(u \otimes u)$, and the bound $p_* \in L^\infty((0, T); L^{3/2}(\mathbb{R}^3))$. We use this same pressure in the local energy equality, so the suitability pressure is identified with the canonical pressure rather than chosen separately on each cylinder. Fix $x_0 \in \mathbb{R}^3$, $t_0 \in (0, T]$, and $\rho > 0$ with $t_0 - \rho^2 > 0$. Define the translated and rescaled fields on $B_1 \times (-1, 0)$ by

$$\begin{aligned} u_\rho(y, s) &= \rho u(x_0 + \rho y, t_0 + \rho^2 s), \\ (Du)_\rho(y, s) &= \rho^2 Du(x_0 + \rho y, t_0 + \rho^2 s), \\ (p_*)_\rho(y, s) &= \rho^2 p_*(x_0 + \rho y, t_0 + \rho^2 s). \end{aligned}$$

By (LH1), these fields have the local energy bounds, weak gradient, and divergence condition required by Theorem 3.12. Lemma 11.1 supplies the pressure identity (S3) with this same canonical pressure, and the global $L_t^\infty L_x^3$ hypothesis supplies the remaining norm bound. These properties are preserved by the displayed change of variables, so the hypotheses of Theorem 3.12 hold for the rescaled fields. Applying Lemma 9.1 with inner radius $3/4$ proves that $(u_\rho, (Du)_\rho, (p_*)_\rho, 0)$ is suitable on $B_{3/4} \times (-1, 0)$. Scaling back, $(u, Du, p_*, 0)$ is suitable on $B_{3\rho/4}(x_0) \times (t_0 - \rho^2, t_0)$.

The function $|u|^3 + |p_*|^{3/2}$ is integrable on every finite slab, and its integral outside B_R tends to zero as $R \rightarrow \infty$.

Fix $r > 0$ with $r^2 < \delta/4$. Since $u \in L^\infty((0, T); L^3)$ and $p_* \in L^\infty((0, T); L^{3/2})$, the function $|u|^3 + |p_*|^{3/2}$ has an integrable spatial tail uniformly after integration over $(\delta/2, T)$. Choose R so large that for every x_0 with $|x_0| > R$,

$$r^{-2} \iint_{B_r(x_0) \times (t_0 - r^2, t_0)} (|u|^3 + |p_*|^{3/2}) dx dt < \varepsilon_0/8$$

whenever $\delta \leq t_0 \leq T$. Indeed all these cylinders lie in $(\delta/2, T)$, and their spatial sections lie outside B_{R-r} ; the integral is bounded by the tail integral over $(\delta/2, T) \times (\mathbb{R}^3 \setminus B_{R-r})$, which tends to zero. For each such t_0 , the preceding rescaled argument with outer scale $2r$ gives suitability on $B_{3r/2}(x_0) \times (t_0 - 4r^2, t_0)$, since $4r^2 < \delta$. This contains the smallness cylinder displayed above. Lemma 8.2 then bounds a representative by $2C_4/r$ on $B_{r/2}(x_0) \times (t_0 - r^2/4, t_0)$, with constants

independent of x_0 and t_0 . These cylinders cover the part of $\{|x| > R+r\} \times (\delta, T)$ with $t \leq T - r^2/8$ by taking $t_0 = t + r^2/8$; the remaining terminal strip is covered by the same estimate with $t_0 = T$. Hence the essential supremum on the unbounded spatial part is at most $2C_4/r$.

On the bounded spatial part, Theorem 3.13 gives a regular point at every (x, t) with $\delta \leq t < T$. The local endpoint theorem, Theorem 3.12, also gives a one-sided Hölder representative at every (x, T) by applying it on a cylinder ending at T . These neighborhoods cover the compact set $\overline{B}_{R+r} \times [\delta, T]$; a finite subcover gives a finite bound there. Taking the larger of this bound and $2C_4/r$ proves the lemma. The dependence is through M , the pressure bound and energy bounds on the slab, the chosen tail radius, and the finite collection of local regularity neighborhoods covering the bounded cylinder; in particular it may depend on the given solution u . $\square$

Lemma 16.2 (Positive-time L^5 integrability). *Under the hypotheses of Theorem 3.13, $u \in L^5(\mathbb{R}^3 \times (\delta, T))$ for every $\delta > 0$.*

Proof. Fix $\delta > 0$ and take $B_{\delta/2}$ from Lemma 16.1. Set $w = |u|^{3/2}$ on $Q_{\delta, T}$. The critical hypothesis gives $\text{ess sup}_{t \in (\delta, T)} \|w(t)\|_2^2 \leq M^3$. The weak chain rule, obtained by applying the Sobolev chain rule to smooth truncations of $z \mapsto |z|^{3/2}$ and passing by monotone convergence, gives

$$\int_{Q_{\delta, T}} |\nabla w|^2 dx dt \leq CB_\delta \int_{Q_{\delta, T}} |Du|^2 dx dt < \infty.$$

Indeed $|\nabla w| \leq (3/2)|u|^{1/2}|Du|$ almost everywhere and $|u| \leq B_\delta$ there. The multiplicative inequality proved in [2, Lemma 8.4], applied to w on this slab, now yields

$$\int_{Q_{\delta, T}} |u|^5 dx dt = \int_{Q_{\delta, T}} |w|^{10/3} dx dt \leq C \left(\text{ess sup}_{t \in (\delta, T)} \|w(t)\|_2 \right)^{4/3} \int_{Q_{\delta, T}} |\nabla w|^2 dx dt < \infty.$$

All constants are finite because the Leray–Hopf energy inequality controls the last integral. This proves the assertion. $\square$

Theorem 16.3 (The L^5 bound and uniqueness). *Let $0 < T < \infty$, $a \in J$, and let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum a . Suppose*

$$\text{ess sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty.$$

Then $u \in L^5(\mathbb{R}^3 \times (0, T))$, and every other Leray–Hopf solution (v, Dv) on the same interval with the same datum satisfies $u = v$ almost everywhere on $\mathbb{R}^3 \times (0, T)$.

Proof. Lemma 12.1 gives $a \in L^2 \cap L^3 \cap J$, so Proposition 14.1 supplies a Leray–Hopf solution U on $[0, \tau]$ with datum a and $U \in L^5(Q_\tau)$. Put $\sigma = \min\{\tau, T\}$ and restrict U to $[0, \sigma]$. Lemma 15.1 on this interval identifies $u = U$ almost everywhere on Q_σ , so $u \in L^5(Q_\sigma)$. Choose $0 < \delta < \sigma$. Lemma 16.2 gives $u \in L^5(\mathbb{R}^3 \times (\delta, T))$. The union of these two cylinders is Q_T , proving the asserted global L^5 bound.

For any other Leray–Hopf solution v with the same datum, apply Lemma 15.1 on $[0, T]$ to u and v . It gives the required almost-everywhere equality. The energy inequality for v is the only energy information used on that side of the comparison. $\square$

Remark 16.4 (Departure from [4]). The proof follows the positive-time reduction (3.5)–(3.7) and the short-time construction of Theorems 7.3–7.4 in [4], printed pp. 221–222 and 244–248. We derive the initial L^3 trace from good times and the strong L^2 trace, prove the special heat estimate rather than use the general mixed-norm assertion (7.4), and write the fixed-point limit in the spaces used here. On positive-time cylinders, the translated and rescaled local energy equality of Lemma 9.1 gives suitability with the canonical pressure; the global velocity bound and L^5 estimate then follow from the explicit spatial tails. Weak–strong uniqueness is proved at exponents (5, 5) by the zero-start energy comparison of Robinson, Rodrigo, and Sadowski [10, (8.12), printed p. 174], with the cross-testing identity derived by spatial mollification instead of their Serrin regularity theorem and testing lemma (Theorem 8.17 and Lemma 8.18). No general mixed-norm estimate from ESS Lemma 7.1(7.4) and no result attributed to Giga is used.

Part VI. The Ladyzhenskaya–Prodi–Serrin theorem

17 Statements

For $3 < s < \infty$ put $\ell(s) = 2s/(s - 3)$. The notation $u \in L^{\ell(s)}(0, T; L^s(\mathbb{R}^3))$ refers to the mixed norm of the velocity on the open slab, whereas the endpoint $s = \infty$ means $u \in L^2(0, T; L^\infty(\mathbb{R}^3))$. In both cases $3/s + 2/\ell = 1$. Equality of two weak solutions below means equality almost everywhere on $\mathbb{R}^3 \times (0, T)$; their prescribed time-slice representatives need not agree at every point. The following is the whole-space theorem in the form of [4, Theorem 1.2, printed p. 213]; that source attributes uniqueness to Prodi and Serrin [9, 12] and smoothness to Ladyzhenskaya [5]. Serrin’s earlier interior regularity result is [11]. A later full-range source is [10, Theorems 8.17 and 8.19, printed pp. 173–174]. Its Leray–Hopf class includes a strong energy inequality from good positive times; the present class supplies the needed restart through Lemma 18.4.

Theorem 17.1 (Ladyzhenskaya–Prodi–Serrin). *Let $0 < T < \infty$, let $a \in J$, and let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum a . Suppose either that $u \in L^{\ell(s)}(0, T; L^s(\mathbb{R}^3))$ for some $3 < s < \infty$, or that $u \in L^2(0, T; L^\infty(\mathbb{R}^3))$. Then every Leray–Hopf solution with datum a on $[0, T]$ equals u almost everywhere on the open slab. There is a representative $\tilde{u} = u$ almost everywhere on that slab which is C^∞ on $\mathbb{R}^3 \times (0, T]$, with all space-time derivatives at T computed within this domain.*

Theorem 16.3 supplies the critical L^5 norm needed to apply Theorem 17.1 at $(s, \ell) = (5, 5)$. Thus the smoothness mentioned in Remark 3.14 is proved here as a separate consequence of that theorem, while its original statement remains the L^5 and uniqueness result.

Corollary 17.2 (Smoothness at the $L^\infty L^3$ endpoint). *Let $0 < T < \infty$, $a \in J$, and let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum a . If $\text{ess sup}_{0 < t < T} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty$, then $u \in L^5(\mathbb{R}^3 \times (0, T))$, it is unique among Leray–Hopf solutions with datum a , and it has a representative that is C^∞ on $\mathbb{R}^3 \times (0, T]$ in the one-sided sense at T .*

Together, Corollary 17.2 and Theorem 17.1 cover the whole range $3 \leq s \leq \infty$ of the regularity criterion.

Corollary 17.3 (The regularity criterion for $3 \leq s \leq \infty$). *Let $0 < T < \infty$, $a \in J$, and let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum a . Suppose that one of the following holds:*

$\operatorname{ess\,sup}_{0 < t < T} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty$; or $u \in L^\ell(0, T; L^s(\mathbb{R}^3))$ for some $3 < s < \infty$ with $\ell = 2s/(s-3)$; or $u \in L^2(0, T; L^\infty(\mathbb{R}^3))$. Then every Leray–Hopf solution with datum a on $[0, T]$ equals u almost everywhere on $\mathbb{R}^3 \times (0, T)$, and u has a representative that is C^∞ on $\mathbb{R}^3 \times (0, T]$ in the one-sided sense at T .

Proof. In the first case apply Corollary 17.2; in the other two, apply Theorem 17.1. $\square$

18 Energy equality and comparison

The weak equation has tests supported away from the time endpoints. A Serrin bound makes the convection term integrable when the solution tests its own equation, and the energy equality is obtained from the mollified cross-testing identity of Part V with the solution in both slots. Definition 3.3 may prescribe nonmeasurable slices on exceptional times. The energy equality and its positive-time restart are therefore asserted only at almost every measurable time, without changing that definition. A whole-space energy identity under L^4 control also appears in [13, Lemma 4.3, printed pp. 72–73]. For comparison, [6, Theorem 12.4, printed pp. 359–361] treats weak–strong uniqueness. The proof below works with almost-every-time slices, as required by the measurability clauses of the Leray–Hopf definition.

Lemma 18.1 (Mixed-norm interpolation). *Under the hypotheses of Theorem 17.1, set $q = 2s/(s-2)$ and $r = 2s/3$ when $s < \infty$, and set $(q, r) = (2, \infty)$ when $s = \infty$. Then $u \in L^r(0, T; L^q(\mathbb{R}^3))$ and $u \in L^4(\mathbb{R}^3 \times (0, T))$. The same energy-class estimate $w \in L_t^r L_x^q$ holds for any Leray–Hopf solution w on $[0, T]$.*

Proof. For almost every t , the weak gradient clause of (LH1) and the whole-space homogeneous Sobolev inequality give $\|w(t)\|_6 \leq C_S \|Dw(t)\|_2$. For finite s , put $\theta = 3/s$. The identities $1/q = (1 - \theta)/2 + \theta/6$ and $r\theta = 2$ give

$$\int_0^T \|w(t)\|_q^r dt \leq C_S^2 \left(\operatorname{ess\,sup}_{0 < t < T} \|w(t)\|_2 \right)^{r-2} \int_0^T \|Dw(t)\|_2^2 dt < \infty. \quad (18.1)$$

The vector Sobolev inequality follows componentwise and the equivalence of the coordinate and Euclidean norms changes only C_S . At $s = \infty$ the assertion is exactly $w \in L_t^\infty L_x^2$. For finite s , $1/s + 1/q = 1/2$ and $1/\ell(s) + 1/r = 1/2$. Hölder’s inequality first in space and then in time therefore gives $\|u\|_{L_{t,x}^2}^2 \leq \|u\|_{L_t^{\ell(s)} L_x^s} \|u\|_{L_t^r L_x^q}$. The same calculation at $s = \infty$ uses $L_t^2 L_x^\infty$ and $L_t^\infty L_x^2$. Hence $u^2 \in L^2$, which is the asserted space-time L^4 bound. $\square$

Lemma 18.2 (Good positive times). *If (u, Du) is a Leray–Hopf solution on $[0, T]$, then there is a full-measure set $G \subset (0, T)$ such that for every $t_0 \in G$, $u(\cdot, t_0) \in H^1(\mathbb{R}^3; \mathbb{R}^3) \cap J$, with weak gradient $Du(\cdot, t_0)$; this slice is measurable as an L^2 field. Every nonempty open subinterval of $(0, T)$ contains a time in G .*

Proof. Fubini applied to (LH1) gives finite L^2 norms of $u(t)$ and $Du(t)$ for almost every t and identifies $Du(t)$ as the weak spatial gradient there. Intersect these times with those on which the weak-divergence clause of (LH1) holds; by Lemma 3.2, $u(t) \in J$. This intersection has full measure. An open interval has positive Lebesgue measure, so it meets that set. $\square$

Lemma 18.3 (Energy chain rule for square-integrable time derivatives). *Let $a < b$, put $Q = \mathbb{R}^3 \times (a, b)$, and let $w, r \in L^2(Q)$ satisfy $\partial_t w = r$ in the sense of distributions on Q .*

(i) The function w has a representative $\bar{w} \in C([a, b]; L^2(\mathbb{R}^3))$, and for $a \leq s \leq t \leq b$

$$\|\bar{w}(t)\|_2^2 - \|\bar{w}(s)\|_2^2 = 2 \int_s^t \int_{\mathbb{R}^3} w r \, dx \, d\tau.$$

(ii) Let $j \in \{1, 2, 3\}$ and suppose in addition that the weak spatial derivatives $g = \partial_j w$ and $h = \partial_j g$ on Q belong to $L^2(Q)$. Then g has a representative $\bar{g} \in C([a, b]; L^2(\mathbb{R}^3))$, and for $a \leq s \leq t \leq b$

$$\|\bar{g}(t)\|_2^2 - \|\bar{g}(s)\|_2^2 = -2 \int_s^t \int_{\mathbb{R}^3} h r \, dx \, d\tau.$$

The same statements hold componentwise for vector fields.

Proof. (i) Regard w as an element of $L^2(a, b; L^2(\mathbb{R}^3))$. Its distributional time derivative r lies in the same space, so $w \in W^{1,2}(a, b; L^2(\mathbb{R}^3))$. Such a function agrees for almost every time with the absolutely continuous curve $\bar{w}(t) = \bar{w}(a) + \int_a^t r \, d\tau$, which is continuous on the closed interval, endpoints included. The product rule for absolutely continuous Hilbert-space valued curves gives the displayed identity.

(ii) Let ρ_ε be a standard spatial mollifier, and for $\varepsilon, \varepsilon' > 0$ put $\kappa = \rho_\varepsilon - \rho_{\varepsilon'}$; convolutions act in x at almost every time. The field $d = \kappa * g = (\partial_j \kappa) * w$ lies in $L^2(Q)$, because $\|\kappa * g(t)\|_2 \leq \|\kappa\|_1 \|g(t)\|_2$, and $\partial_t d = (\partial_j \kappa) * r = \partial_j(\kappa * r) \in L^2(Q)$. Part (i) applies to d , with representative $\bar{d}$: for $s \leq t$,

$$\|\bar{d}(t)\|_2^2 - \|\bar{d}(s)\|_2^2 = 2 \int_s^t \int \partial_j(\kappa * r) (\kappa * g) \, dx \, d\tau = -2 \int_s^t \int (\kappa * r) (\kappa * h) \, dx \, d\tau,$$

where $\kappa * g \in H^1$ has $\partial_j(\kappa * g) = \kappa * h$ at almost every time, and the derivative was moved onto this factor. Hence $|\|\bar{d}(t)\|_2^2 - \|\bar{d}(s)\|_2^2| \leq 2\|\kappa * r\|_{L^2(Q)} \|\kappa * h\|_{L^2(Q)}$. Averaging over $s \in (a, b)$ gives for every $t \in [a, b]$

$$\|\bar{d}(t)\|_2^2 \leq (b-a)^{-1} \|d\|_{L^2(Q)}^2 + 2\|\kappa * r\|_{L^2(Q)} \|\kappa * h\|_{L^2(Q)}.$$

As $\varepsilon, \varepsilon' \rightarrow 0$ the fields $\rho_\varepsilon * g$, $\rho_\varepsilon * r$ and $\rho_\varepsilon * h$ converge in $L^2(Q)$ to g , r and h , so the right side tends to zero. The curves $\bar{g}_\varepsilon$ representing $\rho_\varepsilon * g$ are therefore uniformly Cauchy in $C([a, b]; L^2(\mathbb{R}^3))$, and their limit $\bar{g}$ is a representative of g . The same computation with κ replaced by ρ_ε gives the identity for $\bar{g}_\varepsilon$, and it passes to the limit $\varepsilon \rightarrow 0$. $\square$

Lemma 18.4 (Energy equality at almost every time). *Under the hypotheses of Theorem 17.1, there is a full-measure set $E \subset (0, T)$ of times at which the slice $u(\cdot, t)$ is a measurable L^2 field such that for every $t \in E$,*

$$\frac{1}{2} \|u(t)\|_2^2 + \int_0^t \|Du(\tau)\|_2^2 \, d\tau = \frac{1}{2} \|a\|_2^2. \quad (18.2)$$

Consequently, for $t_0, t \in E$ with $t_0 < t$,

$$\frac{1}{2} \|u(t)\|_2^2 + \int_{t_0}^t \|Du(\tau)\|_2^2 \, d\tau = \frac{1}{2} \|u(t_0)\|_2^2. \quad (18.3)$$

No equality is asserted for arbitrary prescribed time slices.

Proof. Put $Q_T = \mathbb{R}^3 \times (0, T)$. Lemma 18.1 gives $u \in L^4(Q_T)$, hence, with $u \in L^2(Q_T)$, also $u \in L^{5/2}(Q_T)$ and $u \in L^3(Q_T)$. By Theorem 3.7 there is $p \in L^{5/3}(Q_T)$ with which u satisfies the momentum equation against compactly supported smooth vector tests, and the pairing of the slices of u with a fixed test field is continuous on $[0, T]$ and equals the pairing with a at $t = 0$ by (LH2) and (LH5). These are the inputs of the mollified cross-testing calculation in the proof of Lemma 15.1, which we apply with u in both slots and with L^4 in place of L^5 . Its products are integrable: $|u|^2|Du| \in L^1(Q_T)$ because $1/4 + 1/4 + 1/2 = 1$, and the pressure pairs integrably with $u \in L^{5/2}(Q_T)$ because $3/5 + 2/5 = 1$. The mollified slices converge in $L^4(Q_T)$ and their gradients in $L^2(Q_T)$, and the cutoff terms tend to zero by the bound C/R on the cutoff derivative and the same integrability. The initial pairing converges to $\|a\|_2^2$. For almost every $t \in (0, T)$ we obtain

$$\|u(t)\|_2^2 - \|a\|_2^2 = -2 \int_0^t \int u_i u_j \partial_j u_i dx d\tau - 2 \int_0^t \|Du(\tau)\|_2^2 d\tau.$$

No time derivative of u in a negative Sobolev space and no chain rule in time are used.

It remains to show that the transport integral vanishes. By Fubini's theorem and Lemma 18.2, for almost every τ the slice $u(\tau)$ lies in $H^1(\mathbb{R}^3) \cap J \cap L^4(\mathbb{R}^3)$, with weak gradient $Du(\tau)$. For such a slice $|u|^2 \in W_{\text{loc}}^{1,4/3}$, because $|u||Du| \in L^{4/3}$. Let χ_R be a smooth cutoff equal to one on B_R , supported in B_{2R} , with $|\nabla \chi_R| \leq C/R$. Since $u \in L^4 = (L^{4/3})^*$, the weak-divergence identity extends from smooth compactly supported tests to $\chi_R|u|^2/2 \in W^{1,4/3}$, and gives $\int \chi_R u_i u_j \partial_j u_i dx = -\frac{1}{2} \int |u|^2 u \cdot \nabla \chi_R dx$. The right side is bounded by $CR^{-1} \int_{B_{2R} \setminus B_R} |u|^3 \leq CR^{-1} |B_{2R}|^{1/4} \|u\|_4^3 \rightarrow 0$. Since $u_i u_j \partial_j u_i \in L^1(\mathbb{R}^3)$ for such a slice, dominated convergence gives $\int u_i u_j \partial_j u_i dx = 0$. The space-time integral of $u_i u_j \partial_j u_i$ is defined because $|u|^2|Du| \in L^1(Q_T)$, and it vanishes because its slice integrals do. Let E be the set of good times of Lemma 18.2 at which the displayed identity holds. Then E has full measure and the identity is (18.2); subtraction gives (18.3). $\square$

The comparison below is stated for two Leray–Hopf solutions with the same datum at time zero. A comparison after a positive restart time is obtained in Lemma 19.4 by concatenating u with the strong solution and applying this lemma to the concatenation. The nonlinear estimate uses the same exponent q as Lemma 18.1; the gradient of the difference remains in L^2 .

Lemma 18.5 (Mixed-norm comparison). *Let $0 < T < \infty$, $a \in J$, and let (u, Du) and (v, Dv) be Leray–Hopf solutions on $\mathbb{R}^3 \times [0, T]$ with datum a . Suppose that u satisfies a mixed-norm hypothesis of Theorem 17.1 on $(0, T)$. Then $v = u$ almost everywhere on $\mathbb{R}^3 \times (0, T)$.*

Proof. Put $Q = \mathbb{R}^3 \times (0, T)$. By Theorem 3.7, u and v have pressures $p_u, p_v \in L^{5/3}(Q)$ with which they satisfy the momentum equation against compactly supported smooth vector tests, and the pairing of their slices with a fixed test field is continuous on $[0, T]$ and equals the pairing with a at $t = 0$. By Lemma 18.4, u satisfies (18.2) for almost every t , and by (LH4) the field v satisfies the corresponding inequality with Dv .

For either energy field $w = u, v$, the interpolation in (18.1) gives $w \in L_t^r L_x^q$. Hölder with $1/s + 1/q + 1/2 = 1$ and $1/\ell + 1/r + 1/2 = 1$ makes each of $|u||v||Du|$ and $|u||v||Dv|$ integrable. It also controls $|u|^2|Du|$; independently, $u \in L^4(Q)$ makes $|u|^2|Dz|$ integrable for $z = v - u$. At $s = \infty$ the same bounds use $(\ell, q, r) = (2, 2, \infty)$. These are precisely the nonlinear factors retained below; no term $|v|^2|Du|$ is needed.

Use the pressure-inclusive weak equations with a time cutoff, spatially mollified fields, and a spatial cutoff χ_R . Before removing the mollifier, write the convection of v tested against u_n in advective form $\int (v \cdot \nabla) v \cdot u_n$, using $\text{div } v = 0$. Thus one never takes a limit in $\int (v \otimes v) : \nabla u_n$. For the u equation tested against v_n , retain $u \otimes u : \nabla v_n$: the tensor is in $L^2(Q)$ because $u \in L^4(Q)$, and

$\nabla v_n \rightarrow \nabla v$ strongly in $L^2(Q)$, for every s . For the v equation tested against u_n , the advective factor $(v \cdot \nabla)v$ belongs to $L_t^{\ell'} L_x^{s'}$ by energy interpolation. When $s < \infty$, spatial mollification gives $u_n \rightarrow u$ strongly in $L_t^{\ell} L_x^s$, so this second term also converges. At $s = \infty$, strong convergence in $L_t^2 L_x^{\infty}$ is not asserted: $u_n \rightarrow u$ almost everywhere and $|u_n(x, t)| \leq \|u(t)\|_{\infty}$. Dominated convergence applies to the second term because $\|u(t)\|_{\infty} \|v(t)\|_2 \|Dv(t)\|_2 \in L_t^1$.

The cutoff remainders also vanish. Besides $|u|^3$ and $|u||v|^2$, the transport integrations produce $|v||u|^2$. Here $u \in L^{5/2}(Q)$ by interpolation of L^2 and L^4 , $v \in L^{10/3}(Q)$ by the energy interpolation, and the mixed Hölder estimate with one u in $L_t^{\infty} L_x^2$ handles $|v||u|^2$ on the finite time interval. The cutoff tests are not divergence free. Their pressure errors are bounded by

$$\frac{C}{R} \int_Q |p_v||u| + |p_u||v|,$$

which is finite because $p_u, p_v \in L^{5/3}(Q)$, while u and v lie in $L^{5/2}(Q)$ (by interpolation of L^2 with L^4 and with $L^{10/3}$ respectively) and $3/5 + 2/5 = 1$. Every cutoff error tends to zero as $R \rightarrow \infty$. The common datum gives the initial pairing $\|a\|_2^2$ by (LH2) and (LH5); take the terminal limit through times where the weak slices hold. The result is, for almost every $t \in (0, T)$,

$$\langle u(t), v(t) \rangle + 2 \int_0^t \int Du : Dv = \|a\|_2^2 - \int_0^t \int ((u \cdot \nabla)u \cdot v + (v \cdot \nabla)v \cdot u).$$

This is the mollified cross-testing calculation of Lemma 15.1 with its integrable products made explicit for the mixed exponents.

Set $z = v - u$ and let I denote the nonlinear integral in the last display. The cancellation is carried out only with integrable terms. First, $\int (u \cdot \nabla)u \cdot u = 0$, using $|u|^2 |Du| \in L^1$ and cutoff remainder $|u|^3$. Second, $\int (v \cdot \nabla)u \cdot u = 0$, using $|v||u||Du| \in L^1$ and remainder $|v||u|^2$. Third, $\int (u \cdot \nabla)z \cdot u = -\int (u \cdot \nabla)u \cdot z$, using $|u|^2 |Dz|$ and $|u||z||Du|$. Expanding $v = u + z$ in I and applying these three identities gives $I = \int_0^t \int (z \cdot \nabla)z \cdot u$, where $(z \cdot \nabla)z \cdot u = \sum_k u_k \sum_j z_j \partial_j z_k$. The cross-testing identity therefore takes the relative form

$$\langle v(t), u(t) \rangle - \|a\|_2^2 = - \int_0^t \int (z \cdot \nabla)z \cdot u \, dx \, d\tau - 2 \int_0^t \int Du : Dv \, dx \, d\tau \quad (18.4)$$

for almost every $t \in (0, T)$. Now expand $\|z(t)\|_2^2 = \|v(t)\|_2^2 - 2\langle v(t), u(t) \rangle + \|u(t)\|_2^2$ and insert, for almost every t , the energy inequality (LH4) for v , (18.4), and the energy equality (18.2) for u :

$$\|z(t)\|_2^2 \leq \left(\|a\|_2^2 - 2 \int_0^t \|Dv\|_2^2 \right) - 2 \left(\|a\|_2^2 - \int_0^t \int (z \cdot \nabla)z \cdot u - 2 \int_0^t \int Du : Dv \right) + \left(\|a\|_2^2 - 2 \int_0^t \|Du\|_2^2 \right),$$

which equals $2 \int_0^t \int (z \cdot \nabla)z \cdot u - 2 \int_0^t \|Dz\|_2^2$. Thus, for almost every t ,

$$\frac{1}{2} \|z(t)\|_2^2 + \int_0^t \|Dz(\tau)\|_2^2 \, d\tau \leq \int_0^t \int_{\mathbb{R}^3} (z \cdot \nabla)z \cdot u \, dx \, d\tau \leq \int_0^t \int_{\mathbb{R}^3} |u| |z| |Dz| \, dx \, d\tau. \quad (18.5)$$

For finite s , whole-space Sobolev interpolation gives $\|z\|_q \leq C \|z\|_2^{1-3/s} \|Dz\|_2^{3/s}$. Young's inequality with conjugate exponents $2/(1+3/s)$ and $2/(1-3/s) = \ell$ yields

$$\|u\|_s \|z\|_q \|Dz\|_2 \leq \frac{1}{2} \|Dz\|_2^2 + C_s \|u\|_s^{\ell} \|z\|_2^2. \quad (18.6)$$

At $s = \infty$, the same result follows from $\|u\|_\infty \|z\|_2 \|Dz\|_2 \leq \frac{1}{2} \|Dz\|_2^2 + C \|u\|_\infty^2 \|z\|_2^2$. Put $m(\tau) = \|u(\tau)\|_s^\ell$ (with $m = \|u\|_\infty^2$ at $s = \infty$) and $\phi(t) = \|z(t)\|_2^2$. By hypothesis $m \in L^1(0, T)$, and ϕ is essentially bounded by the energy class. Inserting the last estimate in (18.5) and absorbing the gradient term gives, in integrated form, $\phi(t) \leq K \int_0^t m \phi d\tau$ for almost every t , with $K = 2C_s$ (an absolute constant at $s = \infty$). The function $\Phi(t) = \int_0^t m \phi d\tau$ is absolutely continuous with $\Phi' = m\phi \leq Km\Phi$ almost everywhere and $\Phi(0) = 0$, so $\Phi = 0$, and then $\phi = 0$ almost everywhere. Hence $z = 0$ almost everywhere on Q . This is the whole-space relative-energy mechanism of [10, Theorem 8.19 and (8.12), printed p. 174]; no smoothness of u was used. $\square$

19 Strong solutions and continuation

The Fourier– L^2 mild construction of the CKN paper [2] supplies smooth solutions when the transport velocity is mollified. The next proposition takes the mollifier to zero while retaining an H^1 bound independent of its radius. This is the whole-space strong-solution construction underlying [10, Theorem 6.15, printed pp. 144–145]; it avoids the expanding-ball compactness passage of that source.

Lemma 19.1 (Regularized Sobolev start). *Let $b \in J$ and $\varepsilon > 0$, let U_ε and p_ε be the solution and pressure of [2, Theorem 9.4] with datum b , and put $V_\varepsilon = J_\varepsilon U_\varepsilon$. For every finite $S > 0$:*

- (a) *for every integer $k \geq 0$, $U_\varepsilon \in C([0, S]; H^k)$ and $U_\varepsilon(0) = J_\varepsilon b$ in H^k ;*
- (b) *for $t > 0$ the time derivative of U_ε exists classically and $\partial_t U_\varepsilon = R_\varepsilon - \nabla p_\varepsilon$, where $R_\varepsilon = \Delta U_\varepsilon - (V_\varepsilon \cdot \nabla) U_\varepsilon$ is a continuous map from $(0, S]$ into $L^2(\mathbb{R}^3; \mathbb{R}^3)$;*
- (c) *the function $y_\varepsilon(t) = \|\nabla U_\varepsilon(t)\|_2^2$ is continuous on $[0, S]$ and differentiable on $(0, S)$, and for $0 < t < S$*

$$y'_\varepsilon(t) = -2\|\nabla^2 U_\varepsilon(t)\|_2^2 + 2\langle (V_\varepsilon(t) \cdot \nabla) U_\varepsilon(t), \Delta U_\varepsilon(t) \rangle. \quad (19.1)$$

The constants of these statements may depend on ε ; the transport bound in Proposition 19.2 does not. Identity (19.1) is the only case of the higher-order energy identities that is used below.

Proof. (a) [2, Lemma 9.2] puts the initial state $J_\varepsilon b$ in every H^k . For each fixed k , the H^k heat and projected Stokes multiplier bounds in the proof of [2, Theorem 9.4], together with [2, (174)], make the mild map contractive on a short interval in $C_t H^k$. The initial heat term converges in H^k at $t = 0$ by dominated convergence of the Fourier multiplier. The Stokes integral tends to zero there because its kernel is bounded by $C_\varepsilon(t-s)^{-1/2}$ and its norm by $2C_\varepsilon \sqrt{t} \sup_{s \leq t} \|U_\varepsilon(s)\|_{H^k}$. Mild uniqueness identifies this path with the L^2 solution. At each restart the L^2 energy bound fixes a positive step length for the regularized equation; the finite restart argument in the proof of [2, Theorem 9.4] extends the H^k path to S .

(b) The pointwise equation is (R3) of [2, Theorem 9.4]. By (a) with $k = 2$, $t \mapsto \Delta U_\varepsilon(t)$ is continuous into L^2 . By [2, Lemma 9.2], $\|V_\varepsilon(t) - V_\varepsilon(s)\|_\infty \leq C_\varepsilon \|U_\varepsilon(t) - U_\varepsilon(s)\|_2$ and $\|V_\varepsilon(s)\|_\infty \leq C_\varepsilon \|b\|_2$, so

$$\|(V_\varepsilon(t) \cdot \nabla) U_\varepsilon(t) - (V_\varepsilon(s) \cdot \nabla) U_\varepsilon(s)\|_2 \leq \|V_\varepsilon(t) - V_\varepsilon(s)\|_\infty \|\nabla U_\varepsilon(t)\|_2 + \|V_\varepsilon(s)\|_\infty \|\nabla U_\varepsilon(t) - \nabla U_\varepsilon(s)\|_2,$$

which tends to zero as $s \rightarrow t$ by (a) with $k = 1$. Hence R_ε is continuous into L^2 .

(c) Continuity of y_ε on $[0, S]$ is (a) with $k = 1$. For $f, g \in H^2$ the symmetry of the Laplacian gives $\|\nabla f\|_2^2 - \|\nabla g\|_2^2 = \langle f - g, -\Delta(f + g) \rangle$. Let $0 < a < c \leq S$, take $f = U_\varepsilon(c)$, $g = U_\varepsilon(a)$, and put $w = -\Delta(U_\varepsilon(a) + U_\varepsilon(c)) \in L^2$; it is weakly divergence free, because both slices are.

By (R2), $\partial_t U_\varepsilon \in L^2(\mathbb{R}^3 \times (a, c))$ is continuous, so $U_\varepsilon(c) - U_\varepsilon(a) = \int_a^c \partial_t U_\varepsilon dr$ pointwise, and Fubini's theorem against the fixed field w gives $\langle U_\varepsilon(c) - U_\varepsilon(a), w \rangle = \int_a^c \langle R_\varepsilon(r) - \nabla p_\varepsilon(r), w \rangle dr$. For almost every r we have $p_\varepsilon(r) \in L^2$ and $\nabla p_\varepsilon(r) = R_\varepsilon(r) - \partial_t U_\varepsilon(r) \in L^2$, so $p_\varepsilon(r) \in H^1$ and $\langle \nabla p_\varepsilon(r), w \rangle = -\langle p_\varepsilon(r), \operatorname{div} w \rangle = 0$. Therefore

$$y_\varepsilon(c) - y_\varepsilon(a) = \int_a^c \langle R_\varepsilon(r), -\Delta(U_\varepsilon(a) + U_\varepsilon(c)) \rangle dr.$$

Only the pressure-free field R_ε needs continuity in time; no time derivative of ∇U_ε is taken. Divide by $c - a$ and let $a \uparrow t$ and $c \downarrow t$ for a fixed $0 < t < S$. Continuity of R_ε from (b) and of ΔU_ε from (a) give $y'_\varepsilon(t) = 2\langle R_\varepsilon(t), -\Delta U_\varepsilon(t) \rangle$, which is (19.1) because $\|\Delta U_\varepsilon(t)\|_2 = \|\nabla^2 U_\varepsilon(t)\|_2$ by Plancherel. This argument starts at $t = 0$ in H^1 because the mollified initial state is in that space; it is not inferred from (R1)–(R2) alone. $\square$

Proposition 19.2 (Local strong solution). *For every $b \in H^1(\mathbb{R}^3; \mathbb{R}^3) \cap J$ there is a $\tau_b > 0$, depending only on $\|b\|_{H^1}$, a solenoidal solution U of the unregularized equation on $[0, \tau_b]$, and a pressure $p \in L^2(\mathbb{R}^3 \times (0, \tau_b))$ for which the momentum equation with pressure holds in the weak sense against compactly supported smooth vector tests, such that*

$$U \in C([0, \tau_b]; H^1) \cap L^2(0, \tau_b; H^2), \quad \partial_t U \in L^2(0, \tau_b; L^2), \quad U(0) = b. \quad (19.2)$$

It has energy equality from every time and defines a Leray–Hopf solution on this interval. A choice of τ_b bounded below by $c(1 + \|b\|_{H^1})^{-4}$ is possible, with absolute $c > 0$.

Proof. Apply [2, Theorem 9.4] with datum b and mollifier radius $\varepsilon \downarrow 0$. Lemma 19.1 adds the H^1 continuity at zero and the differentiated gradient-energy identity (19.1) needed here; these are not clauses (R1)–(R5) of that theorem. Write $y = \|\nabla U_\varepsilon\|_2^2$ and $d = \|\nabla^2 U_\varepsilon\|_2^2 = \|\Delta U_\varepsilon\|_2^2$. The transport velocity $J_\varepsilon U_\varepsilon$ is divergence free, and Young's inequality for convolution and the homogeneous $H^1 \rightarrow L^6$ inequality give $\|J_\varepsilon U_\varepsilon\|_6 \leq C\|\nabla U_\varepsilon\|_2$. By Hölder's inequality and the interpolation of ∇U_ε between L^2 and L^6 , whose L^6 norm is controlled by $\|\nabla^2 U_\varepsilon\|_2$,

$$|\langle (J_\varepsilon U_\varepsilon \cdot \nabla) U_\varepsilon, \Delta U_\varepsilon \rangle| \leq \|J_\varepsilon U_\varepsilon\|_6 \|\nabla U_\varepsilon\|_3 \|\Delta U_\varepsilon\|_2 \leq C y^{3/4} d^{3/4}$$

with an absolute constant, independent of ε and of the mollifier profile. Inserting this in (19.1) and using Young's inequality with exponents $4/3$ and 4 gives, for $t > 0$,

$$\frac{d}{dt} \|\nabla U_\varepsilon\|_2^2 + \|\Delta U_\varepsilon\|_2^2 \leq C \|\nabla U_\varepsilon\|_2^6. \quad (19.3)$$

The initial field is $J_\varepsilon b$, so its L^2 and H^1 norms do not exceed those of b . Set $X_0 = 1 + \|b\|_{H^1}^2$. Since y is continuous on $[0, S]$ and satisfies $y' \leq C y^3$ on $(0, S)$ for every S , comparison of $X' = C X^3$ with $X(0) = X_0$ (applied on $[s, \tau]$ and letting $s \downarrow 0$) shows that for $\tau_b = c X_0^{-2}$, with c chosen so that $2cC < 1/2$, $\sup_{0 \leq t \leq \tau_b} \|\nabla U_\varepsilon(t)\|_2^2 \leq 2X_0$. Integrating (19.3) over $[s, \tau_b]$, using this bound and letting $s \downarrow 0$ controls $\int_0^{\tau_b} \|\Delta U_\varepsilon\|_2^2$ independently of ε . The global L^2 energy identity adds $\sup_t \|U_\varepsilon(t)\|_2^2 \leq \|b\|_2^2$.

The nonlinear term belongs uniformly to $L_t^2 L_x^2$: $\|(J_\varepsilon U_\varepsilon \cdot \nabla) U_\varepsilon\|_2 \leq \|J_\varepsilon U_\varepsilon\|_\infty \|\nabla U_\varepsilon\|_2 \leq C \|U_\varepsilon\|_{H^2} \|U_\varepsilon\|_{H^1}$. The projected equation therefore bounds $\partial_t U_\varepsilon$ in $L_t^2 L_x^2$; the Leray projection has operator norm one on L^2 . [2, Proposition 12.1], applied to this family with datum b , supplies a subsequence converging strongly in L^2 on bounded space-time cylinders and weakly in the energy gradients. Its compactness input is [2, Lemma 10.6], which gives the required time-pairing

modulus; the present $L_t^2 L_x^2$ time-derivative bound is an additional estimate, not a replacement for that input. The uniform H^2 bound improves local convergence to strong $L_t^2 H_x^1$: for a cutoff χ and $W = \chi(U_\varepsilon - U)$, integration by parts gives $\|\nabla W\|_2^2 \leq \|W\|_2 \|\Delta W\|_2$; the first factor tends to zero and the second is bounded. Weak compactness retains $U \in L_t^\infty H_x^1 \cap L_t^2 H_x^2$ and $\partial_t U \in L_t^2 L_x^2$. The convergence $J_\varepsilon U_\varepsilon - U \rightarrow 0$ locally in $L_t^2 H_x^1$ follows by writing it as $J_\varepsilon(U_\varepsilon - U) + (J_\varepsilon U - U)$ and using the uniform derivative bound and approximate-identity convergence. The quadratic transport term therefore converges in distributions to $(U \cdot \nabla)U$; pressure is recovered by the fixed double-Riesz formula of [2, Section 8], and lies in $L^2(\mathbb{R}^3 \times (0, \tau_b))$ because $U \otimes U \in L_t^\infty L_x^2$ and the Riesz projection is bounded on L^2 . The initial value follows from the uniform time-pairing modulus and $J_\varepsilon b \rightarrow b$ in L^2 .

For the time continuity, apply Lemma 18.3 componentwise. Here $U \in L_t^2 H_x^2$, so the weak derivatives $\partial_j U$ and $\partial_j \partial_j U$ lie in L^2 , and $\partial_t U \in L_t^2 L_x^2$. Part (i) of the lemma gives the L^2 continuity of U together with $\|U(t)\|_2^2 - \|U(s)\|_2^2 = 2 \int_s^t \int U \cdot \partial_t U$. Part (ii), applied to each derivative $\partial_j U_i$ and summed over i, j , gives the L^2 continuity of ∇U together with $\|\nabla U(t)\|_2^2 - \|\nabla U(s)\|_2^2 = -2 \int_s^t \int \Delta U \cdot \partial_t U$. Thus $U \in C_t H_x^1$. Choose this continuous representative for the strong solution; it agrees almost everywhere with the limit field of [2, Proposition 12.1]. The resulting energy identity gives the L^2 trace and, after pairing the projected equation with $-\Delta U$, the H^1 trace. The nonlinearity is in $L_t^2 L_x^2$, so the equation and spatial cutoffs justify the energy equality for every subinterval. These properties, weak continuity, and the weak equation verify (LH1)–(LH5) with datum b ; in particular no positive-time inequality is inserted as an extra hypothesis. $\square$

Lemma 19.3 (Serrin control of the H^1 norm). *Let U be a strong solution on $[t_0, t_1]$, that is, a solenoidal field in the class (19.2) with an L^2 pressure as in Proposition 19.2, with $t_0 < t_1 \leq T$, and suppose it has either mixed norm in Theorem 17.1 there. For finite s , with $\ell = 2s/(s-3)$, there is a constant C_s depending only on s and the whole-space Sobolev constant such that*

$$\|\nabla U(t)\|_2^2 + \int_{t_0}^t \|\Delta U(\tau)\|_2^2 d\tau \leq \|\nabla U(t_0)\|_2^2 \exp \left(C_s \int_{t_0}^t \|U(\tau)\|_s^\ell d\tau \right) \quad (19.4)$$

for $t \in [t_0, t_1]$. At $s = \infty$ the same estimate holds with $\ell = 2$ and an absolute constant.

Proof. Pair the momentum equation with $-\Delta U$. This is legitimate for $U \in L^2 H^2$ and $\partial_t U \in L^2 L^2$ by part (ii) of Lemma 18.3, which gives $\|\nabla U(t)\|_2^2 - \|\nabla U(s)\|_2^2 = -2 \int_s^t \langle \partial_t U, \Delta U \rangle$. At almost every time $\partial_t U = \Delta U - (U \cdot \nabla)U - \nabla p$ in L^2 , and $\langle \nabla p, \Delta U \rangle = 0$ because ΔU is solenoidal and $\nabla p \in L^2$ is a gradient of an L^2 function; hence $\frac{d}{dt} \|\nabla U\|_2^2 = -2 \|\Delta U\|_2^2 + 2 \langle (U \cdot \nabla)U, \Delta U \rangle$ for almost every t . If $3 < s < \infty$, let $q = 2s/(s-2)$. Hölder and homogeneous Sobolev interpolation applied to ∇U , whose L^6 norm is controlled by the full Hessian norm $\|\nabla^2 U\|_2 = \|\Delta U\|_2$ (Plancherel), yield

$$\left| \int (U \cdot \nabla)U \cdot \Delta U \right| \leq \|U\|_s \|\nabla U\|_q \|\Delta U\|_2 \leq C_s \|U\|_s \|\nabla U\|_2^{1-3/s} \|\Delta U\|_2^{1+3/s}. \quad (19.5)$$

Young's inequality with powers $2/(1+3/s)$ and $2/(1-3/s) = \ell$ bounds the last expression by $\|\Delta U\|_2^2/2 + C_s \|U\|_s^\ell \|\nabla U\|_2^2$. At $s = \infty$, use $\|U\|_\infty \|\nabla U\|_2 \|\Delta U\|_2$ and Young with powers two. The coefficients are integrable on the interval. Indeed, $Y(t) = \|\nabla U(t)\|_2^2 + \int_{t_0}^t \|\Delta U(\tau)\|_2^2 d\tau$ satisfies $Y'(t) \leq C_s \|U(t)\|_s^\ell Y(t)$ for almost every t (with $\ell = 2$ at $s = \infty$). Gronwall gives the displayed estimate with this same C_s . This is the whole-space calculation of [10, Lemma 8.16, printed pp. 171–172], including its separately displayed $L_t^2 L_x^\infty$ case. $\square$

Lemma 19.4 (Continuation beyond the comparison interval). *Let (u, Du) satisfy the hypotheses of Theorem 17.1, and choose $t_0 < T$ in the intersection of the good-time set of Lemma 18.2 and the*

equality set of Lemma 18.4. Let S be the set of $t' > t_0$ for which there is a strong solution U on $[t_0, t']$ with $U(t_0) = u(t_0)$ in L^2 and $U = u$ almost everywhere on $\mathbb{R}^3 \times (t_0, \min\{t', T\})$. Then S contains a time larger than T .

Proof. Step 1: comparison after a restart by concatenation. Let $t' > t_0$ and let W be a strong solution on $[t_0, t']$ with $W(t_0) = u(t_0)$ in L^2 . We claim that $W = u$ almost everywhere on $\mathbb{R}^3 \times (t_0, \sigma)$, $\sigma = \min\{t', T\}$. Let V be the field equal to u at times $t < t_0$ and to W at times $t \geq t_0$, and let DV be defined in the same way from Du and DW . We check that (V, DV) is a Leray–Hopf solution with datum a on $[0, \sigma]$.

(LH1): V and DV are measurable and square integrable on $\mathbb{R}^3 \times (0, \sigma)$, being so for u on $(0, t_0)$ and for W on (t_0, σ) ; $\|V(t)\|_2$ is essentially bounded, because u is and $W \in C([t_0, \sigma]; L^2)$; the weak-gradient and weak-divergence clauses hold at almost every time from those of u and the strong class. (LH2): the pairings with a fixed $w \in L^2$ are continuous for $t \leq t_0$ by (LH2) for u , and continuous on $[t_0, \sigma]$ by the strong L^2 continuity of W ; they match at t_0 because $W(t_0) = u(t_0)$, a measurable slice since t_0 is a good time. (LH3): let $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, \sigma); \mathbb{R}^3)$ be divergence free. Testing the equation of u with φ times a time cutoff tending to the indicator of $(0, t_0)$, and using (LH2), shows that the weak-equation integral of u over $(0, t_0)$ equals $-\langle u(t_0), \varphi(t_0) \rangle$. The strong equation of W (pressure terms drop against divergence-free tests) and $W \in C([t_0, \sigma]; L^2)$ show that the integral of W over (t_0, σ) equals $+\langle W(t_0), \varphi(t_0) \rangle$. The two boundary pairings agree and cancel, which is the weak equation for V on $(0, \sigma)$ with no mass at t_0 . (LH4): for $t \leq t_0$ this is (LH4) for u . For $t \geq t_0$, the energy identity of the strong solution (valid for every strong solution: apply part (i) of Lemma 18.3 with $r = \partial_t W = \Delta W - (W \cdot \nabla)W - \nabla p$, and use $\langle \Delta W, W \rangle = -\|\nabla W\|_2^2$, $\langle (W \cdot \nabla)W, W \rangle = 0$, $\langle \nabla p, W \rangle = 0$), $\frac{1}{2}\|W(t)\|_2^2 + \int_{t_0}^t \|DW\|_2^2 = \frac{1}{2}\|W(t_0)\|_2^2$, together with (18.2) at $t_0 \in E$ and $W(t_0) = u(t_0)$, gives $\frac{1}{2}\|V(t)\|_2^2 + \int_0^t \|DV\|_2^2 = \frac{1}{2}\|a\|_2^2$. (LH5): the strong trace at zero is that of u .

The restriction of (u, Du) to $[0, \sigma]$ is a Leray–Hopf solution with the same datum, and it satisfies the mixed-norm hypothesis on $(0, \sigma)$ since $\sigma \leq T$. Lemma 18.5 on $[0, \sigma]$, with u in the Serrin role and V as the rival solution, gives $V = u$ almost everywhere on $\mathbb{R}^3 \times (0, \sigma)$, hence $W = u$ almost everywhere on $\mathbb{R}^3 \times (t_0, \sigma)$. The pressures entering the comparison are those of Theorem 3.7 for u and for V ; the strong solution enters only through the Leray–Hopf class of V .

Step 2: the set S is nonempty. Proposition 19.2 translated to t_0 gives a strong solution W on $[t_0, t_0 + \tau_b]$ with $W(t_0) = u(t_0) \in H^1 \cap J$. By Step 1, $W = u$ almost everywhere on the common interval, so $t_0 + \tau_b \in S$.

Step 3: extension. Suppose every $t' \in S$ satisfies $t' \leq T$. On any such interval, $U = u$ almost everywhere, so Lemma 19.3 gives

$$\sup_{t_0 \leq t \leq t'} \|U(t)\|_{H^1}^2 + \int_{t_0}^{t'} \|\Delta U(t)\|_2^2 dt \leq M,$$

where M depends only on $\|u(t_0)\|_{H^1}$, $\|u(t_0)\|_2$, and the Serrin integral on (t_0, T) , not on t' . The L^2 part is bounded by the strong energy identity. Proposition 19.2, translated to any endpoint $t' \in S$, therefore provides a lifespan $\tau_* > 0$ bounded below in terms of M alone. Restart from $U(t') \in H^1 \cap J$ and glue the two strong fields. Continuity in H^1 at the junction eliminates a distributional time jump; the two L^2 time derivatives remain integrable on the union, and the pressures are glued in the same way. The glued field is a strong solution on $[t_0, t' + \tau_*]$ with the same value $u(t_0)$ at t_0 , so Step 1 shows that it agrees with u almost everywhere on $\mathbb{R}^3 \times (t_0, \min\{t' + \tau_*, T\})$. Thus $t' + \tau_* \in S$. If $\sup S \leq T$, choose $t' \in S$ with $t' > \sup S - \tau_*/2$; the preceding extension exceeds $\sup S$, a contradiction. Hence some member of S exceeds T . This uses the continuation remark

after [10, Corollary 6.9, printed p. 137]; on $\mathbb{R}^3$ the local construction is the one underlying [10, Theorem 6.15, printed pp. 144–145]. $\square$

20 Smooth representatives

The strong solution identifies the weak field on every interval after a good time. On a positive-time slab it then gains spatial regularity by a direct linear ladder: a bound for $\partial_t U$ in L_x^2 gives the H^2 regularity of every slice, and the whole-space heat gain, applied to the projected convection term as a source, gives one further order at a time. The projected equation supplies all time derivatives. The endpoint t_1 is included by working on closed time intervals throughout. On $\mathbb{R}^3$ there is no boundary compatibility condition.

Lemma 20.1 (Joint smoothness from Sobolev curves). *Let $t_0 < t_1$ and let V be a space-time L_{loc}^2 field on $\mathbb{R}^3 \times (t_0, t_1)$. Suppose that for every $\delta \in (0, t_1 - t_0)$ there are curves $Z^j \in C([t_0 + \delta, t_1]; H^m(\mathbb{R}^3))$, for all integers $j, m \geq 0$ (each $Z^j(t)$ being one element of $\bigcap_m H^m$), such that $Z^0(t) = V(\cdot, t)$ for almost every t , and for every $\psi \in C_c^\infty(\mathbb{R}^3)$ and $j \geq 0$ the function $t \mapsto \langle Z^j(t), \psi \rangle$ is differentiable on $[t_0 + \delta, t_1]$, one-sidedly at the endpoints, with derivative $\langle Z^{j+1}(t), \psi \rangle$. Then V has a representative that is jointly C^∞ on $\mathbb{R}^3 \times (t_0, t_1]$, with one-sided time derivatives at t_1 .*

Proof. Fix δ . The continuous Sobolev embedding $H^m \hookrightarrow C_b^k$, $m > k + 3/2$, turns each curve Z^j into a continuous curve in C_b^k for every k . Since Z^{j+1} is continuous with values in H^m , the differentiability of the pairings with all test functions gives $Z^j(t) - Z^j(s) = \int_s^t Z^{j+1}(\tau) d\tau$ as a Bochner integral in H^m , hence in C_b^k , so Z^j is a C^1 curve into C_b^k with derivative Z^{j+1} (one-sided at the endpoints). Evaluation of a C_b^k function at x is jointly C^k in the function and the x variable. Composing the evaluation with the curves shows that $(x, t) \mapsto Z^0(t)(x)$ has continuous mixed partial derivatives $\partial_t^j \partial_x^\alpha = \partial_x^\alpha Z^j(t)(x)$ of every order $j + |\alpha|$, jointly in (x, t) on $\mathbb{R}^3 \times [t_0 + \delta, t_1]$; it is therefore C^∞ there, with one-sided derivatives at the ends of the interval. It equals V almost everywhere. For $\delta' < \delta$ the two smooth functions agree almost everywhere on the common set, hence everywhere by continuity, so they patch as $\delta \downarrow 0$ to a representative on $\mathbb{R}^3 \times (t_0, t_1]$. For formalization, the embedding into C_b^k can be constructed by Fourier inversion and differentiation under the integral, as in the $H^4 \hookrightarrow C_b^2$ argument of the CKN paper [2], iterated to higher orders. $\square$

Proposition 20.2 (Smoothing of strong solutions). *If U is a strong solution on $[t_0, t_1]$ with $t_0 < t_1$, that is, a solenoidal field in the class (19.2) with an L^2 pressure as in Proposition 19.2, then for every $\delta > 0$ with $t_0 + \delta < t_1$ and every integer $m \geq 0$, $U \in C([t_0 + \delta, t_1]; H^m(\mathbb{R}^3))$ and $U \in L^2(t_0 + \delta, t_1; H^{m+1}(\mathbb{R}^3))$. It has a representative smooth jointly in space and time on $\mathbb{R}^3 \times (t_0, t_1]$, with one-sided time derivatives at t_1 .*

Proof. Write $Q = \mathbb{R}^3 \times (t_0, t_1)$. The strong class gives $U, \nabla U, \nabla^2 U, \partial_t U \in L^2(Q)$ and $U \in C([t_0, t_1]; H^1)$. At almost every time $\partial_t U = \Delta U - (U \cdot \nabla)U - \nabla p$ with all terms in L^2 . The fields $\partial_t U$ and ΔU are solenoidal and ∇p is a gradient, so the Helmholtz decomposition gives the projected equation

$$\partial_t U - \Delta U = -\mathbb{P}B, \quad B = (U \cdot \nabla)U, \quad (20.1)$$

in which the pressure no longer appears.

Step 1: $\partial_t U$ is bounded in L_x^2 after positive time. Let $M(t) = \|U(t)\|_{H^2}^2$, so $M \in L^1(t_0, t_1)$, and let K be the constant of the embedding $H^2 \hookrightarrow L^\infty$. For small $h > 0$ put $v_h(t) = h^{-1}(U(t+h) - U(t))$ on $[t_0, t_1 - h]$. Since $U(t+h) \otimes U(t+h) - U(t) \otimes U(t)$ equals $U(t+h) \otimes (U(t+h) - U(t)) + (U(t+h) - U(t)) \otimes U(t)$, the difference of the equations at times $t+h$ and t , divided by h , states that v_h

is solenoidal and solves $\partial_t v_h - \Delta v_h + \nabla \cdot F_h + \nabla q_h = 0$, with flux $|F_h| \leq (|U(t+h)| + |U(t)|)|v_h|$. We have $v_h \in L^2 H^2$ and $\partial_t v_h \in L^2 L^2$, so part (i) of Lemma 18.3 gives the energy identity for $E_h = \|v_h\|_2^2$; the pressure term vanishes against the solenoidal v_h , and

$$E'_h \leq -2\|\nabla v_h\|_2^2 + (\|U(t+h)\|_\infty + \|U(t)\|_\infty) 2\|v_h\|_2 \|\nabla v_h\|_2 \leq -\|\nabla v_h\|_2^2 + \kappa_h E_h,$$

where $\kappa_h(t) = 2K^2(M(t) + M(t+h))$ has integral at most $4K^2\|M\|_{L^1}$, uniformly in h . Moreover $\|v_h(s)\|_2^2 \leq h^{-1} \int_0^h \|\partial_t U(s+\theta)\|_2^2 d\theta$, so $\int_{t_0}^{t_0+\delta} E_h \leq \|\partial_t U\|_{L^2(Q)}^2$. Averaging the Gronwall inequality $E_h(t) \leq E_h(s) \exp \int \kappa_h$ over $s \in (t_0, t_0 + \delta)$ gives, for $t \in [t_0 + \delta, t_1 - h]$,

$$E_h(t) \leq B_\delta := \delta^{-1} e^{4K^2\|M\|_{L^1}} \|\partial_t U\|_{L^2(Q)}^2,$$

independently of h . Since $v_h \rightarrow \partial_t U$ in $L^2(Q)$, weak lower semicontinuity gives $\|\partial_t U(t)\|_2^2 \leq B_\delta$ for almost every $t \in (t_0 + \delta, t_1)$.

Step 2: H^2 bound of the slices and the base of the ladder. At almost every time $U(t) \in H^2$ solves the stationary system $\Delta U = \partial_t U + (U \cdot \nabla)U + \nabla p$. Test it with ΔU : the pressure drops because ΔU is solenoidal, and $|\langle (U \cdot \nabla)U, \Delta U \rangle| \leq \|U\|_6 \|\nabla U\|_3 \|\Delta U\|_2 \leq C \|\nabla U\|_2^{3/2} \|\nabla^2 U\|_2^{3/2}$ by Sobolev interpolation, with $\|\nabla^2 U\|_2 = \|\Delta U\|_2$. Young's inequality gives $\|\nabla^2 U(t)\|_2 \leq 2\|\partial_t U(t)\|_2 + C\|\nabla U(t)\|_2^3$. By Step 1 and $U \in C([t_0, t_1]; H^1)$, $\sup \|U(t)\|_{H^2} \leq C_\delta$ for almost every $t \in (t_0 + \delta, t_1)$. Consequently $B = (U \cdot \nabla)U$ is bounded in H^1 there: $|\nabla B| \leq |\nabla U|^2 + |U| |\nabla^2 U|$, with $\|\nabla U\|_4^2 \leq C\|U\|_{H^2}^2$ and $\|U\|_\infty \|\nabla^2 U\|_2 \leq C\|U\|_{H^2}^2$.

We use the whole-space heat gain. Let $m \geq 1$, $a < a + \sigma < a + \beta$, let $z \in L^2((a, a + \beta); H^m(\mathbb{R}^3))$ and $G \in L^2((a, a + \beta); H^{m-1}(\mathbb{R}^3))$ satisfy $\partial_t z - \Delta z = G$ in distributions. Then $z \in C([a + \sigma, a + \beta]; H^m) \cap L^2((a + \sigma, a + \beta); H^{m+1})$ with norm at most $C(\|z\|_{L^2 H^m} + \|G\|_{L^2 H^{m-1}})$, where C depends only on m , σ and β . This is the whole-space form of Lemma 8.11: multiply by a time cutoff vanishing for $t \leq a + \sigma/2$ and equal to one for $t \geq a + \sigma$; each derivative ∂^γ , $|\gamma| \leq m$, of the product solves a heat equation with square-integrable source or divergence-form source and zero initial trace, and the energy estimate of Lemma 8.10 with $v = 0$ applies with no spatial cutoff. The estimate includes the value at the terminal time $a + \beta$ in its supremum, so the closed terminal interval is covered.

Apply it, using the bounds above with $\delta/2$ in place of δ , with $m = 2$ on $(a, a + \beta) = (t_0 + \delta/2, t_1)$ and $\sigma = \delta/2$ to $z = U$ and $G = -\mathbb{P}B$; here $\|\mathbb{P}B\|_{H^1} \leq C\|B\|_{H^1}$ because $\mathbb{P}$ is a bounded Fourier multiplier. This gives $U \in C([t_0 + \delta, t_1]; H^2) \cap L^2(t_0 + \delta, t_1; H^3)$ for every δ .

Step 3: the ladder. We show by induction on m that for every $\delta > 0$ with $t_0 + \delta < t_1$, $U \in C([t_0 + \delta, t_1]; H^m) \cap L^2(t_0 + \delta, t_1; H^{m+1})$. The cases $m \leq 1$ are contained in the strong class and $m = 2$ is Step 2. Let $m \geq 2$ and assume the statement for m . Apply it with $\delta/2$ in place of δ . Then U is bounded in H^m and $\nabla U \in L^2 H^m$ on $(t_0 + \delta/2, t_1)$, so by the algebra inequality $\|fg\|_{H^m} \leq C_m \|f\|_{H^m} \|g\|_{H^m}$ (valid since $m > 3/2$), $B = \sum_j U_j \partial_j U \in L^2 H^m$ there, and $\mathbb{P}B \in L^2 H^m$. Now $U \in L^2 H^{m+1}$ solves (20.1) with source $-\mathbb{P}B \in L^2 H^m$. The heat gain of order $m+1$ with $(a, a + \beta) = (t_0 + \delta/2, t_1)$ and $\sigma = \delta/2$ gives $U \in C([t_0 + \delta, t_1]; H^{m+1}) \cap L^2(t_0 + \delta, t_1; H^{m+2})$. This proves all the spatial assertions on the closed interval $[t_0 + \delta, t_1]$.

Step 4: the time chain. Fix δ and put $I = [t_0 + \delta, t_1]$. Define $Z^0 = U$ and, recursively,

$$Z^{j+1} = \Delta Z^j - \mathbb{P} \sum_{i=0}^j \binom{j}{i} (Z^i \cdot \nabla) Z^{j-i}.$$

If $Z^0, \dots, Z^j \in C(I; H^m)$ for every m , then $Z^{j+1} \in C(I; H^m)$ for every m , by the algebra property and the boundedness of $\mathbb{P}$ on H^m ; so by Step 3 and induction all Z^j belong to $C(I; H^m)$ for every

m . By (20.1), $\partial_t Z^0 = Z^1$ in the sense of distributions on $(t_0 + \delta, t_1)$. If $\partial_t Z^i = Z^{i+1}$ for $i \leq j$, then the product rule for products of curves that are continuous into H^m and have continuous distributional time derivatives (by time mollification), together with Pascal's rule for the binomial coefficients, gives $\partial_t Z^{j+1} = Z^{j+2}$. For each $\psi \in C_c^\infty(\mathbb{R}^3)$ the pairing $\langle Z^j(t), \psi \rangle$ is thus a C^1 function on the open interval with the continuous derivative $\langle Z^{j+1}(t), \psi \rangle$, and therefore is differentiable on the closed interval I , one-sidedly at the ends, with this derivative.

Step 5: joint smoothness. Step 4 gives exactly the hypotheses of Lemma 20.1, for every δ , with $V = U$; the identification $Z^0(t) = U(t)$ for almost every t was established in Step 3. The lemma produces a representative smooth jointly on $\mathbb{R}^3 \times (t_0, t_1]$ with one-sided derivatives at t_1 . No continuation beyond t_1 is used for the endpoint: all estimates hold on the closed intervals $[t_0 + \delta, t_1]$. This is the whole-space version of [10, Theorems 7.1, 7.3 and 7.5, printed pp. 149–154]; the argument above uses a direct heat-gain ladder and the Fourier multipliers of the CKN paper [2] in place of the periodic Galerkin basis used for the detailed source proof. $\square$

Remark 20.3 (An a posteriori energy inequality). Once Proposition 20.2 is proved, $U \in C^1(I; H^m)$, $I = [t_0 + \delta, t_1]$, for every m and δ by Step 4, and for $m \geq 2$ with $\|U\|_{H^m}^2 = \sum_{|\alpha| \leq m} \|\partial^\alpha U\|_2^2$ pairing the projected equation with $\partial^\alpha U$ and integrating one derivative by parts in the divergence form of the convection term gives

$$\frac{d}{dt} \|U\|_{H^m}^2 + \|\nabla U\|_{H^m}^2 \leq C_m \|U\|_{H^m}^4. \quad (20.2)$$

Indeed the heat pairing gives $-\|\nabla U\|_{H^m}^2$, the projection is orthogonal and drops out against the solenoidal $\partial^\alpha U$, and the tensor-gradient pairing is bounded by $C_m \|U\|_{H^m}^2 \|\nabla U\|_{H^m}$ by the algebra property, after which Young's inequality is used. This inequality is not used in the proof of the proposition, which does not pass through regularized solutions at higher order.

Proof of Theorem 17.1. Lemmas 18.1 and 18.4 give energy equality for u at almost every measurable time. At $t_0 = 0$, the strong initial trace (LH5) gives the common L^2 trace a through those times for u and for any rival Leray–Hopf solution. Theorem 3.11 and the L^4 estimate give the L^2 canonical pressure for u ; Theorem 3.7 gives an $L^{5/3}$ pressure for the rival. Lemma 18.5 with $t_0 = 0$ identifies them almost everywhere. The rival's (LH4) supplies its energy inequality.

Choose t_0 in the intersection of the full-measure good-time and energy-equality sets of Lemmas 18.2 and 18.4. Subtracting two instances of (18.2) gives the equality from t_0 for almost every later time. The trace assertion of the energy lemma identifies the strong L^2 trace there with $u(t_0)$; no positive-time energy clause is added to the stated Leray–Hopf definition. Proposition 19.2 constructs a strong solution from this datum. It belongs to $L_t^2 L_x^\infty$ by $H^2 \hookrightarrow L^\infty$ and its $L_t^2 H^2$ bound. Lemma 18.5, with that strong field in the Serrin role and the original weak field in the energy role, identifies them on their common interval. Lemma 19.4 repeats this identification at each restart and extends the strong field beyond T .

For any $\delta \in (0, T)$ choose such a $t_0 < \delta/2$. Proposition 20.2 gives a smooth representative on $\mathbb{R}^3 \times [\delta, T]$. Representatives from two choices of good time agree almost everywhere on the overlap. Since both are continuous, they agree everywhere there: a nonempty open set on which they differ would have positive measure. The representatives thus patch as $\delta \downarrow 0$. The strong continuation past T supplies the one-sided space-time derivatives there. $\square$

Proof of Corollary 17.2. Theorem 16.3 gives $u \in L^5(Q_T)$ and its uniqueness conclusion. Fubini identifies this spacetime norm with $L^5(0, T; L^5(\mathbb{R}^3))$, and $3/5 + 2/5 = 1$. Apply Theorem 17.1 with $(s, \ell) = (5, 5)$ to obtain the smooth representative through the top time. $\square$

A Corrections to the sources

Escauriaza–Seregin–Šverák (2003)

1. **Cylinder-sign typo.** Theorem 1.4, printed p. 214, writes the ambient cylinder as $B_1 \times (0, 1)$, while its Sobolev hypothesis and target backward cylinder use negative times. The coherent cylinder is $B_1 \times (-1, 0)$. The same inconsistency occurs in the IMA preprint. The manuscript states the negative-time domain in Theorem 3.12 and proves the shifted-cylinder argument in Lemma 8.8.
2. **Test-space mismatch.** Definition 2.1, printed p. 215, names spatial tests $\phi \in C_0^\infty(\mathbb{R}^3)$, but its local energy formula uses $\phi(x, t)$ and $\partial_t \phi$. The IMA preprint repeats the mismatch. The precise space-time test class is not fixed there. The manuscript fixes its suitable-solution test clauses in Definition 3.1.
3. **Higher-order regularity delegated.** Lemma 2.2, printed p. 216, asserts bounds for every spatial derivative order but refers $k > 1$ to an external Stokes regularity result instead of proving the induction. Its proof ends on printed p. 220; §3 begins on p. 221. Constants are indexed by k , but their dependence is not developed. Theorem 8.14 supplies the finite weak-to-regular bootstrap needed here, but does not reproduce the source’s all-orders theorem or supply the all-orders exterior estimates in (3.31), (3.38), and (3.42)–(3.46).
4. **Compressed Campanato compactness.** Propositions 2.4–2.6 and the proof of Lemma 2.2, printed pp. 216–220, appeal to “well-known compactness” and leave the relevant function-space operators and pressure-limit details compressed. No result of the manuscript relies on this argument; the manuscript uses the CKN regularity criterion [2] and proves the compactness it needs in [2, Lemma 11.1].
5. **Time representative and all-time L^3 bound.** The assertion following (3.1), printed p. 221, extends weak continuity from the Leray–Hopf representative and the essential $L_t^\infty L_x^3$ bound to every time, but does not spell out the density and lower-semicontinuity argument. The manuscript writes that representative and modulus in Lemma 9.2.
6. **Strong regularity estimates are used without a detailed proof.** The local estimate (3.10), printed p. 222, is cited through mixed-norm coercive estimates and supplies much more than the weak continuity and L^3 -slice bound needed there. The manuscript replaces those conclusions by the momentum-pairing modulus in Lemma 9.2; this does not claim the full derivative package in (3.10).
7. **Singular-point lower bound is externally imported.** The lower bound (3.13), printed p. 223, is attributed to Seregin–Šverák [39], a result not used here. The contrapositive of CKN Theorem A does not replace it directly: Theorem A gives full-neighborhood regularity below the top face but only one-sided Hölder control at the top, so its contrapositive does not yield ESS’s lower bound at a full-neighborhood singular point. The manuscript avoids the singular-point formulation. Lemma 8.2 proves top-boundary ε -regularity with threshold $\varepsilon_0/8$, and Definition 8.3 defines good points by that smallness. Lemma 8.4 shows that the solution is Hölder on a compact set of good points, and that a point which is not good satisfies the lower bound (8.3) at every admissible scale. The blow-up argument of Proposition 9.5 uses only this lower bound, so no input from [39] is needed.
8. **Blow-up compactness conclusion is stronger than the replacement lemma.** The

- estimates (3.21)–(3.23), printed pp. 224–225, invoke nonstationary Stokes coercivity to obtain derivative bounds, strong L^3 convergence, and $C([a, b]; L^2_{\text{loc}})$ convergence. The cited nonstationary Stokes coercivity is not proved in detail. The manuscript compactness lemma [2, Lemma 11.1] gives strong local space-time L^2 , weak gradients, and weak slice convergence at every time; it does not claim the stronger $C_t L^2_{\text{loc}}$ topology. The blow-up proof here instead handles its endpoint and strong- L^3 passages using pairing continuity, weak slice convergence, lower semicontinuity, and time-strip estimates.
9. **Pressure-splitting normalization matters.** Equations (3.11)–(3.19) begin on printed p. 223 and end on p. 224. In these equations, p_1 is defined by a local Dirichlet problem and p_2 is that split’s harmonic remainder. The decay in (3.19) belongs to this fixed remainder; it cannot be transferred to a different harmonic remainder chosen after rescaling. The manuscript’s Lemma 9.3 instead fixes a whole-space Riesz pressure of the zero-extended tensor and its harmonic difference, then proves the interior bound and rescaled vanishing directly. It does not transfer (3.19) to that changed split or redefine the remainder after rescaling.
 10. **Time reversal in the vorticity equation.** The §3 vorticity equation and (3.32), printed pp. 226–227, use $\partial_t - \Delta$; §5 is written for $\partial_t + \Delta$ in its forward variable. Reversing time reconciles them. This is a convention to state when applying Theorem 6.7, not a sign correction to either displayed equation.
 11. **Exterior continuation proof is abbreviated.** Equations (3.38)–(3.46) are printed on pp. 227–228, and the closing text continues on p. 229. They use all-order velocity, time, and pressure bounds, a divergence correction via stationary Stokes, local strong solvability, and linear regularity; the closing harmonic L^3 Liouville step is only described. The manuscript does not follow this route. Theorem 8.14 and Lemma 10.2 give a bounded, continuous vorticity of class $W^{2,1}_2$ with zero trace, which is the class of Theorem 5.2 and Theorem 6.7. The infinite-order vanishing hypothesis of Theorem 5.2 follows from Lemma 10.1, and the Liouville step is Lemma 10.4. No higher-order regularity is used.
 12. **Initial-time L^5 estimate.** Lemma 7.1, printed pp. 243–244, explicitly cites an external theorem for its general mixed-norm estimate (7.4); Theorem 7.4, printed pp. 246–248, abbreviates the contraction and limit details. These are inputs to the L^5 and uniqueness conclusions of the source’s Theorem 1.3, not to its local Theorem 1.4. The manuscript proves those conclusions in Theorem 16.3 without the general estimate (7.4): Lemma 12.3 proves the special heat estimate that is needed, and Proposition 14.1 carries out the fixed-point construction and its limit (see Remark 16.4).
 13. **Pointwise boundary trace and dimension conventions.** Theorem 4.1, equations (4.1)–(4.3), printed p. 230, evaluates $u(x, 0)$ although the cylinder is open and does not specify a trace or representative. It states a $W^{2,1}_2$ hypothesis and a differential inequality but concludes pointwise vanishing. The source uses a spatial $\mathbb{R}^3$ cylinder, writes an $\mathbb{R}^n$ -valued field, and uses a ρ^{n-1} factor in the proof; it also concludes on the whole ball although Lemma 4.3 estimates only an inner ball before saying the extension is immediate. The notation $A_0(R, T)$ in §4 and the small absolute A_0 in §5 are unrelated. Theorem 5.2 fixes the continuous trace convention and dimension three; its proof uses the Gaussian-box estimate in Lemma 5.1 and gives the explicit radius iteration that enlarges the inner-ball estimate to the whole ball.
 14. **Gaussian propagation omits closure and radius steps.** Lemma 4.3, printed pp. 230–233, applies Proposition 6.1 to a cutoff field in $W^{2,1}_2$, while that proposition is stated for smooth

compactly supported fields; the density passage and the singular $\varepsilon \downarrow 0$ bound are not supplied. It also moves from an inner-ball estimate to the whole target ball without the radius iteration and cites the local L^2 -to-pointwise estimate without proof. The manuscript supplies the smooth-to- $W_2^{2,1}$ passage in Lemma 4.4, proves the Gaussian-box estimate in Lemma 5.1, and carries out the explicit radius iteration in Theorem 5.2.

15. **Half-space trace and weak-gradient hypotheses.** Theorem 5.1, equations (5.2)–(5.4), printed p. 234, does not define the zero initial trace needed for extension across $t = 0$; without it, the extension can create a boundary distribution. Condition (5.4) also omits the weak spatial gradient assumed explicitly in Seregin’s corresponding class. The source sets the spatial cylinder in $\mathbb{R}_+^n$ and the field target in $\mathbb{R}^n$, then states that the result also holds for every finite target dimension m . The manuscript and the formalization fix $n = 3$. The source’s repeated symbol A_0 denotes unrelated quantities in §§4 and 5. Theorem 6.7 makes the continuous trace and weak-gradient clauses explicit.
16. **Omitted quantitative choice and noncompact test field.** Lemma 5.2, printed pp. 234–236, existentially chooses $A_0 < 1/32$; it does not display a numerical choice small enough for the later bounds, although the stated existential choice permits one. This is an omitted explicit quantitative choice, not a threshold contradiction. Lemma 5.3, printed pp. 236–238, applies Proposition 6.2 to a noncompact Sobolev cutoff without a cutoff/tail limit or density argument and compresses the propagation step from Theorem 4.1. Its general-dimensional notation writes $\mathbb{R}_+^3$ where $\mathbb{R}_+^n$ is required. The manuscript computes the absorption constants in Proposition 4.2, extends the estimate by density in Lemma 4.4, and carries out the needed cutoff and iteration in Theorem 6.7.
17. **Final large-growth iteration is compressed.** The proof of Lemma 5.4 and Theorem 5.1, printed p. 238, invokes the small-growth lemma with a parameter $M > A_0$, outside that lemma’s stated hypothesis; the trace and rescaling steps are also skipped. A valid repetition requires translating and rescaling each slab and justifying its trace. The manuscript gives the corresponding explicit iteration in Theorem 6.7; the source’s compressed step is not a counterexample to Theorem 5.1.
18. **Smooth Carleman estimates applied beyond their stated class.** Propositions 6.1–6.2, printed pp. 238–243, are stated for smooth compactly supported fields, but the preceding applications require Sobolev cutoff fields. Proposition 6.1 asserts an absolute positive constant c_0 but gives no numerical value; Proposition 6.2 compresses the final Cauchy–Schwarz absorption without recording the uniform constants. The manuscript computes explicit constants in Proposition 4.1 and Proposition 4.2, and supplies the density extension in Lemma 4.4.

Sources of the existence and pressure results

The corrections to Ożański–Pooley, Tsai, Robinson–Rodrigo–Sadowski and Lemarié–Rieusset, and the corrections to Theorem 1.1 and equations (3.2)–(3.4) of Escauriaza–Seregin–Šverák, concern the existence and associated-pressure results of Section 7. They are recorded in the appendix of the CKN paper [2].

Seregin lecture notes

1. Appendix A.1, Proposition 1.1 and Proposition 1.2, printed pp. 229–234, give the matching Carleman calculations but also compress the final L^2 absorption/Cauchy–Schwarz step after (A.1.26); in particular the uniform constant is not derived there. The manuscript computes the constants in Proposition 4.1 and Proposition 4.2.
2. Appendix A.2, Theorem 2.1 and Lemma 2.2, repeats the pointwise Gaussian route and compressed inner-to-whole-ball continuation; Appendix A.3 repeats the half-space proof’s final iteration without fully stating its trace/rescaling details. The manuscript instead supplies an integral-box estimate and explicit radius/large-growth iterations in Lemma 5.1, Theorem 5.2, and Theorem 6.7.
3. Appendix A.3, condition (A.3.4), explicitly includes weak $\nabla u \in L^2_{\text{loc}}$, which ESS’s (5.4) omits. The appendix’s initial trace convention is still unstated. Its Q_1^+ notation uses $\mathbb{R}_+^3 + e_n$ in an n -dimensional argument; it should be $\mathbb{R}_+^n$. The manuscript fixes spatial and target dimension three and makes the weak-gradient and continuous trace clauses explicit in Theorem 6.7. The notes also describe the first Carleman cutoff with support outside a larger parabolic cylinder; after the separate cutoff at the initial time, the estimates agree with the published proof. Appendix A.3 has the ordinary editorial typos “result result” and “fist integral,” which do not change the argument.

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